6.2.5 The Effect of Non-normality on $\bar{x}$ and R Charts

There often is an assumption that links **normality** and **control charts** in the development of the performance properties of $\bar{x}$ and R control charts; that is, that the underlying distribution of the quality characteristic is **normal.** In many situations, we may have reason to doubt the validity of this assumption. For example, we may know that the underlying distribution is not normal, because we have collected extensive data that indicate the normality assumption is inappropriate. Now if we know the form of the underlying distribution, it is possible to derive the sampling distributions of $\bar{x}$ and R (or some other measure of process variability) and to obtain exact probability limits for the control charts. This approach could be difficult in some cases, and most analysts would probably prefer to use the standard approach based on the normality assumption if they believed that the effect of departure from this assumption was not serious. However, we may know nothing about the form of the underlying distribution, and then our only choice may be to use the normal theory results. Obviously, in either case, we would be interested in knowing the effect of departures from normality on the usual control charts for $\bar{x}$ and R.

Several authors have investigated the effect of departures from normality on control charts. Burr (1967) notes that the usual normal theory control limit constants are very robust to the normality assumption and can be employed unless the population is extremely non-normal. Schilling and Nelson (1976), Chan, Hapuarachchi, and Macpherson (1988), and Yourstone and Zimmer (1992) have also studied the effect of non-normality on the control limits of the $\bar{x}$ chart. Schilling and Nelson investigated the uniform, right triangular, gamma (with $\lambda = 1$ and $r = \frac{1}{2}$, 1, 2, 3, and 4), and two bimodal distributions formed as mixtures of two normal distributions. Their study indicates that, in most cases, samples of size 4 or 5 are sufficient to ensure reasonable robustness to the normality assumption.

The worst cases observed were for small values of r in the gamma distribution [$r = \frac{1}{2}$ and $r = 1$ (the exponential distribution)]. For example, they report the actual α-risk to be 0.014 or less if $n \geq 4$ for the gamma distribution with $r = \frac{1}{2}$, as opposed to a theoretical value of 0.0027 for the normal distribution.

While the use of three-sigma control limits on the $\bar{x}$ chart will produce an α-risk of 0.0027 if the underlying distribution is normal, the same is not true for the R chart. The sampling distribution of R is not symmetric, even when sampling from the normal distribution, and the long tail of the distribution is on the high or positive side. Thus, symmetric three-sigma limits are only an approximation, and the α-risk on such an R chart is *not* 0.0027. (In fact, for $n = 4$, it is $\alpha = 0.00461$.) Furthermore, the R chart is more sensitive to departures from normality than the $\bar{x}$ chart.

Once again, it is important to remember the role of theory and assumptions such as normality and independence. These are fundamental to the study of the performance of the $\bar{x}$ control chart which is very useful for assessing its suitability for phase II, but plays a much less important role in phase I. In fact, these considerations are not a primary concern in phase I.

6.2.6 The Operating-Characteristic Function

The ability of the $\bar{x}$ and R charts to detect shifts in process quality is described by their operating-characteristic (OC) curves. In this section, we present these OC curves for charts used for phase II monitoring of a process.

Consider the **operating-characteristic (OC) curve for an $\bar{x}$ control chart.** The standard deviation σ is assumed known and constant. If the mean shifts from the in-control value—say, μ_0—to another value $\mu_1 = \mu_0 + k\sigma$, the probability of *not* detecting this shift on the first subsequent sample or the β-risk is

$$\beta = P\left\{\text{LCL} \leq \bar{x} \leq \text{UCL} \middle| \mu = \mu_1 = \mu_0 + k\sigma\right\} \tag{6.18}$$

Since $\bar{x} \sim N(\mu, \sigma^2/n)$, and the upper and lower control limits are UCL $= \mu_0 + L\sigma/\sqrt{n}$ and LCL $= \mu_0 - L\sigma/\sqrt{n}$, we may write equation (6.18) as

$$\beta = \Phi\left[\frac{\text{UCL} - (\mu_0 + k\sigma)}{\sigma/\sqrt{n}}\right] - \Phi\left[\frac{\text{LCL} - (\mu_0 + k\sigma)}{\sigma/\sqrt{n}}\right]$$
$$= \Phi\left[\frac{\mu_0 + L\sigma/\sqrt{n} - (\mu_0 + k\sigma)}{\sigma/\sqrt{n}}\right] - \Phi\left[\frac{\mu_0 - L\sigma/\sqrt{n} - (\mu_0 + k\sigma)}{\sigma/\sqrt{n}}\right]$$

where Φ denotes the standard normal cumulative distribution function. This reduces to

$$\beta = \Phi\left(L - k\sqrt{n}\right) - \Phi\left(-L - k\sqrt{n}\right) \tag{6.19}$$

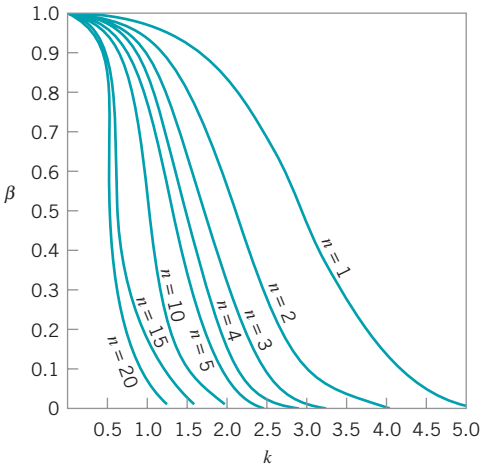
To illustrate the use of equation (6.19), suppose that we are using an $\bar{x}$ chart with $L = 3$ (the usual three-sigma limits), the sample size $n = 5$, and we wish to determine the probability of detecting a shift to $\mu_1 = \mu_0 + 2\sigma$ on the first sample following the shift. Then, since $L = 3$, $k = 2$, and $n = 5$, we have

$$\beta = \Phi\left[3 - 2\sqrt{5}\right] - \Phi\left[-3 - 2\sqrt{5}\right]$$
$$= \Phi(-1.47) - \Phi(-7.37)$$
$$\cong 0.0708$$

This is the β-risk, or the probability of not detecting such a shift. The probability that such a shift *will* be detected on the first subsequent sample is $1 - \beta = 1 - 0.0708 = 0.9292$.

To construct the OC curve for the $\bar{x}$ chart, plot the β-risk against the magnitude of the shift we wish to detect expressed in standard deviation units for various sample sizes n. These probabilities may be evaluated directly from equation (6.19). This OC curve is shown in Fig. 6.13 for the case of three-sigma limits ($L = 3$).

Figure 6.13 indicates that for the typical sample sizes 4, 5, and 6, the $\bar{x}$ chart is not particularly effective in detecting a small shift—say, those on the order of 1.5σ or less—on the first sample following the shift. For example, if the shift is 1.0σ and $n = 5$, then from Fig. 6.13, we have $\beta = 0.75$, approximately. Thus, the probability that the shift will be detected on the first sample is only $1 - \beta = 0.25$. However, the probability that the shift is detected on



■ **FIGURE 6.13** Operating-characteristic curves for the $\bar{x}$ chart with three-sigma limits. $\beta = P$ (not detecting a shift of $k\sigma$ in the mean on the first sample following the shift).

the second sample is $\beta(1-\beta) = 0.75(0.25) = 0.19$, whereas the probability that it is detected on the third sample is $\beta^2(1-\beta) = (0.75^2)0.25 = 0.14$. Thus, the probability that the shift will be detected on the rth subsequent sample is simply $1-\beta$ times the probability of not detecting the shift on each of the initial $r-1$ samples, or

$$\beta^{r-1}(1-\beta)$$

In general, the expected number of samples taken before the shift is detected is simply the **average run length,** or

$$\text{ARL} = \sum_{r=1}^{\infty} r\beta^{r-1}(1-\beta) = \frac{1}{1-\beta}$$

Therefore, in our example, we have

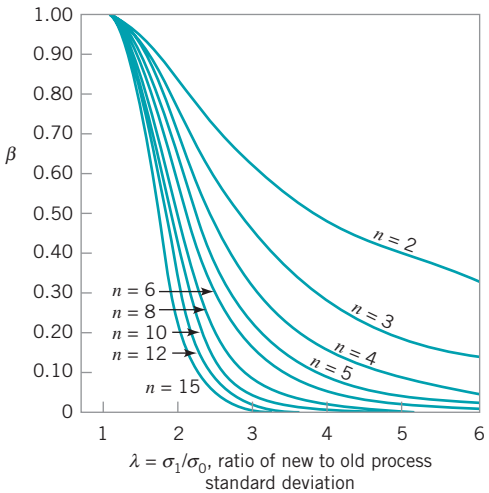
$$\text{ARL} = \frac{1}{1-\beta} = \frac{1}{0.25} = 4$$

In other words, the expected number of samples taken to detect a shift of 1.0σ with $n = 5$ is four.

The above discussion provides a supportive argument for the use of small sample sizes on the $\bar{x}$ chart. Even though small sample sizes often result in a relatively large β-risk, because samples are collected and tested periodically, there is a very good chance that the shift will be detected reasonably quickly, although perhaps not on the first sample following the shift.

To construct the OC curve for the R chart, the distribution of the relative range $W = R/\sigma$ is employed. Suppose that the in-control value of the standard deviation is σ_0. Then the OC curve plots the probability of not detecting a shift to a new value of σ—say, $\sigma_1 > \sigma_0$—on the first sample following the shift. Figure 6.14 presents the OC curve, in which β is plotted against $\lambda = \sigma_1/\sigma_0$ (the ratio of new to old process standard deviation) for various values of n.

From examining Fig. 6.14, we observe that the R chart is not very effective in detecting process shifts for small sample sizes. For example, if the process standard deviation doubles (i.e., $\lambda = \sigma_1/\sigma_0 = 2$), which is a fairly large shift, then samples of size 5 have only about a 40% chance of detecting this shift on each subsequent sample. Most quality engineers believe that the R chart is insensitive to small or moderate shifts for the usual subgroup sizes of $n = 4$, 5, or 6. If $n > 10$ or 12, the s chart discussed in Section 6.3.1 should generally be used instead of the R chart.



■ **FIGURE 6.14** Operating-characteristic curves for the R chart with three-sigma limits. (Adapted from A. J. Duncan, "Operating Characteristics of R Charts," *Industrial Quality Control*, vol. 7, no. 5, pp. 40–41, 1951, with permission of the American Society for Quality Control.)

The OC curves in Figs. 6.13 and 6.14 assume that the $\bar{x}$ and R charts are used for online process monitoring; that is, phase II process monitoring. It is occasionally useful to study the statistical performance of a chart used to analyze past data (phase I). This can give some indication of how the number of preliminary subgroups used to establish the control chart affects the chart's ability to detect out-of-control conditions that existed when the data were collected. It is from such analytical studies, as well as practical experience, that the recommendation to use at least 20 to 25 preliminary subgroups in establishing $\bar{x}$ and R charts has evolved.

6.2.7 The Average Run Length for the $\bar{x}$ Chart

For any **Shewhart control chart,** we have noted previously that the ARL can be expressed as

$$\text{ARL} = \frac{1}{P(\text{one point plots out of control})}$$

or

$$\text{ARL}_0 = \frac{1}{\alpha} \tag{6.20}$$

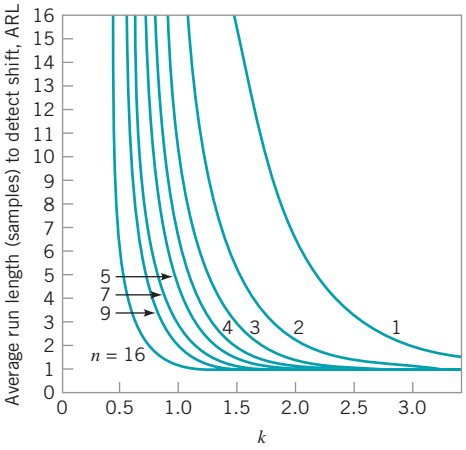
for the in-control ARL and

$$\text{ARL}_1 = \frac{1}{1-\beta} \tag{6.21}$$

for the out-of-control ARL. These results are actually rather intuitive. If the observations plotted on the control chart are independent, then the number of points that must be plotted until the first point exceeds a control limit is a geometric random variable with parameter p (see Chapter 3). The mean of this geometric distribution is simply $1/p$, the average run length.

Since it is relatively easy to develop a general expression for β for the $\bar{x}$ control chart to detect a shift in the mean of $k\sigma$ [see equation (6.19)], then it is not difficult to construct a set of ARL curves for the $\bar{x}$ chart. Figure 6.15 presents these ARL curves for sample sizes of n = 1, 2, 3, 4, 5, 7, 9, and 16 for the $\bar{x}$ control chart, where the ARL is in terms of the expected number of *samples* taken in order to detect the shift. To illustrate the use of Fig. 6.15, note that if we wish to detect a shift of 1.5σ using a sample size of $n = 3$, then the average number of samples required will be $\text{ARL}_1 = 3$. Note also that we could reduce the ARL_1 to approximately 1 if we increased the sample size to $n = 16$.

Recall our discussion in Chapter 5 (Section 5.3.3) indicating that ARLs are subject to some criticism as performance measures for control charts. We noted there (and above as well) that the distribution of run length for a Shewhart control chart is geometric and that this can be a very skewed distribution, so the average (that is, the ARL) may not be the best measure of a typical run length. Another issue concerning ARL relates to the fact that the computations for a specific control chart are usually based on **estimates** of the process



■ **FIGURE 6.15** Average run length (samples) for the $\bar{x}$ chart with three-sigma limits, where the process mean shifts by $k\sigma$. (Adapted from *Modern Methods for Quality Control and Improvement*, by H. M. Wadsworth, K. S. Stephens, and A. B. Godfrey, 2nd edition, John Wiley & Sons, 2002.)

parameters. This results in inflation of both ARL_0 and ARL_1. For example, suppose that the center line of the chart is estimated perfectly, but the process standard deviation is overestimated by 10%. This would result in $ARL_0 = 517$, considerably longer than the nominal or "theoretical" value of 370. With a normally distributed process, we are just as likely to underestimate the process standard deviation by 10%, and this would result in $ARL_0 = 268$, a value considerably shorter than 370. The average is $(268 + 517)/2 = 392.5$, suggesting that errors in estimating the process standard deviation result in ARLs that are overestimated. There is a very comprehensive literature review and discussion of this topic in the paper by Jensen et al. (2006).

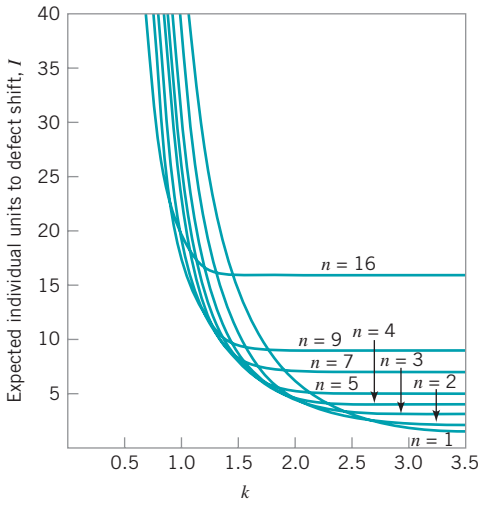
Two other performance measures based on ARL are sometimes of interest. The average time to signal is the number of time periods that occur until a signal is generated on the control chart. If samples are taken at equally spaced intervals of time h, then the **average time to signal** or the ATS is

$$\text{ATS} = \text{ARL}\, h \tag{6.22}$$

It may also be useful to express the ARL in terms of the expected number of individual *units* sampled—say, I—rather than the number of samples taken to detect a shift. If the sample size is n, the relationship between I and ARL is

$$I = n\,\text{ARL} \tag{6.23}$$

Figure 6.16 presents a set of curves that plot the expected number of individual units I that must be sampled for the $\bar{x}$ chart to detect a shift of $k\sigma$. To illustrate the use of the curve, note that to detect a shift of 1.5σ, an $\bar{x}$ chart with $n = 16$ will require that approximately 16 units be sampled, whereas if the sample size is $n = 3$, only about 9 units will be required, on the average.



■ **FIGURE 6.16** Average run length (individual units) for the $\bar{x}$ chart with three-sigma limits, where the process mean shifts by $k\sigma$. (Adapted from *Modern Methods for Quality Control and Improvement*, by H. M. Wadsworth, K. S. Stephens, and A. B. Godfrey, 2nd edition, John Wiley & Sons, 2002.)

6.3 Control Charts for $\bar{x}$ and s

Although $\bar{x}$ and R charts are widely used, it is occasionally desirable to estimate the process standard deviation directly instead of indirectly through the use of the range R. This leads to control charts for $\bar{x}$ and s, where s is the sample standard deviation.[1] Generally, $\bar{x}$ and s charts are preferable to their more familiar counterparts, $\bar{x}$ and R charts, when either

1. The sample size n is moderately large—say, $n > 10$ or 12. (Recall that the range method for estimating σ loses statistical efficiency for moderate to large samples.)
2. The sample size n is variable.

In this section, we illustrate the construction and operation of $\bar{x}$ and s control charts. We also show how to deal with variable sample size and discuss an alternative to the s chart.

6.3.1 Construction and Operation of $\bar{x}$ and s Charts

Setting up and operating control charts for $\bar{x}$ and s requires about the same sequence of steps as those for $\bar{x}$ and R charts, except that for each sample we must calculate the sample average $\bar{x}$ and the sample standard deviation s. Table 6.3 presents the inside diameter measurements of forged automobile engine piston rings. Each sample or subgroup consists of five piston rings. We have calculated the sample average and sample standard deviation for each of the 25 samples. We will use these data to illustrate the construction and operation of $\bar{x}$ and s charts.

If σ^2 is the unknown variance of a probability distribution, then an unbiased estimator of σ^2 is the sample variance

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$

[1]Some authors refer to the s chart as the σ chart.

■ TABLE 6.3
Inside Diameter Measurements (mm) for Automobile Engine Piston Rings

Sample Number	Observations					$\bar{x}_i$	s_i
1	74.030	74.002	74.019	73.992	74.008	74.010	0.0148
2	73.995	73.992	74.001	74.011	74.004	74.001	0.0075
3	73.988	74.024	74.021	74.005	74.002	74.008	0.0147
4	74.002	73.996	73.993	74.015	74.009	74.003	0.0091
5	73.992	74.007	74.015	73.989	74.014	74.003	0.0122
6	74.009	73.994	73.997	73.985	73.993	73.996	0.0087
7	73.995	74.006	73.994	74.000	74.005	74.000	0.0055
8	73.985	74.003	73.993	74.015	73.988	73.997	0.0123
9	74.008	73.995	74.009	74.005	74.004	74.004	0.0055
10	73.998	74.000	73.990	74.007	73.995	73.998	0.0063
11	73.994	73.998	73.994	73.995	73.990	73.994	0.0029
12	74.004	74.000	74.007	74.000	73.996	74.001	0.0042
13	73.983	74.002	73.998	73.997	74.012	73.998	0.0105
14	74.006	73.967	73.994	74.000	73.984	73.990	0.0153
15	74.012	74.014	73.998	73.999	74.007	74.006	0.0073
16	74.000	73.984	74.005	73.998	73.996	73.997	0.0078
17	73.994	74.012	73.986	74.005	74.007	74.001	0.0106
18	74.006	74.010	74.018	74.003	74.000	74.007	0.0070
19	73.984	74.002	74.003	74.005	73.997	73.998	0.0085
20	74.000	74.010	74.013	74.020	74.003	74.009	0.0080
21	73.982	74.001	74.015	74.005	73.996	74.000	0.0122
22	74.004	73.999	73.990	74.006	74.009	74.002	0.0074
23	74.010	73.989	73.990	74.009	74.014	74.002	0.0119
24	74.015	74.008	73.993	74.000	74.010	74.005	0.0087
25	73.982	73.984	73.995	74.017	74.013	73.998	0.0162
						$\Sigma =$ 1850.028	0.2351
						$\bar{\bar{x}} =$ 74.001	$\bar{s} = 0.0094$

However, the sample standard deviation s is *not* an unbiased estimator of σ. In Chapter 4 (Section 4.2) we observed that if the underlying distribution is normal, then s actually estimates $c_4\sigma$, where c_4 is a constant that depends on the sample size n. Furthermore, the standard deviation of s is $\sigma\sqrt{1 - c_4^2}$. This information can be used to establish control charts on $\bar{x}$ and s.

Consider the case where a standard value is given for σ. Since $E(s) = c_4\sigma$, the center line for the chart is $c_4\sigma$. The three-sigma control limits for s are then

$$\text{UCL} = c_4\sigma + 3\sigma\sqrt{1 - c_4^2}$$

$$\text{LCL} = c_4\sigma - 3\sigma\sqrt{1 - c_4^2}$$

It is customary to define the two constants

$$B_5 = c_4 - 3\sqrt{1 - c_4^2} \quad \text{and} \quad B_6 = c_4 + 3\sqrt{1 - c_4^2} \tag{6.24}$$

Consequently, the parameters of the s chart with a standard value for σ given become

$$\begin{aligned} \text{UCL} &= B_6\sigma \\ \text{Center line} &= c_4\sigma \\ \text{LCL} &= B_5\sigma \end{aligned} \tag{6.25}$$

Values of B_5 and B_6 are tabulated for various sample sizes in Appendix Table VI. The parameters of the corresponding $\bar{x}$ chart are given in equation (6.15), Section 6.2.3.

If no standard is given for σ, then it must be estimated by analyzing past data. Suppose that m preliminary samples are available, each of size n, and let s_i be the standard deviation of the ith sample. The average of the m standard deviations is

$$\bar{s} = \frac{1}{m}\sum_{i=1}^{m} s_i$$

The statistic $\bar{s}/c_4$ is an unbiased estimator of σ. Therefore, the parameters of the s chart would be

$$\begin{aligned} \text{UCL} &= \bar{s} + 3\frac{\bar{s}}{c_4}\sqrt{1-c_4^2} \\ \text{Center line} &= \bar{s} \\ \text{LCL} &= \bar{s} - 3\frac{\bar{s}}{c_4}\sqrt{1-c_4^2} \end{aligned}$$

We usually define the constants

$$B_3 = 1 - \frac{3}{c_4}\sqrt{1-c_4^2} \quad \text{and} \quad B_4 = 1 + \frac{3}{c_4}\sqrt{1-c_4^2} \tag{6.26}$$

Consequently, we may write the parameters of the s chart as

$$\begin{aligned} \text{UCL} &= B_4\bar{s} \\ \text{Center line} &= \bar{s} \\ \text{LCL} &= B_3\bar{s} \end{aligned} \tag{6.27}$$

Note that $B_4 = B_6/c_4$ and $B_3 = B_5/c_4$.

When $\bar{s}/c_4$ is used to estimate σ, we may define the control limits on the corresponding $\bar{x}$ chart as

$$\begin{aligned} \text{UCL} &= \bar{\bar{x}} + \frac{3\bar{s}}{c_4\sqrt{n}} \\ \text{Center line} &= \bar{\bar{x}} \\ \text{LCL} &= \bar{\bar{x}} - \frac{3\bar{s}}{c_4\sqrt{n}} \end{aligned}$$

Let the constant $A_3 = 3/(c_4\sqrt{n})$. Then the $\bar{x}$ chart parameters become

$$\begin{aligned} \text{UCL} &= \bar{\bar{x}} + A_3\bar{s} \\ \text{Center line} &= \bar{\bar{x}} \\ \text{LCL} &= \bar{\bar{x}} - A_3\bar{s} \end{aligned} \tag{6.28}$$

The constants B_3, B_4, and A_3 for construction of $\bar{x}$ and s charts from past data are listed in Appendix Table VI for various sample sizes.

Note that we have assumed that the sample standard deviation is defined as

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}} \tag{6.29}$$

Some authors define s with n in the denominator of equation (6.29) instead of $n - 1$. When this is the case, the definitions of the constants c_4, B_3, B_4, and A_3 are altered. The corresponding constants based on the use of n in calculating s are called c_2, B_1, B_2, and A_1, respectively. See Bowker and Lieberman (1972) for their definitions.

Traditionally, quality engineers have preferred the R chart to the s chart because of the simplicity of calculating R from each sample. The availability of hand-held calculators with automatic calculation of s and computers at workstations to implement control charts on site have eliminated any computational difficulty.

EXAMPLE 6.3 $\bar{x}$ and s Charts for the Piston Ring Data

Construct and interpret $\bar{x}$ and s charts using the piston ring inside diameter measurements in Table 6.3.

SOLUTION

The grand average and the average standard deviation are

$$\bar{\bar{x}} = \frac{1}{25}\sum_{i=1}^{25}\bar{x}_i = \frac{1}{25}(1850.028) = 74.001$$

and

$$\bar{s} = \frac{1}{25}\sum_{i=1}^{25}s_i = \frac{1}{25}(0.2351) = 0.0094$$

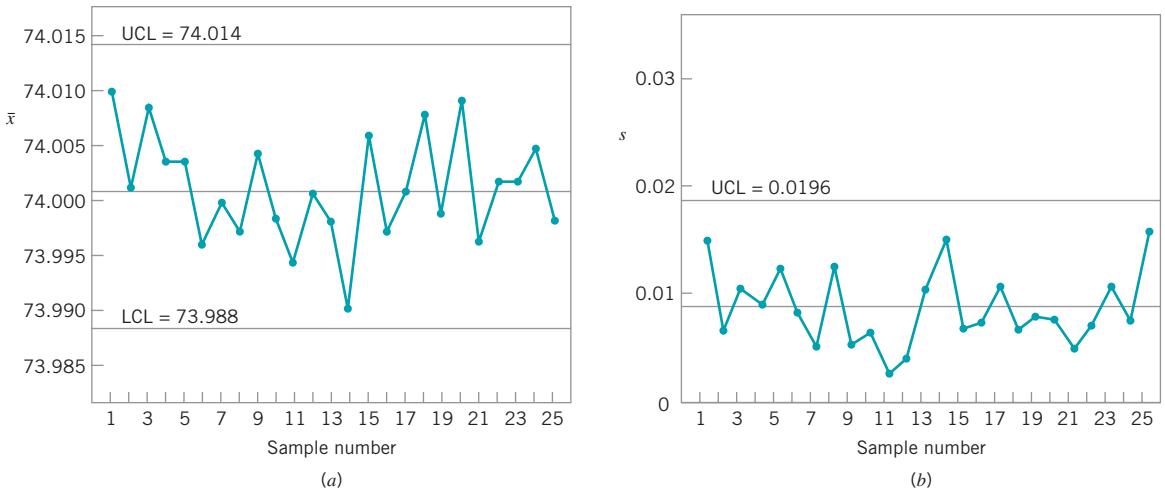
respectively. Consequently, the parameters for the $\bar{x}$ chart are

$$\begin{aligned} \text{UCL} &= \bar{\bar{x}} + A_3\bar{s} = 74.001 + (1.427)(0.0094) = 74.014 \\ \text{CL} &= \bar{\bar{x}} = 74.001 \\ \text{LCL} &= \bar{\bar{x}} - A_3\bar{s} = 74.001 - (1.427)(0.0094) = 73.988 \end{aligned}$$

and for the s chart

$$\begin{aligned} \text{UCL} &= B_4\bar{s} = (2.089)(0.0094) = 0.0196 \\ \text{CL} &= \bar{s} = 0.0094 \\ \text{LCL} &= B_3\bar{s} = (0)(0.0094) = 0 \end{aligned}$$

The control charts are shown in Fig. 6.17. There is no indication that the process is out of control, so those limits could be adopted for phase II monitoring of the process.



■ **FIGURE 6.17** The $\bar{x}$ and s control charts for Example 6.3. (*a*) The $\bar{x}$ chart with control limits based on $\bar{s}$. (*b*) The s control chart.

Estimation of σ. We can estimate the process standard deviation using the fact that s/c_4 is an unbiased estimate of σ. Therefore, since $c_4 = 0.9400$ for samples of size 5, our estimate of the process standard deviation is

$$\hat{\sigma} = \frac{\bar{s}}{c_4} = \frac{0.0094}{0.9400} = 0.01$$

6.3.2 The $\bar{x}$ and s Control Charts with Variable Sample Size

The $\bar{x}$ and s control charts are relatively easy to apply in cases where the sample sizes are variable. In this case, we should use a weighted average approach in calculating $\bar{\bar{x}}$ and $\bar{s}$. If n_i is the number of observations in the ith sample, then use

$$\bar{\bar{x}} = \frac{\sum_{i=1}^{m} n_i \bar{x}_i}{\sum_{i=1}^{m} n_i} \tag{6.30}$$

and

$$\bar{s} = \left[\frac{\sum_{i=1}^{m} (n_i - 1) s_i^2}{\sum_{i=1}^{m} n_i - m} \right]^{1/2} \tag{6.31}$$

as the center lines on the $\bar{x}$ and s control charts, respectively. The control limits would be calculated from equations (6.27) and (6.28), respectively, but the constants A_3, B_3, and B_4 will depend on the sample size used in each individual subgroup.

EXAMPLE 6.4 $\bar{x}$ and s Chart for the Piston Rings, Variable Sample Size

Consider the data in Table 6.4, which is a modification of the piston-ring data used in Example 6.3. Note that the sample sizes vary from $n = 3$ to $n = 5$. Use the procedure described on page 255 to set up the $\bar{x}$ and s control charts.

SOLUTION

The weighted grand mean and weighted average standard deviation are computed from equations (6.30) and (6.31) as follows:

$$\bar{\bar{x}} = \frac{\sum_{i=1}^{25} n_i \bar{x}_i}{\sum_{i=1}^{25} n_i} = \frac{5(74.010) + 3(73.996) + \cdots + 5(73.998)}{5 + 3 + \cdots + 5}$$

$$= \frac{8362.075}{113} = 74.001$$

■ TABLE 6.4
Inside Diameter Measurements (mm) on Automobile Engine Piston Rings

Sample Number	Observations					$\bar{x}_i$	s_i
1	74.030	74.002	74.019	73.992	74.008	74.010	0.0148
2	73.995	73.992	74.001			73.996	0.0046
3	73.988	74.024	74.021	74.005	74.002	74.008	0.0147
4	74.002	73.996	73.993	74.015	74.009	74.003	0.0091
5	73.992	74.007	74.015	73.989	74.014	74.003	0.0122
6	74.009	73.994	73.997	73.985		73.996	0.0099
7	73.995	74.006	73.994	74.000		73.999	0.0055
8	73.985	74.003	73.993	74.015	73.988	73.997	0.0123
9	74.008	73.995	74.009	74.005		74.004	0.0064
10	73.998	74.000	73.990	74.007	73.995	73.998	0.0063
11	73.994	73.998	73.994	73.995	73.990	73.994	0.0029
12	74.004	74.000	74.007	74.000	73.996	74.001	0.0042
13	73.983	74.002	73.998			73.994	0.0100
14	74.006	73.967	73.994	74.000	73.984	73.990	0.0153
15	74.012	74.014	73.998			74.008	0.0087
16	74.000	73.984	74.005	73.998	73.996	73.997	0.0078
17	73.994	74.012	73.986	74.005		73.999	0.0115
18	74.006	74.010	74.018	74.003	74.000	74.007	0.0070
19	73.984	74.002	74.003	74.005	73.997	73.998	0.0085
20	74.000	74.010	74.013			74.008	0.0068
21	73.982	74.001	74.015	74.005	73.996	74.000	0.0122
22	74.004	73.999	73.990	74.006	74.009	74.002	0.0074
23	74.010	73.989	73.990	74.009	74.014	74.002	0.0119
24	74.015	74.008	73.993	74.000	74.010	74.005	0.0087
25	73.982	73.984	73.995	74.017	74.013	73.998	0.0162

and

$$\bar{s} = \left[\frac{\sum_{i=1}^{25}(n_i - 1)s_i^2}{\sum_{i=1}^{25} n_i - 25}\right]^{1/2} = \left[\frac{4(0.0148)^2 + 2(0.0046)^2 + \cdots + 4(0.0162)^2}{5 + 3 + \cdots + 5 - 25}\right]^{1/2}$$
$$= \left[\frac{0.009324}{88}\right]^{1/2} = 0.0103$$

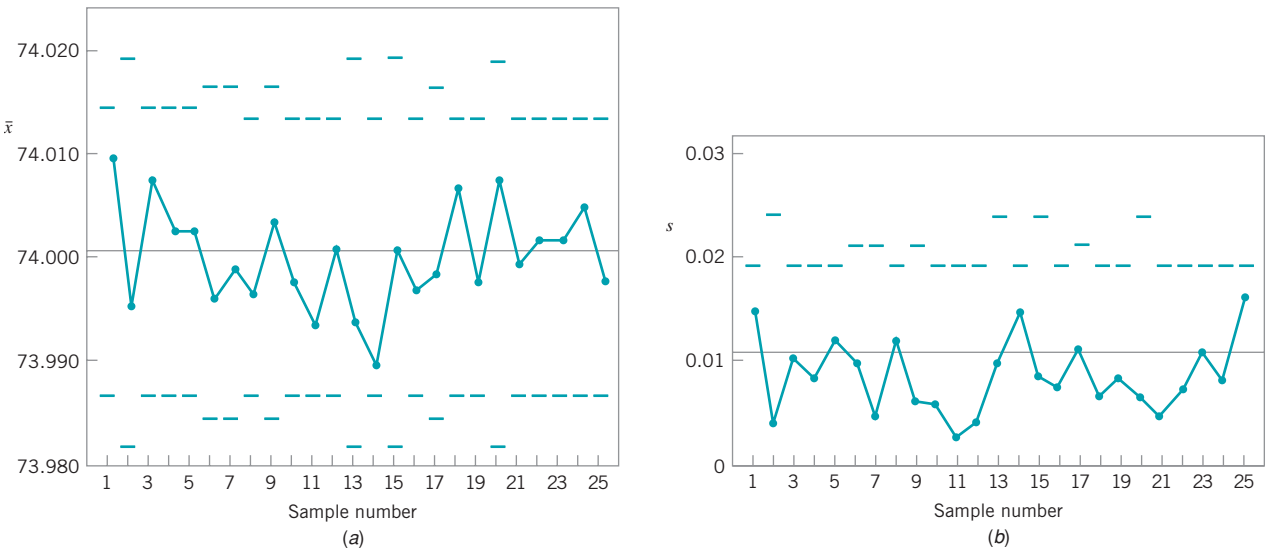
Therefore, the center line of the $\bar{x}$ chart is $\bar{\bar{x}} = 74.001$, and the center line of the s chart is $\bar{s} = 0.0103$. The control limits may now be easily calculated. To illustrate, consider the first sample. The limits for the $\bar{x}$ chart are

$$\text{UCL} = 74.001 \quad 1.427 \quad 0.0103 \quad = 74.016$$
$$\text{CL} = 74.001$$
$$\text{LCL} = 74.001 - \ 1.427 \quad 0.0103 \quad = 73.986$$

The control limits for the s chart are

$$\text{UCL} = \ 2.089 \quad 0.0103 \quad = 0.022$$
$$\text{CL} = 0.0103$$
$$\text{LCL} = 0 \ 0.0103 \quad = 0$$

Note that we have used the values of A_3, B_3, and B_4 for $n_1 = 5$. The limits for the second sample would use the values of these constants for $n_2 = 3$. The control limit calculations for all 25 samples are summarized in Table 6.5. The control charts are plotted in Fig. 6.18.



■ **FIGURE 6.18** The (*a*) $\bar{x}$ and (*b*) s control charts for piston-ring data with variable sample size, Example 6.4.

(*continued*)

■ TABLE 6.5
Computation of Control Limits for $\bar{x}$ and s Charts with Variable Sample Size

					$\bar{x}$ Chart				s Chart	
Sample	n	$\bar{x}$	s	A_3	LCL	UCL	B_3	B_4	LCL	UCL
1	5	74.010	0.0148	1.427	73.986	74.016	0	2.089	0	0.022
2	3	73.996	0.0046	1.954	73.981	74.021	0	2.568	0	0.026
3	5	74.008	0.0147	1.427	73.986	74.016	0	2.089	0	0.022
4	5	74.003	0.0091	1.427	73.986	74.016	0	2.089	0	0.022
5	5	74.003	0.0122	1.427	73.986	74.016	0	2.089	0	0.022
6	4	73.996	0.0099	1.628	73.984	74.018	0	2.266	0	0.023
7	4	73.999	0.0055	1.628	73.984	74.018	0	2.266	0	0.023
8	5	73.997	0.0123	1.427	73.986	74.016	0	2.089	0	0.022
9	4	74.004	0.0064	1.628	73.984	74.018	0	2.266	0	0.023
10	5	73.998	0.0063	1.427	73.986	74.016	0	2.089	0	0.022
11	5	73.994	0.0029	1.427	73.986	74.016	0	2.089	0	0.022
12	5	74.001	0.0042	1.427	73.986	74.016	0	2.089	0	0.022
13	3	73.994	0.0100	1.954	73.981	74.021	0	2.568	0	0.026
14	5	73.990	0.0153	1.427	73.986	74.016	0	2.089	0	0.022
15	3	74.008	0.0087	1.954	73.981	74.021	0	2.568	0	0.026
16	5	73.997	0.0078	1.427	73.986	74.016	0	2.089	0	0.022
17	4	73.999	0.0115	1.628	73.984	74.018	0	2.226	0	0.023
18	5	74.007	0.0070	1.427	73.986	74.016	0	2.089	0	0.022
19	5	73.998	0.0085	1.427	73.986	74.016	0	2.089	0	0.022
20	3	74.008	0.0068	1.954	73.981	74.021	0	2.568	0	0.026
21	5	74.000	0.0122	1.427	73.986	74.016	0	2.089	0	0.022
22	5	74.002	0.0074	1.427	73.986	74.016	0	2.089	0	0.022
23	5	74.002	0.0119	1.427	73.986	74.016	0	2.089	0	0.022
24	5	74.005	0.0087	1.427	73.986	74.016	0	2.089	0	0.022
25	5	73.998	0.0162	1.427	73.986	74.016	0	2.089	0	0.022

An alternative to using variable-width control limits on the $\bar{x}$ and s control charts is to base the control limit calculations on an **average sample size $\bar{n}$.** If the n_i are not very different, this approach may be satisfactory in some situations; it is particularly helpful if the charts are to be used in a presentation to management. Since the average sample size n_i may not be an integer, a useful alternative is to base these approximate control limits on a modal (most common) sample size.

Estimation of σ. We may estimate the process standard deviation, σ, from the individual sample values s_i. First, average all the values of s_i for which $n_i = 5$ (the most frequently occurring value of n_i). This gives

$$\bar{s} = \frac{0.1715}{17} = 0.0101$$

The estimate of the process σ is then

$$\hat{\sigma} = \frac{\bar{s}}{c_4} = \frac{0.0101}{0.9400} = 0.01$$

where the value of c_4 used is for samples of size $n = 5$.

6.3.3 The s^2 Control Chart

Most quality engineers use either the R chart or the s chart to monitor process variability, with s preferable to R for moderate to large sample sizes. Some practitioners recommend a control chart based directly on the sample variance s^2, the ***s*² control chart.** The parameters for the s^2 control chart are

$$\begin{aligned} \text{UCL} &= \frac{\bar{s}^2}{n-1}\chi^2_{\alpha/2,n-1} \\ \text{Center line} &= \bar{s}^2 \\ \text{LCL} &= \frac{\bar{s}^2}{n-1}\chi^2_{1-(\alpha/2),n-1} \end{aligned} \tag{6.32}$$

where $\chi^2_{\alpha/2,}$ and $\chi^2_{(\alpha/2),n-1}$ denote the upper and lower $\alpha/2$ percentage points of the chi-square distribution with $n - 1$ degrees of freedom, and $\bar{s}^2$ is an average sample variance obtained from the analysis of preliminary data. A standard value σ^2 could be used in equation (6.32) instead of $\bar{s}^2$ if one were available. Note that this control chart is defined with probability limits.

6.4 The Shewhart Control Chart for Individual Measurements

There are many situations in which the sample size used for process monitoring is $n = 1$; that is, the sample consists of an individual unit. Some examples of these situations are as follows:

1. Automated inspection and measurement technology is used, and every unit manufactured is analyzed so there is no basis for rational subgrouping.
2. Data comes available relatively slowly, and it is inconvenient to allow sample sizes of $n > 1$ to accumulate before analysis. The long interval between observations will cause problems with rational subgrouping. This occurs frequently in both manufacturing and nonmanufacturing situations.
3. Repeat measurements on the process differ only because of laboratory or analysis error, as in many chemical processes.
4. Multiple measurements are taken on the same unit of product, such as measuring oxide thickness at several different locations on a wafer in semiconductor manufacturing.
5. In process plants, such as papermaking, measurements on some parameter such as coating thickness *across* the roll will differ very little and produce a standard deviation that is much too small if the objective is to control coating thickness *along* the roll.

In such situations, the **control chart for individual units** is useful. (The cumulative sum and exponentially weighted moving-average control charts discussed in Chapter 9 will be a better alternative in phase II or when the magnitude of the shift in process mean that is of interest is small.) In many applications of the *individuals control chart*, we use the *moving range* two successive observations as the basis of estimating the process variability. The moving range is defined as

$$MR_i = |x_i - x_{i-1}|$$

It is also possible to establish a **moving range control chart.** The procedure is illustrated in the following example.

EXAMPLE 6.5 Loan Processing Costs

The mortgage loan processing unit of a bank monitors the costs of processing loan applications. The quantity tracked is the average weekly processing costs, obtained by dividing total weekly costs by the number of loans processed during the week. The processing costs for the most recent 20 weeks are shown in Table 6.6. Set up individual and moving range control charts for these data.

SOLUTION

To set up the control chart for individual observations, note that the sample average cost of the 20 observations is $\bar{x} = 300.5$ and that the average of the moving ranges of two observations is $\overline{MR} = 7.79$. To set up the moving range chart, we use $D_3 = 0$ and $D_4 = 3.267$ for $n = 2$. Therefore, the moving range chart has center line $\overline{MR} = 7.79$, LCL = 0, and UCL $= D_4\overline{MR} = (3.267)7.79 = 25.45$. The control chart (from Minitab) is shown in Fig. 6.19*b*. Notice that no points are out of control.

For the control chart for individual measurements, the parameters are

$$\begin{aligned} \text{UCL} &= \bar{x} + 3\frac{\overline{MR}}{d_2} \\ \text{Center line} &= \bar{x} \\ \text{LCL} &= \bar{x} - 3\frac{\overline{MR}}{d_2} \end{aligned} \qquad (6.33)$$

■ TABLE 6.6
Costs of Processing Mortgage Loan Applications

Weeks	Cost x	Moving Range MR
1	310	
2	288	22
3	297	9
4	298	1
5	307	9
6	303	4
7	294	9
8	297	3
9	308	11
10	306	2
11	294	12
12	299	5
13	297	2
14	299	2
15	314	15
16	295	19
17	293	2
18	306	13
19	301	5
20	304	3
	$\bar{x} = 300.5$	$\overline{MR} = 7.79$

If a moving range of $n = 2$ observations is used, then $d_2 = 1.128$. For the data in Table 6.6, we have

$$\text{UCL} = \bar{x} + 3\frac{\overline{\text{MR}}}{d_2} = 300.5 + 3\frac{7.79}{1.128} = 321.22$$

$$\text{Center line} = \bar{x} = 34.088$$

$$\text{LCL} = \bar{x} - 3\frac{\overline{\text{MR}}}{d_2} = 300.5 - 3\frac{7.79}{1.128} = 279.78$$

The control chart for individual cost values is shown in Fig. 6.19*a*. There are no out of control observations on the individuals control chart.

The interpretation of the individuals control chart is very similar to the interpretation of the ordinary $\bar{x}$ control chart. A shift in the process mean will result in a single point or a series of points that plot outside the control limits on the control chart for individuals. Sometimes a point will plot outside the control limits on both the individuals chart and the moving range chart. This will often occur because a large value of x will also lead to a large value of the moving range for that sample. This is very typical behavior for the individuals and moving range control charts. It is most likely an indication that the mean is out of control and not an indication that both the mean and the variance of the process are out of control.

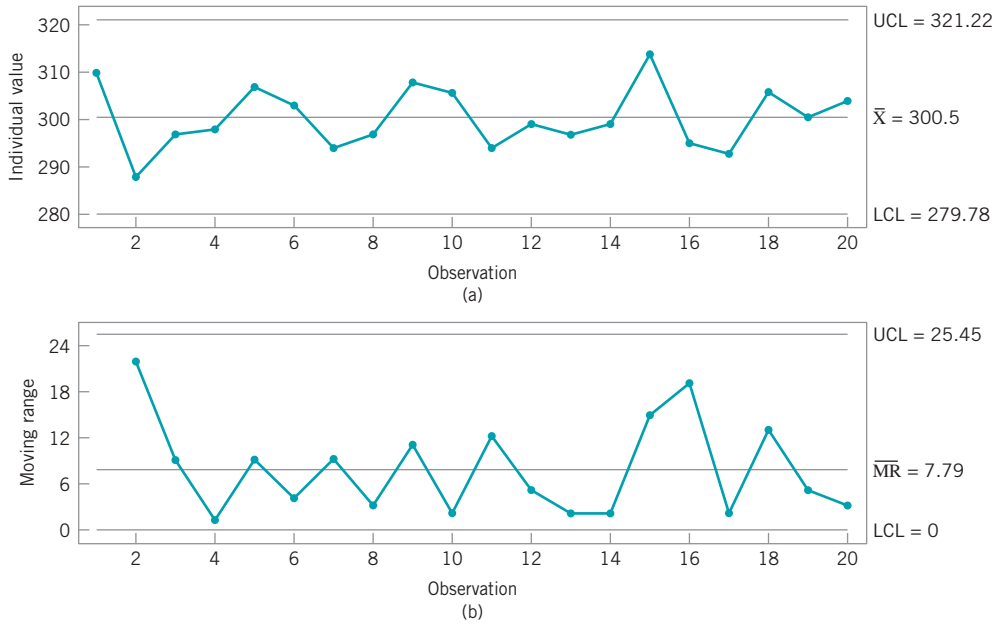


FIGURE 6.19 Control charts for (*a*) individual observations on cost and for (*b*) the moving range.

Phase II Operation and Interpretation of the Charts. Table 6.7 contains data on mortgage application processing costs for weeks 21–40. These data are plotted in Fig. 6.20 on the continuation of the control chart for individuals and the moving range control chart developed in Example 6.5. As this figure makes clear, an upward shift in cost has occurred around week 39, since there is an obvious "shift in process level" pattern on the chart for individuals followed by another out-of-control signal at week 40. Note that the moving range chart also reacts to this level shift with a single large spike at week 39. This spike on the moving range chart is sometimes helpful in identifying exactly where a process shift in the mean has occurred. Clearly one should look for possible assignable causes around weeks 39. Possible causes could include an unusual number of applications requiring additional manual underwriting work, or possibly new underwriters working in the process, or possibly temporary underwriters replacing regular employees taking vacations.

Some care should be exercised in interpreting patterns on the moving range chart. The moving ranges are correlated, and this correlation may often induce a pattern of runs or cycles



■ **FIGURE 6.20** Continuation of the control chart for individuals and the moving range using the additional data in Table 6.7.

■ **TABLE 6.7**
Costs of Processing Mortgage Loan Applications, Weeks 21–40

Week	Cost x	Week	Cost x
21	305	31	310
22	282	32	292
23	305	33	305
24	296	34	299
25	314	35	304
26	295	36	310
27	287	37	304
28	301	38	305
29	298	39	333
30	311	40	328

on the chart. Such a pattern is evident on the moving range chart in Fig. 6.21. The individual measurements on the x chart are assumed to be uncorrelated, however, and any apparent pattern on this chart should be carefully investigated.

Some authorities have recommended not constructing and plotting the moving range chart. They point out (correctly, we add) that the moving range chart cannot really provide useful information about a shift in process variability. For example, see Rigdon, Cruthis, and Champ (1994). In effect, shifts in the process mean also show up on the moving range chart. Our feeling is that as long as the analyst is careful in interpretation and relies primarily on the individuals chart, little trouble will ensue from plotting both charts.

Average Run Lengths. Crowder (1987b) has studied the average run length of the combined control chart for individuals and moving range chart. He produces ARLs for various settings of the control limits and shifts in the process mean and standard deviation. In general, his work shows that the ARL_0 of the combined procedure will generally be much less than the ARL_0 of a standard Shewhart control chart when the process is in control (recall that ARL_0 for a

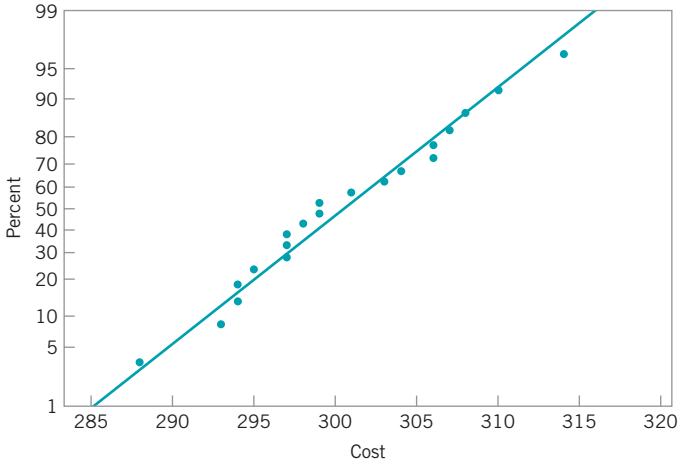


FIGURE 6.21
Normal probability plot of the mortgage application processing cost data from Table 6.6, Example 6.5.

Shewhart chart is 370 samples), if we use the conventional three-sigma limits on the charts. In general, results closer to the Shewhart in-control ARL are obtained if we use three-sigma limits on the chart for individuals and compute the upper control limit on the moving range chart from

$$\text{UCL} = D\overline{\text{MR}}$$

where the constant D should be chosen such that $4 \le D \le 5$.

One can get a very good idea about the ability of the individuals control chart to detect process shifts by looking at the OC curves in Fig. 6.13 or the ARL curves in Fig. 6.15. For an individuals control chart with three-sigma limits, we can compute the following:

Size of Shift	β	ARL_1
1σ	0.9772	43.96
2σ	0.8413	6.30
3σ	0.5000	2.00

Note that the ability of the individuals control chart to detect small shifts is very poor. For instance, consider a continuous chemical process in which samples are taken every hour. If a shift in the process mean of about one standard deviation occurs, the information above tells us that it will take about 44 samples, on the average, to detect the shift. This is nearly two full days of continuous production in the out-of-control state, a situation that has potentially devastating economic consequences. This limits the usefulness of the individuals control chart in phase II process monitoring.

Some individuals have suggested that control limits narrower than three-sigma be used on the chart for individuals to enhance its ability to detect small process shifts. This is a dangerous suggestion, as narrower limits will dramatically reduce the value of ARL_0 and increase the occurrence of false alarms to the point where the charts are ignored and hence become useless. If we are interested in detecting small shifts in phase II, then the correct approach is to use either the cumulative sum control chart or the exponentially weighted moving average control chart (see Chapter 9).

Normality. Our discussion in this section has made an assumption that the observations follow a normal distribution. Borror, Montgomery, and Runger (1999) have studied the phase II performance of the Shewhart control chart for individuals when the process data are not normal. They investigated various gamma distributions to represent skewed process data and t distributions to represent symmetric normal-like data. They found that the in-control ARL is dramatically

affected by non-normal data. For example, if the individuals chart has three-sigma limits so that for normal data $ARL_0 = 370$, the actual ARL_0 for gamma-distributed data is between 45 and 97, depending on the shape of the gamma distribution (more highly skewed distributions yield poorer performance). For the t distribution, the ARL_0 values range from 76 to 283 as the degrees of freedom increase from 4 to 50 (that is, as the t becomes more like the normal distribution).

In the face of these results, we conclude that if the process shows evidence of even moderate departure from normality, the control limits given here may be entirely inappropriate for phase II process monitoring. One approach to dealing with the problem of non-normality would be to determine the control limits for the individuals control chart based on the percentiles of the correct underlying distribution. These percentiles could be obtained from a histogram if a large sample (at least 100 but preferably 200 observations) were available, or from a probability distribution fit to the data. See Willemain and Runger (1996) for details on designing control charts from empirical reference distributions. Another approach would be to transform the original variable to a new variable that is approximately normally distributed, and then apply control charts to the new variable. Borror, Montgomery, and Runger (1999) show how a properly designed EWMA control chart is very insensitive to the normality assumption. This approach will be discussed in Chapter 9.

It is important to check the normality assumption when using the control chart for individuals. A simple way to do this is with the normal probability plot. Figure 6.21 is the normal probability plot for the mortgage application processing cost data in Table 6.6. There is no obvious problem with the normality assumption in these data. However, remember that the normal probability plot is but a **crude check** of the normality assumption and the individuals control chart is very sensitive to non-normality. Furthermore, mean shifts could show up as a problem with normality on the normal probability plot. Process stability is needed to properly interpret the plot. We suggest that the Shewhart individuals chart be used with extreme caution, particularly in phase II process monitoring.

EXAMPLE 6.6 The Use of Transformations

Table 6.8 presents consecutive measurements on the resistivity of 25 silicon wafers after an epitaxial layer is deposited in a single-wafer deposition process. Construct an individuals control chart for this process.

TABLE 6.8
Resistivity Data for Example 6.6

Sample, i	Resistivity (x_i)	ln (x_i)	MR	Sample, i	Resistivity (x_i)	ln (x_i)	MR
1	216	5.37528		14	242	5.48894	0.23791
2	290	5.66988	0.29460	15	168	5.12396	0.36498
3	236	5.46383	0.20605	16	360	5.88610	0.76214
4	228	5.42935	0.03448	17	226	5.42053	0.46557
5	244	5.49717	0.06782	18	253	5.53339	0.11286
6	210	5.34711	0.15006	19	380	5.94017	0.40678
7	139	4.93447	0.41264	20	131	4.87520	1.06497
8	310	5.73657	0.80210	21	173	5.15329	0.27809
9	240	5.48064	0.25593	22	224	5.41165	0.25836
10	211	5.35186	0.12878	23	195	5.27300	0.13865
11	175	5.16479	0.18707	24	199	5.29330	0.02030
12	447	6.10256	0.93777	25	226	5.42053	0.12723
13	307	5.72685	0.37571				
						$\overline{\ln(x_i)} = 5.44402$	$\overline{MR} = 0.33712$

SOLUTION

A normal probability plot of the resistivity measurements is shown in Fig. 6.22. This plot was constructed by Minitab, which fits the line to the points by least squares (not the best method). It is clear from inspection of the normal probability plot that the normality assumption for resistivity is at best questionable, so it would be dangerous to apply an individuals control chart to the original process data.

Figure 6.22 indicates that the distribution of resistivity has a long tail to the right, and consequently we would expect the log transformation (or a similar transformation) to produce a distribution that is closer to normal. The natural log of resistivity is shown in column three of Table 6.8, and the normal probability plot of the natural log of resistivity is shown in Fig. 6.23. Clearly the log transformation has resulted in a new variable that is more nearly approximated by a normal distribution than were the original resistivity measurements.

The last column of Table 6.8 shows the moving ranges of the natural log of resistivity. Figure 6.24 presents the individuals and moving range control charts for the natural log of resistivity. Note that there is no indication of an out-of-control process.

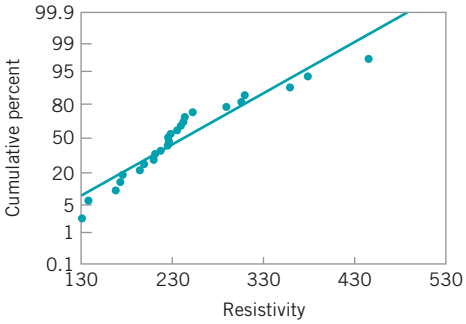


FIGURE 6.22 Normal probability plot of resistivity.

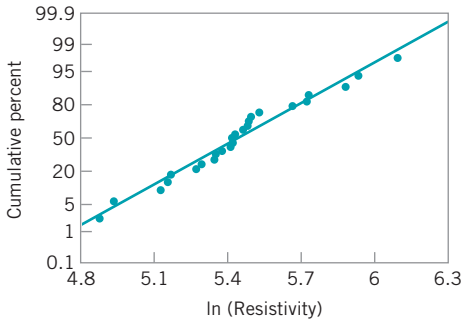


FIGURE 6.23 Normal probability plot of ln (resistivity).

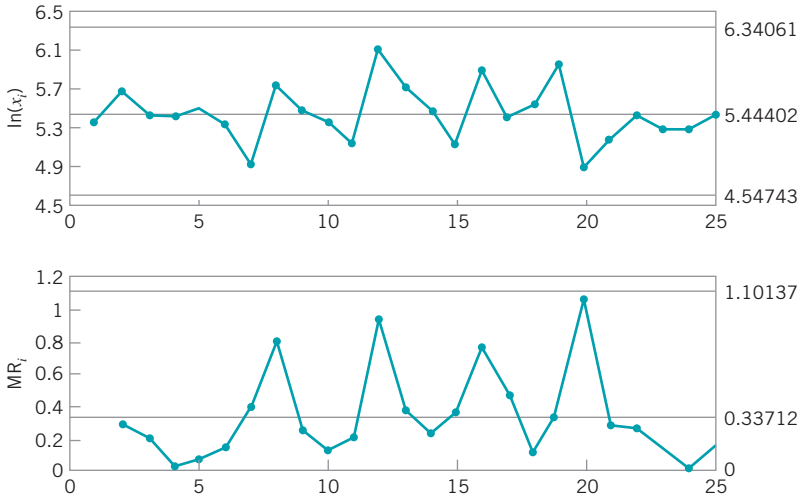


FIGURE 6.24 Individuals and moving range control charts on ln (resistivity), Example 6.6

More about Estimating σ. Very often in practice we use moving ranges in estimating σ for the individuals control chart. Recall that moving ranges are defined as $MR_i = |(x_i - x_{i-1})|$, $i = 2, 3 \ldots, m$. More properly, this statistic should be called a moving range of **span two** since the number of observations used to calculate the range in the moving window could be increased. The most common estimator is the one we used in Example 6.5, based on the **average moving range** $\overline{MR} = \sum_{i=2}^{m} MR_i/(m-1)$ and can be written as

$$\hat{\sigma}_1 = 0.8865\overline{MR}$$

where the constant 0.8865 is the reciprocal of d_2 for samples of size 2. For in-control processes, Cryer and Ryan (1990), among others, have pointed out that a more efficient estimator is one based on the sample standard deviation

$$\hat{\sigma}_2 = \frac{s}{c_4}$$

Both of these estimators are unbiased, assuming that no assignable causes are present in the sequence of m individual observations.

If assignable causes are present, then both $\hat{\sigma}_1$ and $\hat{\sigma}_2$ result in biased estimates of the process standard deviation. To illustrate, suppose that in the sequence of individual observations

$$x_1, x_2, \ldots, x_t, x_{t+1}, \ldots, x_m$$

the process is in control with mean μ_0 and standard deviation σ for the first t observations, but between x_t and x_{t+1} an assignable cause occurs that results in a sustained shift in the process mean to a new level $\mu = \mu_0 + \delta\sigma$ and the mean remains at this new level for the remaining sample observations $x_t + 1, \ldots, x_m$. Under these conditions, Woodall and Montgomery (2000–01) show that

$$E\left(s^2\right) = \sigma^2 + \frac{t(m-1)}{m(m-1)}(\delta\sigma)^2$$

In fact, this result holds for any case in which the mean of t of the observations is μ_0 and the mean of the remaining observations is $\mu_0 + \delta\sigma$, since the order of the observations is not relevant in computing s^2. Therefore, s^2 is biased upwards, and consequently $\hat{\sigma}_2 = S/c_4$ will tend to overestimate σ. Note that the extent of the bias in $\hat{\sigma}_2$ depends on the magnitude of the shift in the mean ($\delta\sigma$), the time period following which the shift occurs (t), and the number of available observations (m). Now the moving range is only impacted by the shift in the mean during one period ($t + 1$), so the bias in $\hat{\sigma}_1$ depends only on the shift magnitude and m. If $1 < t < m - 1$, the bias in $\hat{\sigma}_1$ will always be smaller than the bias in $\hat{\sigma}_2$. Cruthis and Rigdon (1992–93) show how the ratio

$$F^* = \left(\frac{\hat{\sigma}_1}{\hat{\sigma}_2}\right)^2$$

can be used to determine whether the process was in control when both estimates were calculated. They use simulation to obtain the approximate 90th, 95th, 99th, and 99.9th percentiles of the distribution of F^* for sample sizes $m = 10(5)100$, assuming that the process is in control. Since this is an empirical reference distribution, observed values of F^* that exceed one of these percentiles is an indication that the process was not in control over the time period during which the m observations were collected.

One way to reduce or possibly eliminate the bias in estimating σ when a sustained shift in the mean is present is to base the estimator on the **median** of the moving ranges of span two, as suggested by Clifford (1959) and Bryce, Gaudard, and Joiner (1997–98). This estimator is

$$\hat{\sigma}_3 = 1.047\overline{\text{MR}}$$

where $\overline{\text{MR}}$ is the median of the span-two moving ranges, and 1.047 is the reciprocal of the control chart constant d_4 for subgroups of size two defined such that $E(\tilde{R}) = d_4\sigma$ and $\tilde{R}$ is the median range. A table of d_4 values is in Wadsworth, Stephens, and Godfrey (2002). Essentially, only one of the span-two moving ranges should be affected by the sustained shift, and this single large moving range will have little impact on the value of the median moving range, certainly much less impact than it will have on the average moving range. Constructing an individuals control chart using the median moving range to estimate σ is an option in Minitab.

Now suppose that the assignable cause affects a single observation rather than causing a sustained shift in the mean. If there is a single observation that has mean $\mu_0 + \delta\sigma$, then

$$E\left(s^2\right) = \sigma^2 + \frac{1}{m}(\delta\sigma)^2$$

and this observation will affect two of the span-two moving ranges. If there are two adjacent such observations, then

$$E\left(s^2\right) = \sigma^2 + \frac{2(m-2)}{m(m-1)}(\delta\sigma)^2$$

and two of the span-two moving ranges will be affected by the out-of-control observations. Thus, when the assignable cause affects one or only a few adjacent observations, we expect the bias in s^2 to be smaller than when a sustained shift occurs. However, if an assignable cause producing a sustained shift in the mean occurs either very early in the sequence of observations or very late, it will produce much the same effect as an assignable cause affecting only one or a few adjacent points.

Some authors have suggested basing the estimate of σ on moving ranges of span greater than two, and some computer programs for control charts offer this as an option. It is easy to show that this will always lead to potentially increased bias in the estimate of σ when assignable causes are present. Note that if one uses a span-three moving range and there is a single observation whose mean is affected by the assignable cause, then this single observation will affect up to three of the moving ranges. Thus, a span-three moving range will result in more bias in the estimate of σ than will the moving range of span two. Furthermore, two span-three moving ranges will be affected by a sustained shift. In general, if one uses a span-w moving range and there is a single observation whose mean is affected by the assignable cause, up to w of these moving ranges will be impacted by this observation. Furthermore, if there is a sustained shift in the mean, up to $w - 1$ of the moving ranges will be affected by the shift in the mean. Consequently, increasing the span of a moving average beyond two results in increasing the bias in the estimate of σ if assignable causes that either produce sustained shifts in the process mean or that affect the mean of a single observation (or a few adjacent ones) are present. In fact, Wetherill and Brown (1991) advise plotting the estimate of σ versus the span of the moving average used to obtain the estimate. A sharply rising curve indicates the presence of assignable causes. For more discussion of using ranges to estimate process variability, see Woodall and Montgomery (2000–01).

■ TABLE 6.9
Formulas for Control Charts, Standards Given

Chart	Center Line	Control Limits
$\bar{x}$ (μ and σ given)	μ	$\mu \pm A\sigma$
R (σ given)	$d_2\sigma$	UCL = $D_2\sigma$, LCL = $D_1\sigma$
s (σ given)	$c_4\sigma$	UCL = $B_6\sigma$, LCL = $B_5\sigma$

■ TABLE 6.10
Formulas for Control Charts, Control Limits Based on Past Data (No Standards Given)

Chart	Center Line	Control Limits
$\bar{x}$ (using R)	$\bar{\bar{x}}$	$\bar{\bar{x}} \pm A_2\bar{R}$
$\bar{x}$ (using s)	$\bar{\bar{x}}$	$\bar{\bar{x}} \pm A_3 S$
R	$\bar{R}$	UCL = $D_4\bar{R}$, LCL = $D_3\bar{R}$
s	$\bar{s}$	UCL = $B_4\bar{s}$, LCL = $B_3\bar{s}$

6.5 Summary of Procedures for $\bar{x}$, R, And s Charts

It is convenient to summarize in one place the various computational formulas for the major types of variables control charts discussed in this chapter. Table 6.9 summarizes the formulas for $\bar{x}$, R, and s charts when standard values for μ and σ are given. Table 6.10 provides the corresponding summary when no standard values are given and trial control limits must be established from analysis of past data. The constants given for the s chart assume that $n - 1$ is used in the denominator of s. All constants are tabulated for various sample sizes in Appendix Table VI.

6.6 Applications of Variables Control Charts

There are many interesting applications of variables control charts. In this section, a few of them will be described to give additional insights into how the control chart works, as well as ideas for further applications.

EXAMPLE 6.7 Using Control Charts to Improve Suppliers' Processes

A large aerospace manufacturer purchased an aircraft component from two suppliers. These components frequently exhibited excessive variability on a key dimension that made it impossible to assemble them into the final product. This problem always resulted in expensive rework costs and occasionally caused delays in finishing the assembly of an airplane.

The materials receiving group performed 100% inspection of these parts in an effort to improve the situation. They maintained $\bar{x}$ and R charts on the dimension of interest for both suppliers. They found that the fraction of nonconforming units was about the same for both suppliers, but for very different reasons. Supplier A could produce parts with mean dimension equal to the required value, but the process was out of statistical control. Supplier B could maintain good statistical control and, in general, produced a part that exhibited considerably less variability than parts from supplier A, but its process was centered so far off the nominal required dimension that many parts were out of specification.

This situation convinced the procurement organization to work with both suppliers, persuading supplier A to implement SPC and to begin working at continuous improvement, and assisting supplier B to find out why his process was consistently centered incorrectly. Supplier B's problem was ultimately tracked to some incorrect code in an NC (numerical-controlled) machine, and the use of SPC at supplier A resulted in considerable reduction in variability over a six-month period. As a result of these actions, the problem with these parts was essentially eliminated.

9 Cumulative Sum and Exponentially Weighted Moving Average Control Charts

Chapter Outline

9.1 THE CUMULATIVE SUM CONTROL CHART
- 9.1.1 Basic Principles: The Cusum Control Chart for Monitoring the Process Mean
- 9.1.2 The Tabular or Algorithmic Cusum for Monitoring the Process Mean
- 9.1.3 Recommendations for Cusum Design
- 9.1.4 The Standardized Cusum
- 9.1.5 Improving Cusum Responsiveness for Large Shifts
- 9.1.6 The Fast Initial Response or Headstart Feature
- 9.1.7 One-Sided Cusums
- 9.1.8 A Cusum for Monitoring Process Variability
- 9.1.9 Rational Subgroups
- 9.1.10 Cusums for Other Sample Statistics
- 9.1.11 The V-Mask Procedure
- 9.1.12 The Self-Starting Cusum

9.2 THE EXPONENTIALLY WEIGHTED MOVING AVERAGE CONTROL CHART
- 9.2.1 The Exponentially Weighted Moving Average Control Chart for Monitoring the Process Mean
- 9.2.2 Design of an EWMA Control Chart
- 9.2.3 Robustness of the EWMA to Non-normality
- 9.2.4 Rational Subgroups
- 9.2.5 Extensions of the EWMA

9.3 THE MOVING AVERAGE CONTROL CHART

Supplemental Material for Chapter 9

- S9.1 The Markov Chain Approach for Finding the ARL for Cusum and EWMA Control Charts
- S9.2 Integral Equation versus Markov Chains for Finding the ARL

The supplemental material is on the textbook Website www.wiley.com/college/montgomery.

Chapter Overview and Learning Objectives

Chapters 5, 6, and 7 have concentrated on basic SPC methods. The control charts featured in these chapters are predominantly **Shewhart control charts.** These charts are extremely useful in phase I implementation of SPC, where the process is likely to be out of control and experiencing assignable causes that result in large shifts in the monitored parameters. Shewhart charts are also very useful in the diagnostic aspects of bringing an unruly process into statistical control, because the patterns on these charts often provide guidance regarding the nature of the assignable cause.

A major disadvantage of a Shewhart control chart is that it uses only the information about the process contained in the last sample observation and it ignores any information given by the entire sequence of points. This feature makes the Shewhart control chart relatively insensitive to small process shifts, say, on the order of about 1.5σ or less. This potentially makes Shewhart control charts less useful in phase II monitoring problems, where the process tends to operate in control, reliable estimates of the process parameters (such as the mean and standard deviation) are available, and assignable causes do not typically result in large process upsets or disturbances. Of course, other criteria, such as warning limits and other sensitizing rules, can be applied to Shewhart control charts in phase II to improve their performance against small shifts. However, the use of these procedures reduces the simplicity and ease of interpretation of the Shewhart control chart, and as we have previously observed, they also dramatically reduce the average run length of the chart when the process is actually in control. This can be very undesirable in phase II process monitoring.

Two very effective alternatives to the Shewhart control chart may be used when small process shifts are of interest: the **cumulative sum** (**cusum**) **control chart**, and the **exponentially weighted moving average (EWMA) control chart.** Cusum and EWMA control charts are excellent alternatives to the Shewhart control chart for phase II process monitoring situations. These control charts are the subject of this chapter.

After careful study of this chapter you should be able to do the following:

1. Set up and use cusum control charts for monitoring the process mean
2. Design a cusum control chart for the mean to obtain specific ARL performance
3. Incorporate a fast initial response feature into the cusum control chart
4. Use a combined Shewhart-cusum monitoring scheme
5. Set up and use EWMA control charts for monitoring the process mean
6. Design an EWMA control chart for the mean to obtain specific ARL performance
7. Understand why the EWMA control chart is robust to the assumption of normality
8. Understand the performance advantage of cusum and EWMA control charts relative to Shewhart control charts
9. Set up and use a control chart based on an ordinary (unweighted) moving average

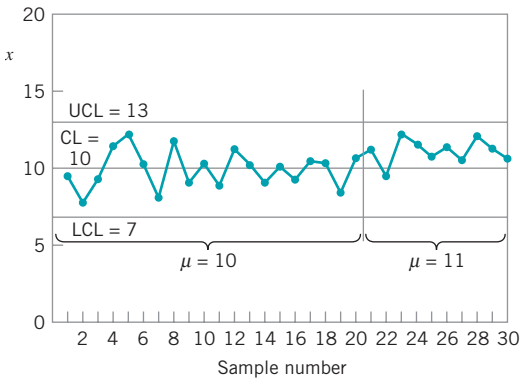
9.1 The Cumulative Sum Control Chart

9.1.1 Basic Principles: The Cusum Control Chart for Monitoring the Process Mean

Consider the data in Table 9.1, column (a). The first 20 of these observations were drawn at random from a normal distribution with mean $\mu = 10$ and standard deviation $\sigma = 1$. These observations have been plotted on a Shewhart control chart in Fig. 9.1. The center line and

■ TABLE 9.1
Data for the Cusum Example

Sample, i	(a) x_i	(b) $x_i - 10$	(c) $C_i = (x_i - 10) + C_{i-1}$
1	9.45	–0.55	–0.55
2	7.99	–2.01	–2.56
3	9.29	–0.71	–3.27
4	11.66	1.66	–1.61
5	12.16	2.16	0.55
6	10.18	0.18	0.73
7	8.04	–1.96	–1.23
8	11.46	1.46	0.23
9	9.20	–0.80	–0.57
10	10.34	0.34	–0.23
11	9.03	–0.97	–1.20
12	11.47	1.47	0.27
13	10.51	0.51	0.78
14	9.40	–0.60	0.18
15	10.08	0.08	0.26
16	9.37	–0.63	–0.37
17	10.62	0.62	0.25
18	10.31	0.31	0.56
19	8.52	–1.48	–0.92
20	10.84	0.84	–0.08
21	10.90	0.90	0.82
22	9.33	–0.67	0.15
23	12.29	2.29	2.44
24	11.50	1.50	3.94
25	10.60	0.60	4.54
26	11.08	1.08	5.62
27	10.38	0.38	6.00
28	11.62	1.62	7.62
29	11.31	1.31	8.93
30	10.52	0.52	9.45



■ FIGURE 9.1 A Shewhart control chart for the data in Table 9.1.

three-sigma control limits on this chart are at

$$\begin{aligned} \text{UCL} &= 13 \\ \text{Center line} &= 10 \\ \text{LCL} &= 7 \end{aligned}$$

Note that all 20 observations plot in control.

The last 10 observations in column (a) of Table 9.1 were drawn from a normal distribution with mean $\mu = 11$ and standard deviation $\sigma = 1$. Consequently, we can think of these last 10 observations as having been drawn from the process when it is out of control—that is, after the process has experienced a shift in the mean of 1σ. These last 10 observations are also plotted on the control chart in Fig. 9.1. None of these points plots outside the control limits, so we have no strong evidence that the process is out of control. Note that there is an indication of a shift in process level for the last 10 points, because all but one of the points plot above the center line. However, if we rely on the traditional signal of an out-of-control process, one or more points beyond a three-sigma control limit, then the Shewhart control chart has failed to detect the shift.

The reason for this failure, of course, is the relatively small magnitude of the shift. The Shewhart chart for averages is very effective if the magnitude of the shift is 1.5σ to 2σ or larger. For smaller shifts, it is not as effective. The cumulative sum (or cusum) control chart is a good alternative when small shifts are important.

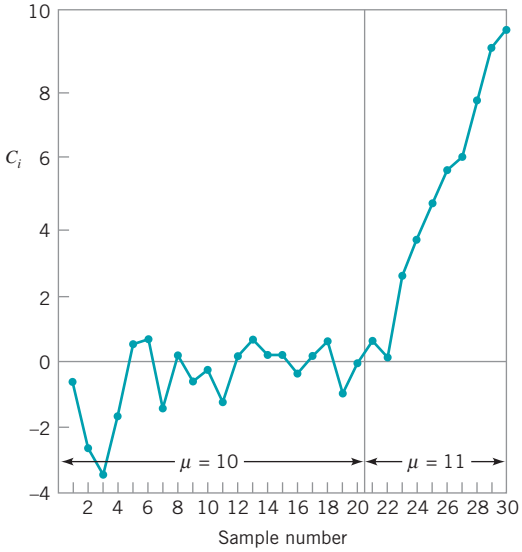
The cusum chart directly incorporates all the information in the sequence of sample values by plotting the cumulative sums of the deviations of the sample values from a target value. For example, suppose that samples of size $n \geq 1$ are collected, and $\bar{x}_j$ is the average of the jth sample. Then if μ_0 is the target for the process mean, the cumulative sum control chart is formed by plotting the quantity

$$C_i = \sum_{j=1}^{i} \left(\bar{x}_j - \mu_0 \right) \tag{9.1}$$

against the sample number i. C_i is called the cumulative sum up to and including the ith sample. Because they combine information from *several* samples, cumulative sum charts are more effective than Shewhart charts for detecting small process shifts. Furthermore, they are particularly effective with samples of size $n = 1$. This makes the cumulative sum control chart a good candidate for use in the chemical and process industries where rational subgroups are frequently of size 1, and in discrete parts manufacturing with automatic measurement of each part and on-line process monitoring directly at the work center.

Cumulative sum control charts were first proposed by Page (1954) and have been studied by many authors; in particular, see Ewan (1963), Page (1961), Gan (1991), Lucas (1976), Hawkins (1981, 1993a), and Woodall and Adams (1993). The book by Hawkins and Olwell (1998) is highly recommended. In this section, we concentrate on the cumulative sum chart for the process mean. It is possible to devise cumulative sum procedures for other variables, such as Poisson and binomial variables for modeling nonconformities and fraction nonconforming. We will show subsequently how the cusum can be used for monitoring process variability.

We note that if the process remains in control at the target value μ_0, the cumulative sum defined in equation (9.1) is a random walk with mean zero. However, if the mean shifts upward to some value $\mu_1 > \mu_0$, say, then an upward or positive drift will develop in the cumulative sum C_i. Conversely, if the mean shifts downward to some $\mu_1 < \mu_0$, then a downward or negative drift in C_i will develop. Therefore, if a significant trend develops in the plotted points either upward or downward, we should consider this as evidence that the process mean has shifted, and a search for some assignable cause should be performed.



■ **FIGURE 9.2** Plot of the cumulative sum from column (c) of Table 9.1.

This theory can be easily demonstrated by using the data in column (a) of Table 9.1 again. To apply the cusum in equation (9.1) to these observations, we would take $\bar{x} = x_i$ (since our sample size is $n = 1$) and let the target value $\mu_0 = 10$. Therefore, the cusum becomes

$$\begin{aligned} C_i &= \sum_{j=1}^{i} (x_j - 10) \\ &= (x_i - 10) + \sum_{j=1}^{i-1} (x_j - 10) \\ &= (x_i - 10) + C_{i-1} \end{aligned}$$

Column (b) of Table 9.1 contains the differences $x_i - 10$, and the cumulative sums are computed in column (c). The starting value for the cusum, C_0, is taken to be zero. Figure 9.2 plots the cusum from column (c) of Table 9.1. Note that for the first 20 observations where $\mu = 10$, the cusum tends to drift slowly, in this case maintaining values near zero. However, in the last 10 observations, where the mean has shifted to $\mu = 11$, a strong upward trend develops.

Of course, the cusum plot in Fig. 9.2 is not a control chart because it lacks statistical control limits. There are two ways to represent cusums, the **tabular** (or **algorithmic**) **cusum,** and the **V-mask form of the cusum.** Of the two representations, the tabular cusum is preferable. We now present the construction and use of the tabular cusum. We will also briefly discuss the V-mask procedure and indicate why it is not the best representation of a cusum.

9.1.2 The Tabular or Algorithmic Cusum for Monitoring the Process Mean

We now show how a tabular cusum may be constructed for monitoring the mean of a process. Cusums may be constructed both for individual observations and for the averages of rational subgroups. The case of individual observations occurs very often in practice, so that situation will be treated first. Later we will see how to modify these results for rational subgroups.

Let x_i be the ith observation on the process. When the process is in control, x_i has a normal distribution with mean μ_0 and standard deviation σ. We assume that either σ is known or that a reliable estimate is available. These assumptions are very consistent with

phase II applications of SPC, the situation in which the cusum is most useful. Later we will discuss monitoring σ with a cusum.

Sometimes we think of μ_0 as a **target value** for the quality characteristic x. This viewpoint is often taken in the chemical and process industries when the objective is to control x (viscosity, say) to a particular target value (such as 2000 centistokes at 100°C). If the process drifts or shifts off this target value, the cusum will signal, and an adjustment is made to some manipulatable variable (such as the catalyst feed rate) to bring the process back on target. Also, in some cases a signal from a cusum indicates the presence of an assignable cause that must be investigated just as in the Shewhart chart case.

The tabular cusum works by accumulating derivations from μ_0 that are above target with one statistic C^+ and accumulating derivations from μ_0 that are below target with another statistic C^-. The statistics C^+ and C^- are called **one-sided upper and lower cusums,** respectively. They are computed as follows:

The Tabular Cusum

$$C_i^+ = \max\left[0, x_i - (\mu_0 + K) + C_{i-1}^+\right] \tag{9.2}$$

$$C_i^- = \max\left[0, (\mu_0 - K) - x_i + C_{i-1}^-\right] \tag{9.3}$$

where the starting values are $C_0^+ = C_0^- = 0$.

In equations (9.2) and (9.3), K is usually called the **reference value** (or the **allowance,** or the **slack value**), and it is often chosen about halfway between the target μ_0 and the out-of-control value of the mean μ_1 that we are interested in detecting quickly.

Thus, if the shift is expressed in standard deviation units as $\mu_1 = \mu_0 + \delta\sigma$ (or $\delta = |\mu_1 - \mu_0|/\sigma$), then K is one-half the magnitude of the shift or

$$K = \frac{\delta}{2}\sigma = \frac{|\mu_1 - \mu_0|}{2} \tag{9.4}$$

Note that C_i^+ and C_i^- accumulate deviations from the target value μ_0 that are greater than K, with both quantities reset to zero on becoming negative. If either C_i^+ or C_i^- exceed the **decision interval** H, the process is considered to be out of control.

We have briefly mentioned how to choose K, but how does one choose H? Actually, the proper selection of these two parameters is quite important, as it has substantial impact on the performance of the cusum. We will talk more about this later, but a reasonable value for H is five times the process standard deviation σ.

EXAMPLE 9.1 A Tabular Cusum

Set up the tabular cusum using the data from Table 9.1.

SOLUTION

Recall that the target value is $\mu_0 = 10$, the subgroup size is $n = 1$, the process standard deviation is $\sigma = 1$, and suppose that the magnitude of the shift we are interested in detecting is $1.0\sigma = 1.0(1.0) = 1.0$. Therefore, the out-of-control value of the process mean is $\mu_1 = 10 + 1 = 11$. We will use a tabular cusum with $K = \frac{1}{2}$ (because the shift size is 1.0σ and $\sigma = 1$) and $H = 5$

■ TABLE 9.2
The Tabular Cusum for Example 9.1

		(a)			(b)		
Period *i*	x_i	$x_i - 10.5$	C_i^+	N^+	$9.5 - x_i$	C_i^-	N^-
1	9.45	−1.05	0	0	0.05	0.05	1
2	7.99	−2.51	0	0	1.51	1.56	2
3	9.29	−1.21	0	0	0.21	1.77	3
4	11.66	1.16	1.16	1	−2.16	0	0
5	12.16	1.66	2.82	2	−2.66	0	0
6	10.18	−0.32	2.50	3	−0.68	0	0
7	8.04	−2.46	0.04	4	1.46	1.46	1
8	11.46	0.96	1.00	5	−1.96	0	0
9	9.20	−1.3	0	0	0.30	0.30	1
10	10.34	−0.16	0	0	−0.84	0	0
11	9.03	−1.47	0	0	0.47	0.47	1
12	11.47	0.97	0.97	1	−1.97	0	0
13	10.51	0.01	0.98	2	−1.01	0	0
14	9.40	−1.10	0	0	0.10	0.10	1
15	10.08	−0.42	0	0	−0.58	0	0
16	9.37	−1.13	0	0	0.13	0.13	1
17	10.62	0.12	0.12	1	−1.12	0	0
18	10.31	−0.19	0	0	−0.81	0	0
19	8.52	−1.98	0	0	0.98	0.98	1
20	10.84	0.34	0.34	1	−1.34	0	0
21	10.90	0.40	0.74	2	−1.40	0	0
22	9.33	−1.17	0	0	0.17	0.17	1
23	12.29	1.79	1.79	1	−2.79	0	0
24	11.50	1.00	2.79	2	−2.00	0	0
25	10.60	0.10	2.89	3	−1.10	0	0
26	11.08	0.58	3.47	4	−1.58	0	0
27	10.38	−0.12	3.35	5	−0.88	0	0
28	11.62	1.12	4.47	6	−2.12	0	0
29	11.31	0.81	5.28	7	−1.81	0	0
30	10.52	0.02	5.30	8	−1.02	0	0

(because the recommended value of the decision interval is $H = 5\sigma = 5(1) = 5$).

Table 9.2 presents the tabular cusum scheme. To illustrate the calculations, consider period 1. The equations for C_i^+ and C_i^- are

$$C_1^+ = \max\left[0, x_1 - 10.5 + C_0^+\right]$$

and

$$C_1^- = \max\left[0, 9.5 - x_1 + C_0^-\right]$$

since $K = 0.5$ and $\mu_0 = 10$. Now $x_1 = 9.45$, so since $C_0^+ = C_0^- = 0$,

$$C_1^+ = \max[0, 9.45 - 10.5 + 0] = 0$$

and

$$C_1^- = \max[0, 9.5 - 9.45 + 0] = 0.05$$

For period 2, we would use

$$C_2^+ = \max\left[0, x_2 - 10.5 + C_1^+\right]$$
$$= \max[0, x_2 - 10.5 + 0]$$

and

$$C_2^- = \max\left[0, 9.5 - x_2 + C_1^-\right]$$
$$= \max[0, 9.5 - x_2 + 0.05]$$

Since $x_2 = 7.99$, we obtain

$$C_2^+ = \max[0, 7.99 - 10.5 + 0] = 0$$

(*continued*)

and

$$C_2^- = \max[0, 9.5 - 7.99 + 0.05] = 1.56$$

Panels (a) and (b) of Table 9.2 summarize the remaining calculations. The quantities N^+ and N^- in Table 9.2 indicate the number of consecutive periods that the cusums C_i^+ or C_i^- have been nonzero.

The cusum calculations in Table 9.2 show that the upper-side cusum at period 29 is $C_{29}^+ = 5.28$. Since this is the first period at which $C_i^+ > H = 5$, we would conclude that the process is out of control at that point. The tabular cusum also indicates when the shift probably occurred. The counter N^+ records the number of consecutive periods since the upper-side cusum C_i^+ rose above the value of zero. Since $N^+ = 7$ at period 29, we would conclude that the process was last in control at period $29 - 7 = 22$, so the shift likely occurred between periods 22 and 23.

It is useful to present a graphical display for the tabular cusum. These charts are sometimes called **cusum status charts.** They are constructed by plotting C_i^+ and C_i^- versus the sample number. Figure 9.3*a* shows the cusum status chart for the data in Example 9.1. Each vertical bar represents the value of C_i^+ and C_i^- in period i. With the decision interval plotted on the chart, the cusum status chart resembles a Shewhart control chart. We have also plotted the observations x_i for each period on the cusum status chart as the solid dots. This frequently helps the user of the control chart to visualize the actual process performance that has led to a particular value of the cusum. Some computer software packages have implemented the cusum status chart. Figure 9.3*b* shows the Minitab version. In Minitab, the lower cusum is defined as

$$C_i^- = \min\left(0, x_i - \mu_0 + k + C_{i-1}^-\right)$$

This results in a lower cusum that is always ≤ 0 [it is the negative of the lower cusum value from equation (9.3)]. Note in Figure 9.3*b* that the values of the lower cusum range from 0 to –5.

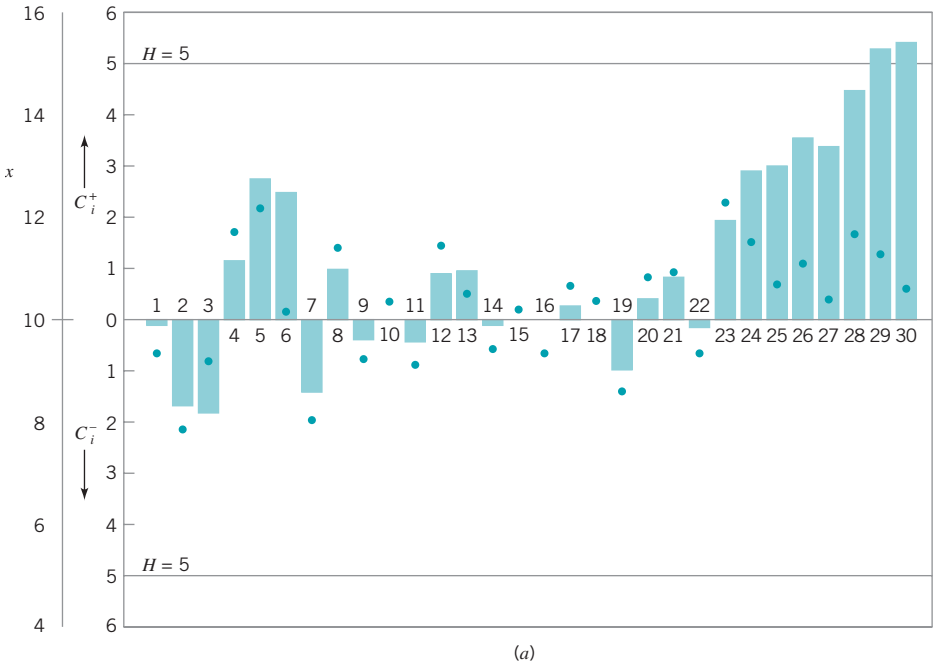
The action taken following an out-of-control signal on a cusum control scheme is identical to that with any control chart; one should search for the assignable cause, take any corrective action required, and then reinitialize the cusum at zero. The cusum is particularly helpful in determining when the assignable cause has occurred; as we noted in the previous example, just count backward from the out-of-control signal to the time period when the cusum lifted above zero to find the first period following the process shift. The counters N^+ and N^- are used in this capacity.

In situations where an adjustment to some manipulatable variable is required in order to bring the process back to the target value μ_0, it may be helpful to have an estimate of the new process mean following the shift. This can be computed from

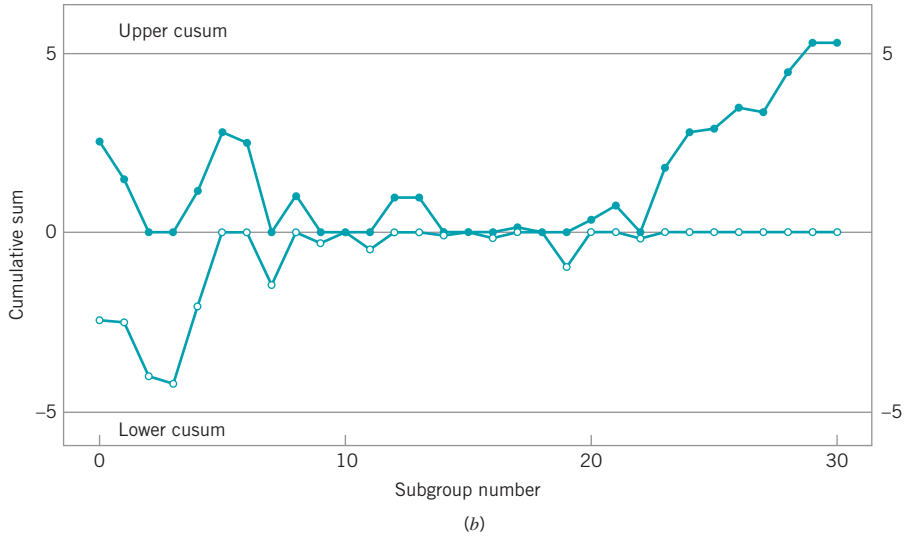
$$\hat{\mu} = \begin{cases} \mu_0 + K + \dfrac{C_i^+}{N^+}, & \text{if } C_i^+ > H \\ \mu_0 - K - \dfrac{C_i^-}{N^-}, & \text{if } C_i^- > H \end{cases} \tag{9.5}$$

To illustrate the use of equation (9.5), consider the cusum in period 29 with $C_{29}^+ = 5.28$. From equation (9.5), we would estimate the new process average as

$$\begin{aligned} \hat{\mu} &= \mu_0 + K + \frac{C_{29}^+}{N^+} \\ &= 10.0 + 0.5 + \frac{5.28}{7} \\ &= 11.25 \end{aligned}$$



(*a*)



(*b*)

■ **FIGURE 9.3** Cusum status charts for Example 9.1. (*a*) Manual chart. (*b*) Minitab chart.

So, for example, if the process characteristic is viscosity, then we would conclude that mean viscosity has shifted from 10 to 11.25, and if the manipulatable variable that affects viscosity is catalyst feed rate, then we would need to make an adjustment in catalyst feed rate that would result in moving the viscosity down by 1.25 units.

Finally, we should note that runs tests, and other sensitizing rules such as the zone rules, cannot be safely applied to the cusum, because successive values of C_i^+ and C_i^- are not independent. In fact, the cusum can be thought of as a weighted average, where the weights are stochastic or random. For example, consider the cusum shown in Table 9.2. The cusum at period 30 is $C_{30}^+ = 5.30$. This can be thought of as a weighted average in which we give equal weight to the last $N^+ = 8$ observations and weight zero to all other observations.

9.1.3 Recommendations for Cusum Design

The tabular cusum is designed by choosing values for the reference value K and the decision interval H. It is usually recommended that these parameters be selected to provide good average run length performance. There have been many analytical studies of cusum ARL performance. Based on these studies, we may give some general recommendations for selecting H and K.

Define $H = h\sigma$ and $K = k\sigma$, where σ is the standard deviation of the sample variable used in forming the cusum. Using $h = 4$ or $h = 5$ and $k = \frac{1}{2}$ will generally provide a cusum that has good ARL properties against a shift of about 1σ in the process mean.

To illustrate how well the recommendations of $h = 4$ or $h = 5$ with $k = \frac{1}{2}$ work, consider the two-sided average run lengths shown in Table 9.3. Note that a 1σ shift would be detected in either 8.38 samples (with $k = \frac{1}{2}$ and $h = 4$) or 10.4 samples (with $k = \frac{1}{2}$ and $h = 5$). By comparison, a Shewhart control chart for individual measurements would require 43.96 samples, on the average, to detect this shift.

Note also from Table 9.3 that $h = 4$ results in an in-control $ARL_0 = 168$ samples, whereas $h = 5$ results in $ARL_0 = 465$ samples. If we choose $h = 4.77$, this will provide a cusum with $ARL_0 = 370$ samples, which matches the ARL_0 value for a Shewhart control chart with the usual 3σ limits.

Generally, we want to choose k relative to the size of the shift we want to detect; that is, $k = \frac{1}{2}\delta$, where δ is the size of the shift in standard deviation units. This approach comes very close to minimizing the ARL_1 value for detecting a shift of size δ for fixed ARL_0. As mentioned earlier, a widely used value in practice is $k = \frac{1}{2}$. Then, once k is selected, you should choose h to give the desired in-control ARL_0 performance. Hawkins (1993a) gives a table of k values and the corresponding h values that will achieve $ARL_0 = 370$. These are reproduced in Table 9.4.

Several techniques can be used to calculate the ARL of a cusum. Vance (1986) provides a very accurate computer program. A number of authors have used an approach to calculating

■ **TABLE 9.3**
ARL Performance of the Tabular Cusum with $k = \frac{1}{2}$ and $h = 4$ or $h = 5$

Shift in Mean (multiple of σ)	$h = 4$	$h = 5$
0	168	465
0.25	74.2	139
0.50	26.6	38.0
0.75	13.3	17.0
1.00	8.38	10.4
1.50	4.75	5.75
2.00	3.34	4.01
2.50	2.62	3.11
3.00	2.19	2.57
4.00	1.71	2.01

■ **TABLE 9.4**
Values of k and the Corresponding Values of h That Give $ARL_0 = 370$ for the Two-Sided Tabular Cusum [from Hawkins (1993a)]

k	0.25	0.5	0.75	1.0	1.25	1.5
h	8.01	4.77	3.34	2.52	1.99	1.61

ARLs due to Brook and Evans (1972) that is based on approximating transitions from the in-control to the out-of-control state with a Markov chain.[1] Hawkins (1992) has provided a simple but very accurate ARL calculation procedure based on an approximating equation. His approximation requires a table of constants to apply and is accurate to within 1%–3% of the true ARL value. Woodall and Adams (1993) recommend the ARL approximation given by Siegmund (1985) because of its simplicity. For a **one-sided cusum** (that is, C_i^+ or C_i^-) with parameters h and k, Siegmund's approximation is

$$\text{ARL} = \frac{\exp(-2\Delta b) + 2\Delta b - 1}{2\Delta^2} \tag{9.6}$$

for $\Delta \neq 0$, where $\Delta = \delta^* - k$ for the upper one-sided cusum C_i^+, $\Delta = -\delta^* - k$ for the lower one-sided cusum C_i^-, $b = h + 1.166$, and $\delta^* = (\mu_1 - \mu_0)/\sigma$. If $\Delta = 0$, one can use $\text{ARL} = b^2$.

The quantity δ^* represents the shift in the mean, in the units of σ, for which the ARL is to be calculated. Therefore, if $\delta^* = 0$, we would calculate ARL_0 from equation (9.6), whereas if $\delta^* \neq 0$, we would calculate the value of ARL_1 corresponding to a shift of size δ^*. To obtain the ARL of the two-sided cusum from the ARLs of the two one-sided statistics—say, ARL^+ and ARL^-—use

$$\frac{1}{\text{ARL}} = \frac{1}{\text{ARL}^+} + \frac{1}{\text{ARL}^-} \tag{9.7}$$

To illustrate, consider the two-sided cusum with $k = \frac{1}{2}$ and $h = 5$. To find ARL_0 we would first calculate the ARL_0 values for the two-sided schemes—say, ARL_0^+ and ARL_0^-. Set $\delta^* = 0$; then $\Delta = \delta^* - k = 0 - \frac{1}{2} = -\frac{1}{2}$, $b = h + 1.166 = 5 + 1.166 = 6.166$, and from equation (9.6)

$$\text{ARL}_0^+ \simeq \frac{\exp\left[-2\left(-\frac{1}{2}\right)(6.166)\right] + 2\left(-\frac{1}{2}\right)(6.166) - 1}{2\left(-\frac{1}{2}\right)^2} = 938.2$$

By symmetry, we have $\text{ARL}_0^- = \text{ARL}_0^+$, and so from equation (9.7), the in-control ARL for the two-sided cusum is

$$\frac{1}{\text{ARL}_0} = \frac{1}{938.2} + \frac{1}{938.2}$$

or

$$\text{ARL}_0 = 469.1$$

This is very close to the true ARL_0 value of 465 shown in Table 9.3. If the mean shifts by 2σ, then $\delta^* = 2$, $\Delta = 1.5$ for the upper one-sided cusum, $\Delta = -2.5$ for the lower one-sided cusum, and from equations (9.6) and (9.7) we can calculate the approximate ARL_1 of the two-sided cusum as $\text{ARL}_1 = 3.89$. The exact value shown in Table 9.3 is 4.01.

[1]The Brook and Evans Markov chain approach is discussed in the supplemental text material for this chapter. Cusum ARLs can also be determined from integral equations. This approach is also discussed in the supplemental material for this chapter.

One could use Siegmund's approximation and trial-and-error arithmetic to give a control limit that would have any desired ARL. Alternatively, numerical root-finding methods would also work well. Woodall and Adams (1993) give an excellent discussion of this approach.

9.1.4 The Standardized Cusum

Many users of the cusum prefer to standardize the variable x_i before performing the calculations. Let

$$y_i = \frac{x_i - \mu_0}{\sigma} \tag{9.8}$$

be the standardized value of x_i. Then the standardized cusums are defined as follows.

The Standardized Two-Sided Cusum

$$C_i^+ = \max\left[0, y_i - k + C_{i-1}^+\right] \tag{9.9}$$

$$C_i^- = \max\left[0, -k - y_i + C_{i-1}^-\right] \tag{9.10}$$

There are two advantages to standardizing the cusum. First, many cusum charts can now have the same values of k and h, and the choices of these parameters are not scale dependent (that is, they do not depend on σ). Second, a standardized cusum leads naturally to a cusum for controlling variability, as we will see in Section 9.1.8.

9.1.5 Improving Cusum Responsiveness for Large Shifts

We have observed that the cusum control chart is very effective in detecting small shifts. However, the cusum control chart is not as effective as the Shewhart chart in detecting large shifts. An approach to improving the ability of the cusum control chart to detect large process shifts is to use a **combined cusum-Shewhart procedure** for on-line control. Adding the Shewhart control is a very simple modification of the cumulative sum control procedure. The Shewhart control limits should be located approximately 3.5 standard deviations from the center line or target value μ_0. An out-of-control signal on either (or both) charts constitutes an action signal. Lucas (1982) gives a good discussion of this technique. Column (a) of Table 9.5 presents the ARLs of the basic cusum with $k = \frac{1}{2}$ and $h = 5$. Column (b) of Table 9.5 presents the ARLs of the cusum with Shewhart limits added to the individual measurements. As suggested above, the Shewhart limits are at 3.5σ. Note from examining these ARL values that the addition of the Shewhart limits has improved the ability of the procedure to detect larger shifts and has only slightly decreased the in-control ARL_0. We conclude that a combined cusum-Shewhart procedure is an effective way to improve cusum responsiveness to large shifts.

9.1.6 The Fast Initial Response or Headstart Feature

This procedure was devised by Lucas and Crosier (1982) to improve the sensitivity of a cusum at process start-up. Increased sensitivity at process start-up would be desirable if the corrective action did not reset the mean to the target value. The **fast initial response (FIR)** or **headstart** essentially just sets the starting values C_0^+ and C_0^- equal to some nonzero value, typically $H/2$. This is called a 50% headstart.

To illustrate the headstart procedure, consider the data in Table 9.6. These data have a target value of 100, $K = 3$, and $H = 12$. We will use a 50% headstart value of $C_0^+ = C_0^- =$

TABLE 9.5

ARL Values for Some Modifications of the Basic Cusum with $k = \frac{1}{2}$ and $h = 5$ (If subgroups of size $n > 1$ are used, then $\sigma = \sigma_{\bar{x}} = \sigma/\sqrt{n}$)

Shift in Mean (multiple of σ)	(a) Basic Cusum	(b) Cusum-Shewhart (Shewhart limits at 3.5σ)	(c) Cusum with FIR	(d) FIR Cusum-Shewhart (Shewhart limits at 3.5σ)
0	465	391	430	360
0.25	139	130.9	122	113.9
0.50	38.0	37.20	28.7	28.1
0.75	17.0	16.80	11.2	11.2
1.00	10.4	10.20	6.35	6.32
1.50	5.75	5.58	3.37	3.37
2.00	4.01	3.77	2.36	2.36
2.50	3.11	2.77	1.86	1.86
3.00	2.57	2.10	1.54	1.54
4.00	2.01	1.34	1.16	1.16

TABLE 9.6

A Cusum with a Headstart, Process Mean Equal to 100

		(a)			(b)		
Period i	x_i	$x_i - 103$	C_i^+	N^+	$97 - x_i$	C_i^-	N^-
1	102	−1	5	1	−5	1	1
2	97	−6	0	0	0	1	2
3	104	1	1	1	−7	0	0
4	93	−6	0	0	4	4	1
5	100	−3	0	0	−3	1	2
6	105	2	2	1	−8	0	0
7	96	−7	0	0	1	1	1
8	98	−5	0	0	−1	0	0
9	105	2	2	1	−8	0	0
10	99	−4	0	0	−2	0	0

$H/2 = 6$. The first 10 samples are in control with mean equal to the target value of 100. Since $x_1 = 102$, the cusums for the first period will be

$$\begin{aligned} C_i^+ &= \max\left[0, x_1 - 103 + C_0^+\right] \\ &= \max[0, 102 - 103 + 6] = 5 \\ C_i &= \max\left[0, 97 - x_1 + C_0^-\right] \\ &= \max[0, 97 - 102 + 6] = 1 \end{aligned}$$

Note that the starting cusum value is the headstart $H/2 = 6$. In addition, we see from panels (a) and (b) of Table 9.6 that both cusums decline rapidly to zero from the starting value. In fact, from period 2 onward C_1^+ is unaffected by the headstart, and from period 3 onward C_1^- is unaffected by the headstart. This has occurred because the process is in control at the target value of 100, and several consecutive observations near the target value were observed.

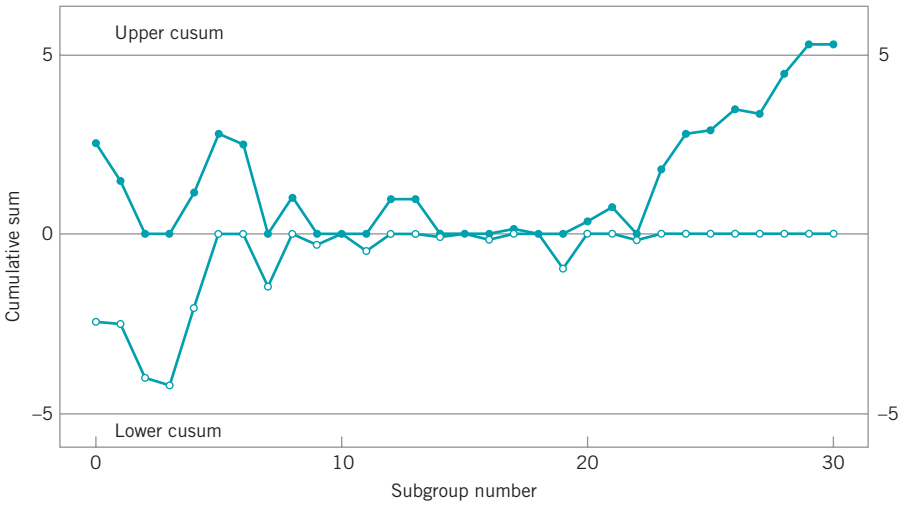
■ TABLE 9.7
A Cusum with a Headstart, Process Mean Equal to 105

		(a)			(b)		
Period i	x_i	$x_i - 103$	C_i^+	N^+	$97 - x_i$	C_i^-	N^-
1	107	4	10	1	−10	0	0
2	102	−1	9	2	−5	0	0
3	109	6	15	3	−12	0	0
4	98	−5	10	4	−1	0	0
5	105	2	12	5	−8	0	0
6	110	7	19	6	−13	0	0
7	101	−2	17	7	−4	0	0
8	103	0	17	8	−6	0	0
9	110	7	24	9	−13	0	0
10	104	1	25	10	−7	0	0

Now suppose the process had been out of control at process start-up, with mean 105. Table 9.7 presents the data that would have been produced by this process and the resulting cusums. Note that the third sample causes C_3^+ to exceed the limit $H = 12$. If no headstart had been used, we would have started with $C_0^+ = 0$, and the cusum would not exceed H until sample number 6.

This example demonstrates the benefits of a headstart. If the process starts in control at the target value, the cusums will quickly drop to zero and the headstart will have little effect on the performance of the cusum procedure. Figure 9.4 illustrates this property of the headstart using the data from Table 9.1. The cusum chart was produced using Minitab. However, if the process starts at some level different from the target value, the headstart will allow the cusum to detect it more quickly, resulting in shorter out-of-control ARL values.

Column (c) of Table 9.5 presents the ARL performance of the basic cusum with the headstart or FIR feature. The ARLs were calculated using a 50% headstart. Note that the ARL values for the FIR cusum are valid for the case when the process is out of control at the time the cusums are reset. When the process is in control, the headstart value quickly drops to zero.



■ FIGURE 9.4 A Minitab cusum status chart for the data in Table 9.1 illustrating the fast initial response or headstart feature.

Thus, if the process is in control when the cusum is reset but shifts out of control later, the more appropriate ARL for such a case should be read from column (a)—that is, the cusum without the FIR feature.

9.1.7 One-Sided Cusums

We have focused primarily on the two-sided cusum. Note that the tabular procedure is constructed by running two one-sided procedures, C_i^+ and C_i^-. There are situations in which only a single one-sided cusum procedure is useful.

For example, consider a chemical process for which the critical quality characteristic is the product viscosity. If viscosity drops below the target ($\mu_0 = 2000$ centistokes at 100°C, say), there is no significant problem, but any increase in viscosity should be detected quickly. A one-sided upper cusum would be an ideal process-monitoring scheme. Siegmund's procedure [equation (9.6)] could be used to calculate the ARLs for the one-sided scheme.

It is also possible to design cusums that have different sensitivity on the upper and lower side. This could be useful in situations where shifts in either direction are of interest, but shifts above the target (say) are more critical than shifts below the target.

9.1.8 A Cusum for Monitoring Process Variability

It is possible to construct cusum control charts for monitoring process variability. Since cusums are usually employed with individual observations, the procedure due to Hawkins (1981) is potentially useful. As before, let x_i be the normally distributed process measurement with mean or target value μ_0 and standard deviation σ. The standardized value of x_i is $y_i = (x_i - \mu_0)/\sigma$. Hawkins (1981, 1993a) suggests creating a new standardized quantity

$$v_i = \frac{\sqrt{|y_i|} - 0.822}{0.349} \tag{9.11}$$

He suggests that the v_i are sensitive to variance changes rather than mean changes. In fact, the statistic v_i is sensitive to both mean and variance changes. Since the in-control distribution of v_i is approximately $N(0,1)$, two one-sided standardized scale (i.e., standard deviation) cusums can be established as follows.

The Scale Cusum

$$S_i^+ = \max\left[0, v_i - k + S_{i-1}^+\right] \tag{9.12}$$

$$S_i^- = \max\left[0, -k - v_i + S_{i-1}^-\right] \tag{9.13}$$

where $S_0^+ = S_0^- = 0$ (unless a FIR feature is used) and the values of k and h are selected as in the cusum for controlling the process mean.

The interpretation of the scale cusum is similar to the interpretation of the cusum for the mean. If the process standard deviation increases, the values of S_i^+ will increase and eventually exceed h, whereas if the standard deviation decreases, the values of S_i^- will increase and eventually exceed h.

Although one could maintain separate cusum status charts for the mean and standard deviation, Hawkins (1993a) suggests plotting them on the same graph. He also provides several excellent examples and further discussion of this procedure. Study of his examples will be of value in improving your ability to detect changes in process variability from the scale cusum. If the scale cusum signals, one would suspect a change in variance, but if both cusums signal, one would suspect a shift in the mean.

9.1.9 Rational Subgroups

Although we have given the development of the tabular cusum for the case of individual observations ($n = 1$), it is easily extended to the case of averages of rational subgroups where the sample size $n > 1$. Simply replace x_i by $\bar{x}_i$ (the sample or subgroup average) in the above formulas, and replace σ with $\sigma_{\bar{x}} = \sigma/\sqrt{n}$.

With Shewhart charts, the use of averages of rational subgroups substantially improves control chart performance. However, this does not always happen with the cusum. If, for example, you have a choice of taking a sample of size $n = 1$ every half hour or a sample consisting of a rational subgroup of size $n = 5$ every 2.5 hours (not that both choices have the same sampling intensity), the cusum will often work best with the choice of $n = 1$ every half hour. For more discussion of this, see Hawkins and Olwell (1998). Only if there is some significant economy of scale or some other valid reason for taking samples of size greater than unity should one consider using $n > 1$ with the cusum.

One practical reason for using rational subgroups of size $n > 1$ is that we could now set up a cusum on the sample **variance** and use it to monitor **process variability.** Cusums for variances are discussed in detail by Hawkins and Olwell (1998); the paper by Chang and Gan (1995) is also recommended. We assume that the observations are normally distributed and that the in-control and out-of-control values are σ_0^2 and σ_1^2, respectively.

Let S_i^2 be the sample variance of the ith subgroup. The cusum for a normal variance is

$$C_i^- = \max\left(0, C_{i-1}^- + S_i^2 - k\right) \tag{9.14}$$

$$C_i^+ = \max\left(0, C_{i-1}^+ + S_i^2 + k\right) \tag{9.15}$$

where $k = [2\ln(\sigma_0/\sigma_1)\sigma_0^2\sigma_1^2/(\sigma_0^2 - \sigma_1^2)]$ with $C_0^- = C_0^+ = 0$. A headstart or FIR feature can also be used with this cusum. Hawkins and Olwell (1998) have a Website with software that supports their book [(the cusum Website of the School of Statistics at the University of Minnesota (www.stat.umn.edu)]. The software provided at this Website can be used for designing this cusum; that is, obtaining the required value of H for a specified target value of ARL_0.

9.1.10 Cusums for Other Sample Statistics

We have concentrated on cusums for sample averages. However, it is possible to develop cusums for other sample statistics such as the ranges and standard deviations of rational subgroups, fractions nonconforming, and defects. These are well-developed procedures and have proven optimality properties for detecting step changes in the parameters. Some of these cusums are discussed in the papers by Lowry, Champ, and Woodall (1995), Gan (1993), Lucas (1985), and White, Keats, and Stanley (1997). The book by Hawkins and Olwell (1998) is an excellent reference.

One variation of the cusum is extremely useful when working with count data and the count rate is very low. In this case, it is frequently more effective to form a cusum using the time between events (TBE). The most common situation encountered in practice is to use the TBE cusum to detect an *increase* in the count rate. This is equivalent to detecting a *decrease* in the time between these events. When the number of counts is generated from a

Poisson distribution, the time between these events will follow an exponential distribution. An appropriate TBE cusum scheme is

$$C_i^- = \max\left[0, K - T_i + C_{i-1}^-\right] \tag{9.16}$$

where K is the reference value and T_i is the time that has elapsed since that last observed count. Lucas (1985) and Bourke (1991) discuss the choice of K and H for this procedure. Borror, Keats, and Montgomery (2003) have examined the robustness of the TBE cusum to the exponential distribution and report that moderate departures from the exponential do not affect its performance.

An alternative and very effective procedure would be to transform the time between observed counts to an approximately normally distributed random variable, as discussed in Section 7.3.5, and use the cusum for monitoring the mean of a normal distribution in Section 9.1.2 instead of equation (9.16).

9.1.11 The V-Mask Procedure

An alternative procedure to the use of a tabular cusum is the V-mask control scheme proposed by Barnard (1959). The V-mask is applied to successive values of the cusum statistic

$$C_i = \sum_{j=1}^{i} y_j = y_i + C_{i-1}$$

where y_i is the standardized observation $y_i = (x_i - \mu_0)/\sigma$. A typical V-mask is shown in Fig. 9.5.

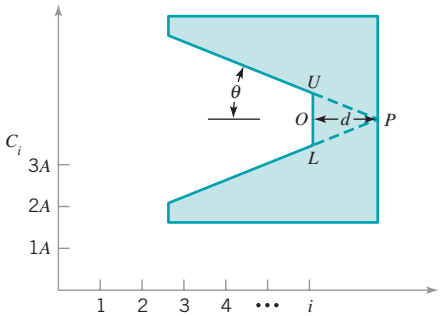
The decision procedure consists of placing the V-mask on the cumulative sum control chart with the point O on the last value of C_i and the line OP parallel to the horizontal axis. If all the previous cumulative sums, $C_1, C_2, \ldots, C_i$ lie within the two arms of the V-mask, the process is in control. However, if any of the cumulative sums lie outside the arms of the mask, the process is considered to be out of control. In actual use, the V-mask would be applied to each new point on the cusum chart as soon as it was plotted, and the arms are assumed to extend backward to the origin. The performance of the V-mask is determined by the lead distance d and the angle θ shown in Fig. 9.5.

The tabular cusum and the V-mask scheme are equivalent if

$$k = A\tan\theta \tag{9.17}$$

and

$$h = A\,d\tan(\theta) = dk \tag{9.18}$$



■ FIGURE 9.5 A typical V-mask.

In these two equations, A is the horizontal distance on the V-mask plot between successive points in terms of unit distance on the vertical scale. Refer to Fig. 9.5. For example, to construct a V-mask equivalent to the tabular cusum scheme used in Example 9.1, where $k = \frac{1}{2}$ and $h = 5$, we would select $A = 1$ (say), and then equations (9.17) and (9.18) would be solved as follows:

$$k = A\tan\theta$$

$$\frac{1}{2} = (1)\tan\theta$$

or

$$\theta = 26.57°$$

and

$$h = dk$$

$$5 = d\left(\frac{1}{2}\right)$$

or

$$d = 10$$

That is, the lead distance of the V-mask would be 10 horizontal plotting positions, and the angle opening on the V-mask would be 26.57°.

Johnson (1961) [also see Johnson and Leone (1962a, b, c)] has suggested a method for designing the V-mask; that is, selecting d and θ. He recommends the V-mask parameters

$$\theta = \tan^{-1}\left(\frac{\delta}{2A}\right) \tag{9.19}$$

and

$$d = \left(\frac{2}{\delta^2}\right)\ln\left(\frac{1-\beta}{\alpha}\right) \tag{9.20}$$

where 2α is the greatest allowable probability of a signal when the process mean is on target (a false alarm) and β is the probability of not detecting a shift of size δ. If β is small, which is usually the case, then

$$d \simeq -2\frac{\ln(\alpha)}{\delta} \tag{9.21}$$

We strongly advise **against using the V-mask procedure.** Some of the disadvantages and problems associated with this scheme are as follows:

1. The headstart feature, which is very useful in practice, cannot be implemented with the V-mask.
2. It is sometimes difficult to determine how far backward the arms of the V-mask should extend, thereby making interpretation difficult for the practitioner.
3. Perhaps the biggest problem with the V-mask is the ambiguity associated with α and β in the Johnson design procedure.

Adams, Lowry, and Woodall (1992) point out that defining 2α as the probability of a false alarm is incorrect. Essentially, 2α cannot be the probability of a false alarm on any single sample, because this probability changes over time on the cusum, nor can 2α be the probability of eventually obtaining a false alarm (this probability is, of course, 1). In fact, 2α must be

TABLE 9.8

Actual Values of ARL_0 for a V-Mask Scheme Designed Using Johnson's Method [Adapted from Table 2 in Woodall and Adams (1993)]

	Values of α [Desired Value of $ARL_0 = 1/(2\alpha)$]	
Shift to Be Detected, δ	**0.00135 (370)**	**0.001 (500)**
1.0	2350.6	3184.5
2.0	1804.5	2435.8
3.0	2194.8	2975.4

the long-run proportion of observations resulting in false alarms. If this is so, then the in-control ARL should be $ARL_0 = 1/(2\alpha)$. However, Johnson's design method produces values of ARL_0 that are substantially larger than $1/(2\alpha)$.

Table 9.8 shows values of ARL_0 for a V-mask scheme designed using Johnson's method. Note that the actual values of ARL_0 are about five times the desired value used in the design procedure. The schemes will also be much less sensitive to shifts in the process mean. Consequently, the use of the V-mask scheme is not a good idea. Unfortunately, it is the default cusum in some SPC software packages.

9.1.12 The Self-Starting Cusum

The cusum is typically used as a phase II procedure; that is, it is applied to monitor a process that has already been through the phase I process and most of the large assignable causes have been removed. In phase II, we typically assume that the process parameters are reasonably well estimated. In practice, this turns out to be a fairly important assumption, as using estimates of the parameters instead of the true values has an effect on the average run length performance of the control chart [this was discussed in Chapter 4; also see the review paper by Jensen et al. (2006)]. Control charts that are designed to detect small shifts are particularly sensitive to this assumption, including the cusum. A Shewhart control chart with the Western Electric rules also would be very sensitive to the estimates of the process parameters. One solution to this is to use a large sample of phase I data to estimate the parameters.

An alternative approach for the cusum is to use a **self-starting cusum** procedure due to Hawkins (1987). The self-starting cusum for the mean of a normally distributed random variable is easy to implement. It can be applied immediately without any need for a phase I sample to estimate the process parameters, in this case the mean μ and the variance σ^2.

Let $\bar{x}_n$ be the average of the first n observations and let $w_n = \sum_{i=1}^{n} (x_i - \bar{x}_n)^2$ be the sum of squared deviations from the average of those observations. Convenient computing formulas to update these quantities after each new observation are

$$\bar{x}_n = \bar{x}_{n-1} + \frac{x_n - \bar{x}_{n-1}}{n}$$

$$w_n = w_{n-1} + \frac{(n-1)(x_n - \bar{x}_{n-1})^2}{n}$$

The sample variance of the first n observations is $s_n^2 = w_n/(n-1)$. Standardize each successive new process observation using

$$T_n = \frac{x_n - \bar{x}_{n-1}}{s_{n-1}}$$

for the case where n is greater than or equal to 3. If the observations are normally distributed, the distribution of $\sqrt{\frac{n-1}{n}}T_n$ is a t distribution with $n-1$ degrees of freedom. The cumulative distribution of T_n is

$$P(T_n \le t) = F_{n-2}\left(t\sqrt{\frac{n-1}{n}}\right)$$

where F_n is the cumulative t distribution with $n-1$ degrees of freedom. It turns out that if the tail area for any continuous random variable is converted to a normal ordinate we obtain a new random variable that is distributed exactly as a standard normal random variable. That is, if Φ^{-1} is the inverse normal cumulative distribution, then the transformation

$$U_n = \Phi^{-1}[F_{n-2}(a_n T_n)] \quad \text{where } a_n = \sqrt{\frac{n-1}{n}}$$

converts the cusum quantity T_n into a standard normal random variable. It turns out that the values of U_n are statistically independent (this isn't obvious, because successive values of U_n share the same data points), so one can plot all values of U_n for $n \ge 3$ on a N(0, 1) cusum. This nicely avoids the problem of using a large sample of phase I data to estimate the process parameters for a conventional cusum.

To illustrate the procedure, consider the data from Table 9.1. The first eight of these observations are shown in Table 9.9. The columns to the right of the data are the running average, the running sum of squared deviations from the average, the running standard deviation, the standardized value of each observation, the t distribution value that corresponds to it, the t distribution tail area, and finally the equivalent inverse normal value. The values of U_n can be handled as any data for a cusum. For example, they could be entered into a tabular cusum and plotted on a cusum status chart.

There is a difference in practical usage of a self-starting cusum in comparison to an ordinary cusum surrounding the out-of-control situation. In a standard or ordinary cusum, the cusum moves upward indefinitely following an upward shift in the mean until the assignable cause is discovered. However, in a self-starting cusum the plotted statistic will begin moving upward following the shift, but as the shifted values are entered into the calculations of the running average and running standard deviation, they move the running mean closer to the new shifted value of the mean and the running standard deviation gets larger. Consequently, if the process is not adjusted and the cusum reset, the self-starting cusum will turn back downward. Therefore, the users of a self-starting cusum should to take investigative and corrective action immediately following an out-of-control signal. When restarting, it is also necessary to remove the out-of-control data from the running average and running standard deviation. Outliers can also impact the self-starting cusum, principally by inflating the running standard

■ **TABLE 9.9**
Calculations for a Self-Starting Cusum

n	x_n	$\bar{x}_n$	w_n	S_n	T_n	a_nT_n	$F_{n-2}(a_nT_n)$	U_n
1	9.45	9.45	0	—	—	—	—	—
2	7.99	8.72	1.07	1.03	—	—	—	—
3	9.29	8.91	1.25	0.80	0.55	0.45	0.6346	0.34406
4	11.66	9.60	6.92	1.52	3.44	2.98	0.9517	1.66157
5	12.16	10.11	12.16	1.75	1.68	1.50	0.8847	1.19881
6	10.18	10.12	12.16	1.56	0.04	0.04	0.5152	0.03811
7	8.04	9.82	15.87	1.63	−1.33	−1.23	0.1324	−1.11512
8	11.46	10.03	18.22	1.62	1.01	0.94	0.8107	0.88048

deviation. Maintaining a Shewhart control chart on the observations can give protection against this problem. Refer to Hawkins and Olwell (1998) for more suggestions and discussion.

9.2 The Exponentially Weighted Moving Average Control Chart

The exponentially weighted moving average (EWMA) control chart is also a good alternative to the Shewhart control chart when we are interested in detecting small shifts. The performance of the EWMA control chart is approximately equivalent to that of the cumulative sum control chart, and in some ways it is easier to set up and operate. As with the cusum, the EWMA is typically used with individual observations, and so, consequently, we will discuss that case first. We will also give the results for rational subgroups of size $n > 1$.

9.2.1 The Exponentially Weighted Moving Average Control Chart for Monitoring the Process Mean

The EWMA control chart was introduced by Roberts (1959). See also Crowder (1987a, 1989) and Lucas and Saccucci (1990) for good discussions of the EWMA. The exponentially weighted moving average is defined as

$$z_i = \lambda x_i + (1-\lambda) z_{i-1} \tag{9.22}$$

where $0 < \lambda \leq 1$ is a constant and the starting value (required with the first sample at $i = 1$) is the process target, so that

$$z_0 = \mu_0$$

Sometimes the average of preliminary data is used as the starting value of the EWMA, so that $z_0 = \bar{x}$.

To demonstrate that the EWMA z_i is a weighted average of all previous sample means, we may substitute for z_{i-1} on the right-hand side of equation (9.22) to obtain

$$\begin{aligned} z_i &= \lambda x_i + (1-\lambda)\left[\lambda x_{i-1} + (1-\lambda) z_{i-2}\right] \\ &= \lambda x_i + \lambda(1-\lambda) x_{i-1} + (1-\lambda)^2 z_{i-2} \end{aligned}$$

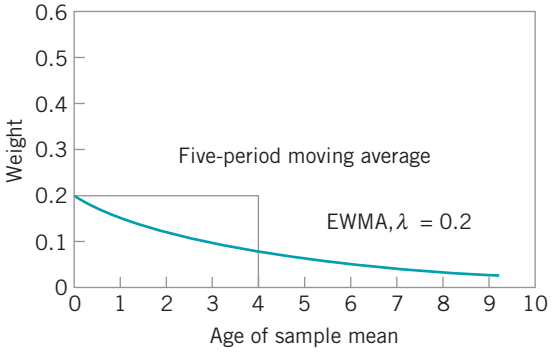
Continuing to substitute recursively for z_{i-j}, $j = 2, 3, \ldots, t$, we obtain

$$z_i = \lambda \sum_{j=0}^{i-1} (1-\lambda)^j x_{i-j} + (1-\lambda)^i z_0 \tag{9.23}$$

The weights $\lambda(1-\lambda)^j$ decrease geometrically with the age of the sample mean. Furthermore, the weights sum to unity, since

$$\lambda \sum_{j=0}^{i-1} (1-\lambda)^j = \lambda \left[\frac{1-(1-\lambda)^i}{1-(1-\lambda)} \right] = 1-(1-\lambda)^i$$

If $\lambda = 0.2$, then the weight assigned to the current sample mean is 0.2 and the weights given to the preceding means are 0.16, 0.128, 0.1024, and so forth. A comparison of these weights with those of a five-period moving average is shown in Fig. 9.6. Because these weights decline geometrically when connected by a smooth curve, the EWMA is sometimes called a **geometric moving average (GMA).** The EWMA is used extensively in time series modeling and in forecasting [see Box, Jenkins, and Reinsel (1994) and Montgomery, Johnson, and Gardiner (1990)]. Since the EWMA can be viewed as a weighted average of all past and current



■ **FIGURE 9.6** Weights of past sample means.

observations, it is very insensitive to the normality assumption. It is therefore an ideal control chart to use with individual observations.

If the observations x_i are independent random variables with variance σ^2, then the variance of z_i is

$$\sigma_{z_i}^2 = \sigma^2\left(\frac{\lambda}{2-\lambda}\right)\left[1-(1-\lambda)^{2i}\right] \tag{9.24}$$

Therefore, the EWMA control chart would be constructed by plotting z_i versus the sample number i (or time). The center line and control limits for the EWMA control chart are as follows.

The EWMA Control Chart

$$\text{UCL} = \mu_0 + L\sigma\sqrt{\frac{\lambda}{(2-\lambda)}\left[1-(1-\lambda)^{2i}\right]} \tag{9.25}$$

$$\text{Center line} = \mu_0$$

$$\text{LCL} = \mu_0 - L\sigma\sqrt{\frac{\lambda}{(2-\lambda)}\left[1-(1-\lambda)^{2i}\right]} \tag{9.26}$$

In equations (9.25) and (9.26), the factor L is the width of the control limits. We will discuss the choice of the parameters L and λ shortly.

Note that the term $[1 - (1 - \lambda)^{2i}]$ in equations (9.25) and (9.26) approaches unity as i gets larger. This means that after the EWMA control chart has been running for several time periods, the control limits will approach steady-state values given by

$$\text{UCL} = \mu_0 + L\sigma\sqrt{\frac{\lambda}{(2-\lambda)}} \tag{9.27}$$

and

$$\text{LCL} = \mu_0 - L\sigma\sqrt{\frac{\lambda}{(2-\lambda)}} \tag{9.28}$$

However, we strongly recommend using the exact control limits in equations (9.25) and (9.26) for small values of i. This will greatly improve the performance of the control chart in detecting an off-target process immediately after the EWMA is started up.

EXAMPLE 9.2 Constructing an EWMA Control Chart

Set up an EWMA control chart with $\lambda = 0.10$ and $L = 2.7$ to the data in Table 9.1.

SOLUTION

Recall that the target value of the mean is $\mu_0 = 10$ and the standard deviation is $\sigma = 1$. The calculations for the EWMA control chart are summarized in Table 9.10, and the control chart (from Minitab) is shown in Fig. 9.7.

To illustrate the calculations, consider the first observation, $x_1 = 9.45$. The first value of the EWMA is

$$\begin{aligned} z_1 &= \lambda x_1 + (1-\lambda) z_0 \\ &= 0.1(9.45) + 0.9(10) \\ &= 9.945 \end{aligned}$$

TABLE 9.10
EWMA Calculations for Example 9.2

Subgroup, i	* = Beyond Limits x_i	EWMA, z_i	Subgroup, i	* = Beyond Limits x_i	EWMA, z_i
1	9.45	9.945	16	9.37	9.98426
2	7.99	9.7495	17	10.62	10.0478
3	9.29	9.70355	18	10.31	10.074
4	11.66	9.8992	19	8.52	9.91864
5	12.16	10.1253	20	10.84	10.0108
6	10.18	10.1307	21	10.9	10.0997
7	8.04	9.92167	22	9.33	10.0227
8	11.46	10.0755	23	12.29	10.2495
9	9.2	9.98796	24	11.5	10.3745
10	10.34	10.0232	25	10.6	10.3971
11	9.03	9.92384	26	11.08	10.4654
12	11.47	10.0785	27	10.38	10.4568
13	10.51	10.1216	28	11.62	10.5731
14	9.4	10.0495	29	11.31	10.6468*
15	10.08	10.0525	30	10.52	10.6341*

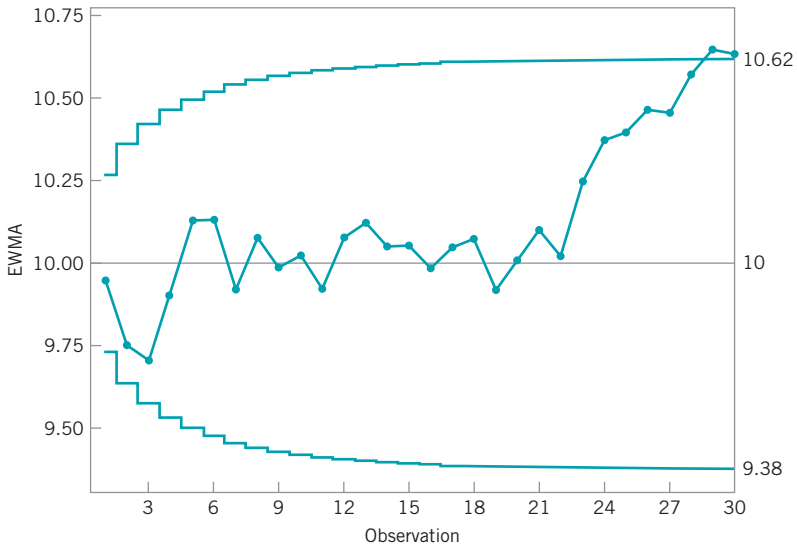


FIGURE 9.7 The EWMA control chart for Example 9.2.

(*continued*)

Therefore, $z_1 = 9.945$ is the first value plotted on the control chart in Fig. 9.7. The second value of the EWMA is

$$\begin{aligned} z_2 &= \lambda x_2 + (1-\lambda)z_1 \\ &= 0.1(7.99) + 0.9(9.945) \\ &= 9.7495 \end{aligned}$$

The other values of the EWMA statistic are computed similarly.

The control limits in Fig. 9.7 are found using equations (9.25) and (9.26). For period $i = 1$,

$$\begin{aligned} \text{UCL} &= \mu_0 + L\sigma\sqrt{\frac{\lambda}{(2-\lambda)}\left[1-(1-\lambda)^{2i}\right]} \\ &= 10 + 2.7(1)\sqrt{\frac{0.1}{(2-0.1)}\left[1-(1-0.1)^{2(1)}\right]} \\ &= 10.27 \end{aligned}$$

and

$$\begin{aligned} \text{LCL} &= \mu_0 - L\sigma\sqrt{\frac{\lambda}{(2-\lambda)}\left[1-(1-\lambda)^{2i}\right]} \\ &= 10 - 2.7(1)\sqrt{\frac{0.1}{(2-0.1)}\left[1-(1-0.1)^{2(1)}\right]} \\ &= 9.73 \end{aligned}$$

For period 2, the limits are

$$\begin{aligned} \text{UCL} &= \mu_0 + L\sigma\sqrt{\frac{\lambda}{(2-\lambda)}\left[1-(1-\lambda)^{2i}\right]} \\ &= 10 + 2.7(1)\sqrt{\frac{0.1}{(2-0.1)}\left[1-(1-0.1)^{2(2)}\right]} \\ &= 10.36 \end{aligned}$$

and

$$\begin{aligned} \text{LCL} &= \mu_0 - L\sigma\sqrt{\frac{\lambda}{(2-\lambda)}\left[1-(1-\lambda)^{2i}\right]} \\ &= 10 - 2.7(1)\sqrt{\frac{0.1}{(2-0.1)}\left[1-(1-0.1)^{2(2)}\right]} \\ &= 9.64 \end{aligned}$$

Note from Fig. 9.7 that the control limits increase in width as i increases from $i = 1, 2, \ldots$, until they stabilize at the steady-state values given by equations (9.27) and (9.28)

$$\begin{aligned} \text{UCL} &= \mu_0 + L\sigma\sqrt{\frac{\lambda}{(2-\lambda)}} \\ &= 10 + 2.7(1)\sqrt{\frac{0.1}{(2-0.1)}} \\ &= 10.62 \end{aligned}$$

and

$$\begin{aligned} \text{LCL} &= \mu_0 - L\sigma\sqrt{\frac{\lambda}{(2-\lambda)}} \\ &= 10 - 2.7(1)\sqrt{\frac{0.1}{(2-0.1)}} \\ &= 9.38 \end{aligned}$$

The EWMA control chart in Fig. 9.7 signals at observation 28, so we would conclude that the process is out of control.

9.2.2 Design of an EWMA Control Chart

The EWMA control chart is very effective against small process shifts. The design parameters of the chart are the multiple of sigma used in the control limits (L) and the value of λ. It is possible to choose these parameters to give ARL performance for the EWMA control chart that closely approximates cusum ARL performance for detecting small shifts.

There have been several theoretical studies of the average run length properties of the EWMA control chart. For example, see the papers by Crowder (1987a, 1989) and Lucas and Saccucci (1990). These studies provide average run length tables or graphs for a range of values of λ and L. The average run length performance for several EWMA control schemes is shown in Table 9.11. The optimal design procedure would consist of specifying the desired in-control and out-of-control average run lengths and the magnitude of the process shift that is anticipated, and then to select the combination of λ and L that provide the desired ARL performance.

TABLE 9.11
Average Run Lengths for Several EWMA Control Schemes [Adapted from Lucas and Saccucci (1990)]

Shift in Mean (multiple of σ)	$L = 3.054$ $\lambda = 0.40$	2.998 0.25	2.962 0.20	2.814 0.10	2.615 0.05
0	500	500	500	500	500
0.25	224	170	150	106	84.1
0.50	71.2	48.2	41.8	31.3	28.8
0.75	28.4	20.1	18.2	15.9	16.4
1.00	14.3	11.1	10.5	10.3	11.4
1.50	5.9	5.5	5.5	6.1	7.1
2.00	3.5	3.6	3.7	4.4	5.2
2.50	2.5	2.7	2.9	3.4	4.2
3.00	2.0	2.3	2.4	2.9	3.5
4.00	1.4	1.7	1.9	2.2	2.7

In general, we have found that values of λ in the interval $0.05 \le \lambda \le 0.25$ work well in practice, with $\lambda = 0.05$, $\lambda = 0.10$, and $\lambda = 0.20$ being popular choices. A good rule of thumb is to use smaller values of λ to detect smaller shifts. We have also found that $L = 3$ (the usual three-sigma limits) works reasonably well, particularly with the larger value of λ, although when λ is small—say, $\lambda \le 0.1$—there is an advantage in reducing the width of the limits by using a value of L between about 2.6 and 2.8. Recall that in Example 9.2, we used $\lambda = 0.1$ and $L = 2.7$. We would expect this choice of parameters to result in an in-control ARL of $ARL_0 \simeq 500$ and an ARL for detecting a shift of one standard deviation in the mean of $ARL_1 = 10.3$. Thus this design is approximately equivalent to the cusum with $h = 5$ and $k = \frac{1}{2}$.

Hunter (1989) has also studied the EWMA and suggested choosing λ so that the weight given to current and previous observations matches as closely as possible the weights given to these observations by a Shewhart chart with the Western Electric rules. This results in a recommended value of $\lambda = 0.4$. If $L = 3.054$, then Table 9.11 indicates that this chart would have $ARL_0 = 500$ and for detecting a shift of one standard deviation in the process mean, the $ARL_1 = 14.3$.

There is one potential concern about an EWMA with a small value of λ. If the value of the EWMA is on one side of the center line when a shift in the mean in the opposite direction occurs, it could take the EWMA several periods to react to the shift, because the small λ does not weight the new data very heavily. This is called the **inertia effect.** It can reduce the effectiveness of the EWMA in shift detection.

Woodall and Mahmoud (2005) have investigated the inertial properties of several different types of control charts. They define the **signal resistance of a control chart** to be the largest standardized deviation of the sample mean from the target or in-control value not leading to an immediate out-of-control signal. For a Shewhart $\bar{x}$ chart, the signal resistance is $SR(\bar{x}) = L$, the multiplier used to obtain the control limits. Thus the signal resistance is constant. For the EWMA control chart, the signal resistance is

$$SR(EWMA) = \frac{L\sqrt{\dfrac{\lambda}{2-\lambda}} - (1-\lambda)w}{\lambda}$$

where w is the value of the EWMA statistic. For the EWMA, the maximum value of the signal resistance averaged over all values of the EWMA statistic is $L\sqrt{(2-\lambda)/\lambda}$, if the chart has the asymptotic limits. These results apply for any sample size, as they are given in terms of shifts expressed as multiples of the standard error.

Clearly the signal resistance of the EWMA control chart depends on the value chosen for λ, with smaller values leading to larger values of the maximum signal resistance. This is in a sense unfortunate, because we almost always want to use the EWMA with a small value of λ as this results in good ARL performance in detecting small shifts. As we will see in Section 9.2.3, small values of λ are also desirable because they make the EWMA chart quite insensitive to normality of the process data. Woodall and Mahmoud (2005) recommend always using a Shewhart chart in conjunction with an EWMA (especially if λ is small) as one way to counteract the signal resistance.

Like the cusum, the EWMA performs well against small shifts but does not react to large shifts as quickly as the Shewhart chart. A good way to further improve the sensitivity of the procedure to large shifts without sacrificing the ability to detect small shifts quickly is to combine a Shewhart chart with the EWMA. These combined Shewhart-EWMA control procedures are effective against both large and small shifts. When using such schemes, we have found it helpful to use slightly wider than usual limits on the Shewhart chart (say, 3.25-sigma, or even 3.5-sigma). It is also possible to plot *both* x_i (or $\bar{x}_i$) and the EWMA statistic z_i on the same control chart along with both the Shewhart and EWMA limits. This produces one chart for the combined control procedure that operators quickly become adept at interpreting. When the plots are computer generated, different colors or plotting symbols can be used for the two sets of control limits and statistics.

9.2.3 Robustness of the EWMA to Non-normality

When discussing the Shewhart control chart for individuals in Chapter 6, we observed that the individuals chart was very sensitive to non-normality in the sense that the actual in-control ARL (ARL_0) would be considerably less than the "advertised" or expected value based on the assumption of a normal distribution. Borror, Montgomery, and Runger (1999) compared the ARL performance of the Shewhart individuals chart and the EWMA control chart for the case of non-normal distributions. Specifically, they used the gamma distribution to represent the case of skewed distributions and the t distribution to represent symmetric distributions with heavier tails than the normal.

The ARL_0 of the Shewhart individuals chart and several EWMA control charts for these non-normal distributions are given in Tables 9.12 and 9.13. Two aspects of the information in these tables are very striking. First, even moderately non-normal distributions have the effect of greatly reducing the in-control ARL of the Shewhart individuals chart. This will, of course, dramatically increase the rate of false alarms. Second, an EWMA with $\lambda = 0.05$ or $\lambda = 0.10$ and an appropriately chosen control limit will perform very well against both normal and non-normal distributions. With $\lambda = 0.05$ and $L = 2.492$ the ARL_0 for the EWMA is within approx-

■ TABLE 9.12
In-Control ARLs for the EWMA and the Individuals Control Charts for Various Gamma Distributions

	EWMA			Shewhart
λ	0.05	0.1	0.2	1
L	2.492	2.703	2.86	3.00
Normal	370.4	370.8	370.5	370.4
Gam(4, 1)	372	341	259	97
Gam(3, 1)	372	332	238	85
Gam(2, 1)	372	315	208	71
Gam(1, 1)	369	274	163	55
Gam(0.5, 1)	357	229	131	45

■ TABLE 9.13
In-Control ARLs for the EWMA and the Individuals Control Charts for Various *t* Distributions

	EWMA			Shewhart
λ	0.05	0.1	0.2	1
L	2.492	2.703	2.86	3.00
Normal	370.4	370.8	370.5	370.4
t_{50}	369	365	353	283
t_{40}	369	363	348	266
t_{30}	368	361	341	242
t_{20}	367	355	325	204
t_{15}	365	349	310	176
t_{10}	361	335	280	137
t_8	358	324	259	117
t_6	351	305	229	96
t_4	343	274	188	76

imately 8% of the advertised normal theory in-control ARL_0 of 370, except in the most extreme cases. Furthermore, the shift detection properties of the EWMA are uniformly superior to the Shewhart chart for individuals.

Based on this information, we would recommend a properly designed EWMA as a control chart for individual measurements in a wide range of applications, particularly phase II process monitoring. It is almost a perfectly nonparametric (distribution-free) procedure.

9.2.4 Rational Subgroups

The EWMA control chart is often used with individual measurements. However, if rational subgroups of size $n > 1$ are taken, then simply replace x_i with $\bar{x}_i$ and σ with $\sigma_{\bar{x}} = \sigma/\sqrt{n}$ in the previous equations.

9.2.5 Extensions of the EWMA

There have been numerous extensions and variations of the basic EWMA control chart. In this section, we describe a few of these procedures.

Fast Initial Response Feature. It is possible to add the fast initial response (FIR) or headstart feature to the EWMA. As with the cusum, the advantage of the procedure would be to more quickly detect a process that is off target at start-up.

Two approaches have been suggested. Rhoads, Montgomery, and Mastrangelo (1996) set up two one-sided EWMA charts and start them off at values midway between the target and the control limit. Both one-sided charts are assumed to have the time-varying limits (readers are asked to develop a one-sided EWMA procedure in Exercise 9.39).

Steiner (1999) uses a single control chart but narrows the time-varying limits even further for the first few sample points. He uses an exponentially decreasing adjustment to further narrow the limits, so that the control limits are located a distance

$$\pm L\sigma\left\{\left(1-(1-f)^{1+a(t-1)}\right)\sqrt{\frac{\lambda}{2-\lambda}\left[1-(1-\lambda)^{2t}\right]}\right\}$$

around the target. The constants f and a are to be determined. Steiner suggests choosing a so that the FIR has little effect after about 20 observations. This leads to choosing $a =$

$[-2/\log(1-f)-1]/19$. For example, if $f = 0.5$, then $a = 0.3$. The choice of $f = 0.5$ is attractive because it mimics the 50% headstart often used with cusums.

Both of these procedures perform very well in reducing the ARL to detect an off-target process at start-up. The Steiner procedure is easier to implement in practice.

Monitoring Variability. MacGregor and Harris (1993) discuss the use of EWMA-based statistics for monitoring the process standard deviation. Let x_i be normally distributed with mean μ and standard deviation σ. The **exponentially weighted mean square error (EWMS)** is defined as

$$S_i^2 = \lambda(x_i - \mu)^2 + (1-\lambda)S_{i-1}^2 \tag{9.29}$$

It can be shown that $E(S_i^2) = \sigma^2$ (for large i) and if the observations are independent and normally distributed, then S_i^2/σ^2 has an approximate chi-square distribution with $v = (2-\lambda)/\lambda$ degrees of freedom. Therefore, if σ_0 represents the in-control or target value of the process standard deviation, we could plot $\sqrt{S_i^2}$ on an exponentially weighted root mean square (EWRMS) control chart with control limits given by

$$\text{UCL} = \sigma_0\sqrt{\frac{\chi^2_{v,\alpha/2}}{v}} \tag{9.30}$$

and

$$\text{LCL} = \sigma_0\sqrt{\frac{\chi^2_{v,1-(\alpha/2)}}{v}} \tag{9.31}$$

MacGregor and Harris (1993) point out that the EWMS statistic can be sensitive to shifts in both the process mean and the standard deviation. They suggest replacing μ in equation (9.27) with an estimate $\hat{\mu}_i$ at each point in time. A logical estimate of μ turns out to be the ordinary EWMA z_i. They derive control limits for the resulting exponentially weighted moving variance (EWMV)

$$S_i^2 = \lambda(x_i - z_i)^2 + (1-\lambda)S_{i-1}^2 \tag{9.32}$$

Another approach to monitoring the process standard deviation with an EWMA is in Crowder and Hamilton (1992).

The EWMA for Poisson Data. Just as the cusum can be used as the basis of an effective control chart for Poisson counts, so can a suitably designed EWMA. Borror, Champ, and Rigdon (1998) describe the procedure, show how to design the control chart, and provide an example. If x_i is a count, then the basic EWMA recursion remains unchanged:

$$z_i = \lambda x_i + (1-\lambda)z_{i-1}$$

with $z_0 = \mu_0$ the in-control or target count rate. The control chart parameters are as follows:

$$\begin{aligned} \text{UCL} &= \mu_0 + A_U\sqrt{\frac{\lambda\mu_0}{2-\lambda}\left[1-(1-\lambda)^{2i}\right]} \\ \text{Center line} &= \mu_0 \\ \text{LCL} &= \mu_0 - A_L\sqrt{\frac{\lambda\mu_0}{2-\lambda}\left[1-(1-\lambda)^{2i}\right]} \end{aligned} \tag{9.33}$$

where A_U and A_L are the upper and lower control limit factors. In many applications we would choose $A_U = A_L = A$. Borror, Champ, and Rigdon (1998) give graphs of the ARL performance

of the Poisson EWMA control chart as a function of λ and A and for various in-control or target count rates μ_0. Once μ_0 is determined and a value is specified for λ, these charts can be used to select the value of A that results in the desired in-control ARL_0. The authors also show that this control chart has considerably better ability to detect assignable causes than the Shewhart c chart. The Poisson EWMA should be used much more widely in practice.

The EWMA as a Predictor of Process Level. Although we have discussed the EWMA primarily as a statistical process-monitoring tool, it actually has a much broader interpretation. From an SPC viewpoint, the EWMA is roughly equivalent to the cusum in its ability to *monitor* a process and detect the presence of assignable causes that result in a process shift. However, the EWMA provides a **forecast** of where the process mean will be at the next time period. That is, z_i is actually a forecast of the value of the process mean μ at time $i + 1$. Thus, the EWMA could be used as the basis for a dynamic process-control algorithm.

In computer-integrated manufacturing where sensors are used to measure every unit manufactured, a forecast of the process mean based on previous behavior would be very useful. If the forecast of the mean is different from target by a critical amount, then either the operator or some electromechanical control system can make the necessary process adjustment. If the operator makes the adjustment, then he or she must exercise caution and not make adjustments too frequently because this will actually cause process variability to increase. The control limits on the EWMA chart can be used to signal *when* an adjustment is necessary, and the difference between the target and the forecast of the mean μ_{i+1} can be used to determine *how much* adjustment is necessary.

The EWMA can be modified to enhance its ability to forecast the mean. Suppose that the process mean trends or drifts steadily away from the target. The forecasting performance of the EWMA can be improved in this case. First, note that the usual EWMA can be written as

$$\begin{aligned} z_i &= \lambda x_i + (1-\lambda) z_{i-1} \\ &= z_{i-1} + \lambda(x_i - z_{i-1}) \end{aligned}$$

and if we view z_{i-1} as a forecast of the process mean in period i, we can think of $x_i - z_{i-1}$ as the forecast error e_i for period i. Therefore,

$$z_i = z_{i-1} + \lambda e_i \tag{9.34}$$

Thus, the EWMA for period i is equal to the EWMA for period $i - 1$ plus a fraction λ of the forecast error for the mean in period i. Now add a second term to this last equation to give

$$z_i = z_{i-1} + \lambda_1 e_i + \lambda_2 \sum_{j=1}^{i} e_j \tag{9.35}$$

where λ_1 and λ_2 are constants that weight the error at time i and the *sum* of the errors accumulated to time i. If we let $\nabla e_i = e_i - e_{i-1}$ be the first difference of the errors, then we can arrive at a final modification of the EWMA:

$$z_i = z_{i-1} + \lambda_1 e_i + \lambda_2 \sum_{j=1}^{i} e_j + \lambda_3 \nabla e_i \tag{9.36}$$

Note that in this empirical control equation the EWMA in period i (which is the forecast of the process mean in period $i + 1$) equals the current estimate of the mean (z_{i-1} estimates μ_i), plus a term proportional to the error, plus a term related to the sum of the errors, plus a term related to the first difference of the errors. These three terms can be thought of as **proportional, integral,** and **differential** adjustments. The parameters λ_1, λ_2, and λ_3 would be chosen to give the best forecasting performance.

Because the EWMA statistic z_i can be viewed as a **forecast** of the mean of the process at time $i + 1$, we often plot the EWMA statistic one time period ahead. That is, we actually plot z_i at time period $i + 1$ on the control chart. This allows the analyst to visually see how much difference there is between the current observation and the estimate of the current mean of the process. In statistical process-control applications where the mean may "wander" over time, this approach has considerable appeal.

9.3 The Moving Average Control Chart

Both the cusum and the EWMA are **time-weighted** control charts. The EWMA chart uses a weighted average as the chart statistic. Occasionally, another type of time-weighted control chart based on a simple, unweighted **moving average** may be of interest.

Suppose that individual observations have been collected, and let $x_1, x_2, \ldots$ denote these observations. The moving average of span w at time i is defined as

$$M_i = \frac{x_i + x_{i-1} + \cdots + x_{i-w+1}}{w} \tag{9.37}$$

That is, at time period i, the oldest observation in the moving average set is dropped and the newest one added to the set. The variance of the moving average M_i is

$$V(M_i) = \frac{1}{w^2} \sum_{j=i-w+1}^{i} V(x_j) = \frac{1}{w^2} \sum_{j=i-w+1}^{i} \sigma^2 = \frac{\sigma^2}{w} \tag{9.38}$$

Therefore, if μ_0 denotes the target value of the mean used as the center line of the control chart, then the three-sigma control limits for M_i are

$$\text{UCL} = \mu_0 + \frac{3\sigma}{\sqrt{w}} \tag{9.39}$$

and

$$\text{LCL} = \mu_0 - \frac{3\sigma}{\sqrt{w}} \tag{9.40}$$

The control procedure would consist of calculating the new moving average M_i as each observation x_i becomes available, plotting M_i on a control chart with upper and lower control limits given by equations (9.39) and (9.40), and concluding that the process is out of control if M_i exceeds the control limits. In general, the magnitude of the shift of interest and w are inversely related; smaller shifts would be guarded against more effectively by longer-span moving averages, at the expense of quick response to large shifts.

EXAMPLE 9.3 A Moving Average Control Chart

Set up a moving average control chart for the data in Table 9.1, using $w = 5$.

SOLUTION

The observations x_i for periods $1 \le i \le 30$ are shown in Table 9.14. The statistic plotted on the moving average control chart will be

$$M_i = \frac{x_i + x_{i-1} + \cdots + x_{i-4}}{5}$$

for periods $i \ge 5$. For time periods $i < 5$ the average of the observations for periods $1, 2, \ldots, i$ is plotted. The values of these moving averages are shown in Table 9.14.

■ TABLE 9.14
Moving Average Chart for Example 9.3

Observation, i	x_i	M_i	Observation, i	x_i	M_i
1	9.45	9.45	16	9.37	10.166
2	7.99	8.72	17	10.62	9.996
3	9.29	8.91	18	10.31	9.956
4	11.66	9.5975	19	8.52	9.78
5	12.16	10.11	20	10.84	9.932
6	10.18	10.256	21	10.9	10.238
7	8.04	10.266	22	9.33	9.98
8	11.46	10.7	23	12.29	10.376
9	9.2	10.208	24	11.5	10.972
10	10.34	9.844	25	10.6	10.924
11	9.03	9.614	26	11.08	10.96
12	11.47	10.3	27	10.38	11.17
13	10.51	10.11	28	11.62	11.036
14	9.4	10.15	29	11.31	10.998
15	10.08	10.098	30	10.52	10.982

The control limits for the moving average control chart may be easily obtained from equations (9.39) and (9.40). Since we know that $\sigma = 1.0$, then

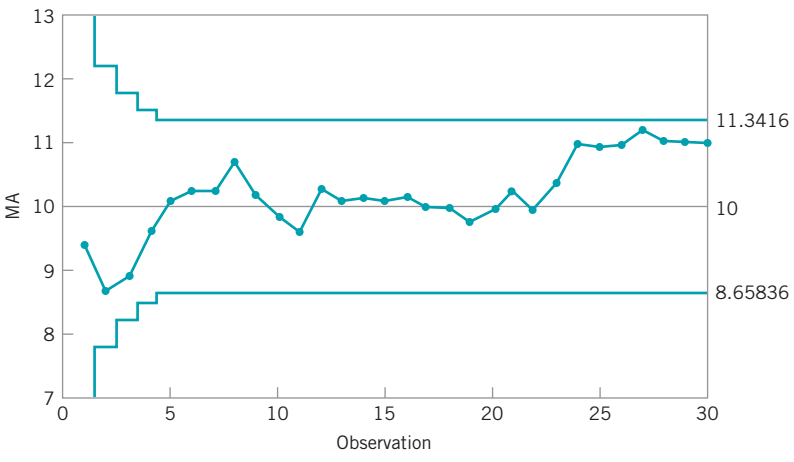
$$\text{UCL} = \mu_0 + \frac{3\sigma}{\sqrt{w}} = 10 + \frac{3(1.0)}{\sqrt{5}} = 11.34$$

and

$$\text{LCL} = \mu_0 - \frac{3\sigma}{\sqrt{w}} = 10 - \frac{3(1.0)}{\sqrt{5}} = 8.66$$

The control limits for M_i apply for periods $i \geq 5$. For periods $0 < i < 5$, the control limits are given by $\mu_0 \pm 3\sigma/\sqrt{i}$. An alternative procedure that avoids using special control limits for periods $i < w$ is to use an ordinary Shewhart chart until at least w sample means have been obtained.

The moving average control chart is shown in Fig. 9.8. No points exceed the control limits. Note that for the initial periods $i < w$ the control limits are wider than their final steady-state value. Moving averages that are less than w periods apart are highly correlated, which often complicates interpreting patterns on the control chart. This is easily seen by examining Fig. 9.8.



■ FIGURE 9.8 Moving average control chart with $w = 5$, Example 9.3.

The moving average control chart is more effective than the Shewhart chart in detecting small process shifts. However, it is generally not as effective against small shifts as either the cusum or the EWMA. The moving average control chart is considered by some to be simpler to implement than the cusum. This author prefers the EWMA to the moving average control chart.

Important Terms and Concepts

ARL calculations for the cusum
Average run length
Combined cusum-Shewhart procedures
Cusum control chart
Cusum status chart
Decision interval
Design of a cusum
Design of an EWMA control chart
EWMA control chart
Fast initial response (FIR) or headstart feature for a cusum
Fast initial response (FIR) or headstart feature for an EWMA
Moving average control chart
One-sided cusums
Poisson EWMA
Reference value
Robustness of the EWMA to normality
Scale cusum
Self-starting cusum
Signal resistance of a control chart
Standardized cusum
Tabular or algorithmic cusum
V-mask form of the cusum

Exercises



The Student Resource Manual presents comprehensive annotated solutions to the odd-numbered exercises included in the Answers to Selected Exercises section in the back of this book.

9.1. The data in Table 9E.1 represent individual observations on molecular weight taken hourly from a chemical process.

The target value of molecular weight is 1050 and the process standard deviation is thought to be about $\sigma = 25$.

(a) Set up a tabular cusum for the mean of this process. Design the cusum to quickly detect a shift of about 1.0σ in the process mean.

■ **TABLE 9E.1**
Molecular Weight

Observation Number	x	Observation Number	x
1	1045	11	1139
2	1055	12	1169
3	1037	13	1151
4	1064	14	1128
5	1095	15	1238
6	1008	16	1125
7	1050	17	1163
8	1087	18	1188
9	1125	19	1146
10	1146	20	1167

(b) Is the estimate of σ used in part (a) of this problem reasonable?

9.2. Rework Exercise 9.1 using a standardized cusum.

9.3. (a) Add a headstart feature to the cusum in Exercise 9.1.

(b) Use a combined Shewhart-cusum scheme on the data in Exercise 9.1. Interpret the results of both charts.

9.4. A machine is used to fill cans with motor oil additive. A single sample can is selected every hour and the weight of the can is obtained. Since the filling process is automated, it has very stable variability, and long experience indicates that $\sigma = 0.05$ oz. The individual observations for 24 hours of operation are shown in Table 9E-2.

(a) Assuming that the process target is 8.02 oz, set up a tabular cusum for this process. Design the cusum using the standardized values $h = 4.77$ and $k = \frac{1}{2}$.

(b) Does the value of $\sigma = 0.05$ seem reasonable for this process?

9.5. Rework Exercise 9.4 using the standardized cusum parameters of $h = 8.01$ and $k = 0.25$. Compare the results with those obtained previously in Exercise 9.4. What can you say about the theoretical performance of those two cusum schemes?

9.6. Reconsider the data in Exercise 9.4. Suppose the data there represent observations taken immediately after a process adjustment that was intended to reset the

■ TABLE 9E.2
Fill Data for Exercise 9.4

Sample Number	x	Sample Number	x
1	8.00	13	8.05
2	8.01	14	8.04
3	8.02	15	8.03
4	8.01	16	8.05
5	8.00	17	8.06
6	8.01	18	8.04
7	8.06	19	8.05
8	8.07	20	8.06
9	8.01	21	8.04
10	8.04	22	8.02
11	8.02	23	8.03
12	8.01	24	8.05

process to a target of $\mu_0 = 8.00$. Set up and apply an FIR cusum to monitor this process.

9.7. The data in Table 9E.3 are temperature readings from a chemical process in °C, taken every two minutes. (Read the observations down, from left.)

The target value for the mean is $\mu_0 = 950$.

(a) Estimate the process standard deviation.

(b) Set up and apply a tabular cusum for this process, using standardized values $h = 5$ and $k = \frac{1}{2}$. Interpret this chart.

9.8. Bath concentrations are measured hourly in a chemical process. Data (in ppm) for the last 32 hours are shown in Table 9E.4 (read down from left).

The process target is $\mu_0 = 175$ ppm.

(a) Estimate the process standard deviation.

(b) Construct a tabular cusum for this process using standardized values of $h = 5$ and $k = \frac{1}{2}$.

9.9. Viscosity measurements on a polymer are made every 10 minutes by an on-line viscometer. Thirty-six observations are shown in Table 9E.5 (read down from left). The target viscosity for this process is $\mu_0 = 3200$.

(a) Estimate the process standard deviation.

(b) Construct a tabular cusum for this process using standardized values of $h = 8.01$ and $k = 0.25$.

(c) Discuss the choice of h and k in part (b) of this problem on cusum performance.

9.10. Set up a tabular cusum scheme for the flow width data used in Example 6.1 (see Tables 6.1 and 6.2). When the procedure is applied to all 45 samples, does the cusum react more quickly than the $\bar{x}$ chart to the shift in the process mean? Use $\sigma = 0.14$ in setting up the cusum, and design the procedure to quickly detect a shift of about 1σ.

9.11. Apply the scale cusum discussed in Section 9.1.8 to the data in Exercise 9.1.

9.12. Apply the scale cusum discussed in Section 9.1.8 to the concentration data in Exercise 9.8.

9.13. Consider a standardized two-sided cusum with $k = 0.2$ and $h = 8$. Use Siegmund's procedure to evaluate the in-control ARL performance of this scheme. Find ARL_1 for $\delta^* = 0.5$.

9.14. Consider the viscosity data in Exercise 9.9. Suppose that the target value of viscosity is $\mu_0 = 3150$ and that it is only important to detect disturbances in the process that result in increased viscosity. Set up and apply an appropriate one-sided cusum for this process.

■ TABLE 9E.3
Chemical Process Temperature Data

953	985	949	937	959	948	958	952
945	973	941	946	939	937	955	931
972	955	966	954	948	955	947	928
945	950	966	935	958	927	941	937
975	948	934	941	963	940	938	950
970	957	937	933	973	962	945	970
959	940	946	960	949	963	963	933
973	933	952	968	942	943	967	960
940	965	935	959	965	950	969	934
936	973	941	956	962	938	981	927

■ TABLE 9E.4
Bath Concentration

160	186	190	206
158	195	189	210
150	179	185	216
151	184	182	212
153	175	181	211
154	192	180	202
158	186	183	205
162	197	186	197

■ TABLE 9E.5
Polymer Viscosity

3169	3205	3185	3188
3173	3203	3187	3183
3162	3209	3192	3175
3154	3208	3199	3174
3139	3211	3197	3171
3145	3214	3193	3180
3160	3215	3190	3179
3172	3209	3183	3175
3175	3203	3197	3174

9.15. Consider the velocity of light data introduced in Exercises 6.59 and 6.60. Use only the 20 observations in Exercise 6.59 to set up a cusum with target value 734.5. Plot all 40 observations from both Exercises 6.59 and 6.60 on this cusum. What conclusions can you draw?

9.16. Rework Exercise 9.15 using an EWMA with $\lambda = 0.10$.

9.17. Rework Exercise 9.1 using an EWMA control chart with $\lambda = 0.1$ and $L = 2.7$. Compare your results to those obtained with the cusum.

9.18. Consider a process with $\mu_0 = 10$ and $\sigma = 1$. Set up the following EWMA control charts:
(a) $\lambda = 0.1, L = 3$
(b) $\lambda = 0.2, L = 3$
(c) $\lambda = 0.4, L = 3$
Discuss the effect of λ on the behavior of the control limits.

9.19. Reconsider the data in Exercise 9.4. Set up an EWMA control chart with $\lambda = 0.2$ and $L = 3$ for this process. Interpret the results.

9.20. Reconstruct the control chart in Exercise 9.19 using $\lambda = 0.1$ and $L = 2.7$. Compare this chart with the one constructed in Exercise 9.19.

9.21. Reconsider the data in Exercise 9.7. Apply an EWMA control chart to these data using $\lambda = 0.1$ and $L = 2.7$.

9.22. Reconstruct the control chart in Exercise 9.21 using $\lambda = 0.4$ and $L = 3$. Compare this chart to the one constructed in Exercise 9.21.

9.23. Reconsider the data in Exercise 9.8. Set up and apply an EWMA control chart to these data using $\lambda = 0.05$ and $L = 2.6$.

9.24. Reconsider the homocide data in Exercise 7.62. Set up an EWMA control chart for this process with $\lambda = 0.1$ and $L = 2.7$. Does potential non-normality in the data pose a concern here?

9.25. Reconsider the data in Exercise 9.9. Set up and apply an EWMA control chart to these data using $\lambda = 0.1$ and $L = 2.7$.

9.26. Analyze the data in Exercise 9.1 using a moving average control chart with $w = 6$. Compare the results obtained with the cumulative sum control chart in Exercise 9.1.

9.27. Analyze the data in Exercise 9.4 using a moving average control chart with $w = 5$. Compare the results obtained with the cumulative sum control chart in Exercise 9.4.

9.28. Analyze the homocide data in Exercise 7.62 using a moving average control chart with $w = 5$. Does potential non-normality in the data pose a concern here?

9.29. Show that if the process is in control at the level μ, the exponentially weighted moving average is an unbiased estimator of the process mean.

9.30. Derive the variance of the exponentially weighted moving average z_i.

9.31. **Equivalence of moving average and exponentially weighted moving average control charts.** Show that if $\lambda = 2/(w + 1)$ for the EWMA control chart, then this chart is equivalent to a w-period moving average control chart in the sense that the control limits are identical in the steady state.

9.32. **Continuation of Exercise 9.31.** Show that if $\lambda = 2/(w + 1)$, then the average "ages" of the data used in computing the statistics z_i and M_i are identical.

9.33. Show how to modify the control limits for the moving average control chart if rational subgroups of size $n > 1$ are observed every period, and the objective of the control chart is to monitor the process mean.

9.34. A Shewhart $\bar{x}$ chart has center line at 10 with UCL = 16 and LCL = 4. Suppose you wish to supplement this chart with an EWMA control chart using $\lambda = 0.1$ and the same control limit width in σ-units as employed on the $\bar{x}$ chart. What are the values of the steady-state upper and lower control limits on the EWMA chart?

9.35. An EWMA control chart uses $\lambda = 0.4$. How wide will the limits be on the Shewhart control chart, expressed as a multiple of the width of the steady-state EWMA limits?

9.36. Consider the valve failure data in Example 7.6. Set up a cusum chart for monitoring the time between events using the transformed variable approach illustrated in that example. Use standardized values of $h = 5$ and $k = \frac{1}{2}$.

9.37. Consider the valve failure data in Example 7.6. Set up a one-sided cusum chart for monitoring and detecting an increase in failure rate of the valve. Assume that the target value of the mean time between failures is 700 hr.

9.38. Set up an appropriate EWMA control chart for the valve failure data in Example 7.6. Use the transformed variable approach illustrated in that example.

9.39. Discuss how you could set up one-sided EWMA control charts.

10.2 Modified and Acceptance Control Charts

In most situations in which control charts are used, the focus is on statistical monitoring or control of the process, reduction of variability, and continuous process improvement. When a high level of process capability has been achieved, it is sometimes useful to relax the level of surveillance provided by the standard control chart. One method for doing this with $\bar{x}$ charts uses **modified** (or **reject**) **control limits,** and the second uses the **acceptance control chart.**

10.2.1 Modified Control Limits for the $\bar{x}$ Chart

Modified control limits are generally used in situations where the natural variability or "spread" of the process is considerably smaller than the spread in the specification limits; that is, C_p or C_{pk} is much greater than 1. This situation occurs occasionally in practice. In fact, this should be the natural eventual result of a successful variability reduction effort—reduction of process variability with a corresponding increase in the process capability ratio. The six-sigma approach to variability reduction focuses on improving processes until the minimum value of C_{pk} is 2.0.

Suppose, for example, that the specification limits on the fill volume of a carbonated beverage container are LSL = 10.00 oz and USL = 10.20 oz, but as the result of a program of engineering and operating refinements, the filling machine can operate with a standard deviation of fill volume of approximately $\sigma = 0.01$ oz. Therefore, the distance USL − LSL is approximately twenty-sigma, or much greater than the six-sigma natural tolerance limits on the process, and the process capability ratio is PCR = (USL − LSL)/6σ = 0.20/[6(0.01)] = 3.33. This is clearly a six-sigma process.

In situations where six-sigma is much smaller than the spread in the specifications (USL–LSL), the process mean can sometimes be allowed to vary over an interval without appreciably affecting the overall performance of the process.[1] For example, see Fig. 10.2. When this situation occurs, we can use a **modified control chart** for $\bar{x}$ instead of the usual $\bar{x}$ chart. The modified $\bar{x}$ control chart is concerned only with detecting whether the true process mean μ is located such that the process is producing a fraction nonconforming in excess of some specified value δ. In effect, μ is allowed to vary over an interval—say, $\mu_L \le \mu \le \mu_U$—where μ_L and μ_U are chosen as the smallest and largest permissible values of μ, respectively, consistent with producing a fraction nonconforming of at most δ. We will assume that the process variability σ is in control. Good general discussions of the modified control chart are in Hill (1956) and Duncan (1986). As noted in Duncan (1986), the procedure is sometimes used when a process is subject to tool wear (see Section 10.7.1).

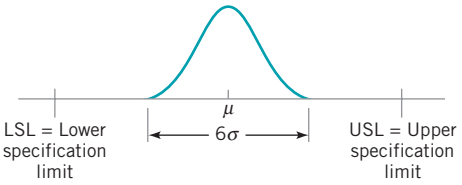
To specify the control limits for a modified $\bar{x}$ chart, we will assume that the process output is normally distributed. For the process fraction nonconforming to be less than δ, we must require that the true process mean is in the interval. Consequently, we see from Fig. 10.3*a* that we must have

$$\mu_L = \text{LSL} + Z_\delta \sigma$$

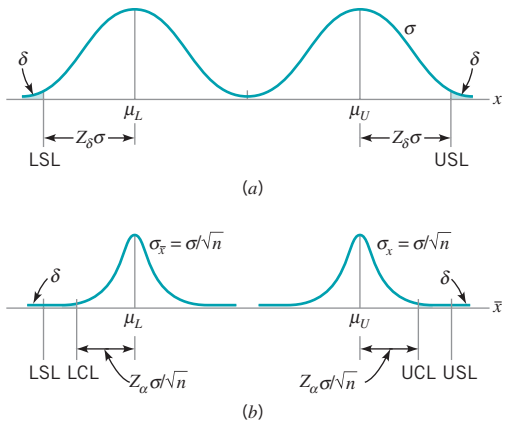
and

$$\mu_U = \text{USL} - Z_\delta \sigma$$

[1]In the original Motorola definition of a six-sigma process, the process mean was assumed to drift about, wandering as far as 1.5σ from the desired target. If this is the actual behavior of the process and this type of behavior is acceptable, then modified control charts are a useful alternative to standard charts. There are also many cases where the mean should not be allowed to vary even if C_p or C_{pk} is large. The conventional control charts should be used in such situations.



■ **FIGURE 10.2** A process with the spread of the natural tolerance limits less than the spread of the specification limits, or $6\sigma < \text{USL} - \text{LSL}$.



■ **FIGURE 10.3** Control limits on a modified control chart. (*a*) Distribution of process output. (*b*) Distribution of the sample mean $\bar{x}$.

where Z_δ is the upper $100(1-\delta)$ percentage point of the standard normal distribution. Now if we specify a type I error of α, the upper and lower control limits are

$$\begin{aligned}\text{UCL} &= \mu_U + \frac{Z_\alpha \sigma}{\sqrt{n}} \\ &= \text{USL} - Z_\delta \sigma + \frac{Z_\alpha \sigma}{\sqrt{n}} \\ &= \text{USL} - \left(Z_\delta - \frac{Z_\alpha}{\sqrt{n}} \right) \sigma \end{aligned} \tag{10.3a}$$

and

$$\begin{aligned}\text{LCL} &= \mu_L - \frac{Z_\alpha \sigma}{\sqrt{n}} \\ &= \text{LSL} + Z_\delta \sigma - \frac{Z_\alpha \sigma}{\sqrt{n}} \\ &= \text{LSL} + \left(Z_\delta - \frac{Z_\alpha}{\sqrt{n}} \right) \sigma \end{aligned} \tag{10.3b}$$

respectively. The control limits are shown on the distribution of $\bar{x}$ in Fig. 10.3*b*. Instead of specifying a type I error, one may use the following:

$$\text{UCL} = \text{USL} - \left(Z_\delta - \frac{3}{\sqrt{n}} \right) \sigma \tag{10.4a}$$

and

$$\text{LCL} = \text{LSL} + \left(Z_\delta - \frac{3}{\sqrt{n}} \right) \sigma \tag{10.4b}$$

Two-sigma limits are sometimes recommended for the modified control chart, based on an argument that the tighter control limits afford better protection (smaller β-risk) against critical shifts in the mean at little loss in the α-risk. A discussion of this subject is in Freund (1957). Note that the modified control chart is equivalent to testing the hypothesis that the process mean lies in the interval $\mu_L \le \mu \le \mu_U$.

To design a modified control chart, we must have a good estimate of σ available. If the process variability shifts, then the modified control limits are not appropriate. Consequently, an R or an s chart should always be used in conjunction with the modified control chart. Furthermore, the initial estimate of σ required to set up the modified control chart would usually be obtained from an R or an s chart.

EXAMPLE 10.1 A Control Chart for a Six-Sigma Process

Consider a normally distributed process with a target value of the mean of 20 and standard deviation $\sigma = 2$. The upper and lower process specifications are at LSL = 8 and USL = 32, so that if the process is centered at the target, $C_p = C_{pk} = 2.0$. This is a process with six-sigma capability. In a six-sigma process it is assumed that the mean may drift as much as 1.5 standard deviations off target without causing serious problems. Set up a control chart for monitoring the mean of this six-sigma process with a sample size of $n = 4$.

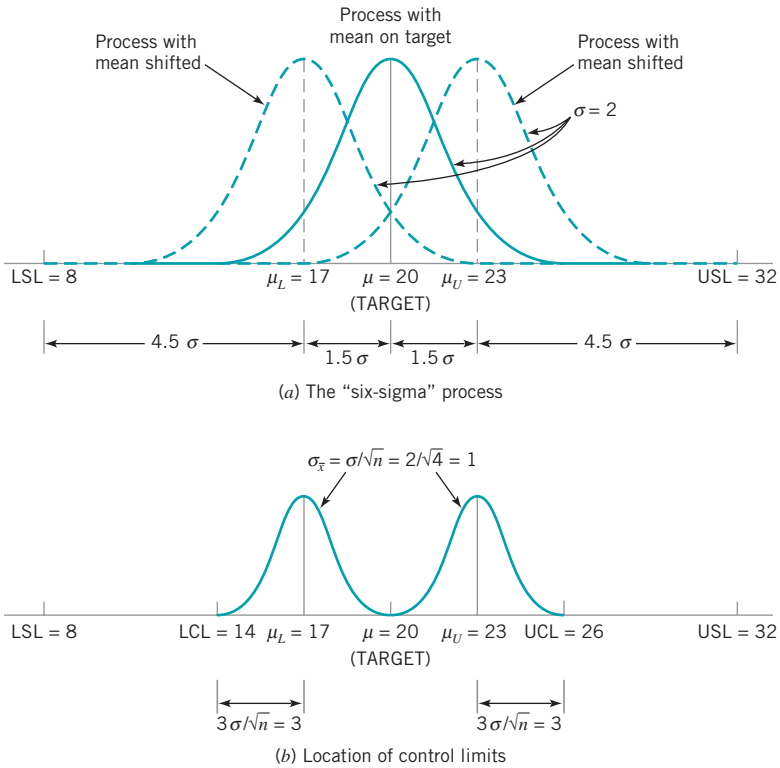
SOLUTION

Figure 10.4*a* shows this six-sigma process and Fig. 10.4*b* illustrates the control limit calculation. Notice that we can use equation (10.4) with Z_δ replaced by 4.5. Therefore, the upper and lower control limits become

$$\begin{aligned}\text{UCL} &= \text{USL} - \left(4.5 - \frac{3}{\sqrt{4}}\right)\sigma \\ &= 32 - (4.5 - 1.5)2 \\ &= 26\end{aligned}$$

and

$$\begin{aligned}\text{LCL} &= \text{LSL} + \left(4.5 - \frac{3}{\sqrt{4}}\right)\sigma \\ &= 8 + (4.5 - 1.5)2 \\ &= 14\end{aligned}$$



■ **FIGURE 10.4** Control limits for a six-sigma process.

10.2.2 Acceptance Control Charts

The second approach to using an $\bar{x}$ chart to monitor the fraction of nonconforming units, or the fraction of units exceeding specifications, is called the acceptance control chart. In the modified control chart design of Section 10.2.1, the chart was based on a specified sample size n, a process fraction nonconforming δ, and type I error probability α. Thus, we interpret δ as a process fraction nonconforming that we will accept with probability $1 - \alpha$. Freund (1957) developed the **acceptance control chart** to take into account both the risk of rejecting a process operating at a satisfactory level (type I error or α-risk) and the risk of accepting a process that is operating at an unsatisfactory level (type II error or β-risk).

There are two ways to design the control chart. In the first approach, we design the control chart based on a specified n and a process fraction nonconforming γ that we would like to reject with probability $1 - \beta$. In this case, the control limits for the chart are

$$\begin{aligned} \text{UCL} &= \mu_U - \frac{Z_\beta \sigma}{\sqrt{n}} \\ &= \text{USL} - Z_\gamma \sigma - \frac{Z_\beta \sigma}{\sqrt{n}} \\ &= \text{USL} - \left(Z_\gamma + \frac{Z_\beta}{\sqrt{n}} \right) \sigma \end{aligned} \tag{10.5a}$$

and

$$\begin{aligned} \text{LCL} &= \mu_L + \frac{Z_\beta \sigma}{\sqrt{n}} \\ &= \text{LSL} + Z_\gamma \sigma + \frac{Z_\beta \sigma}{\sqrt{n}} \\ &= \text{LSL} + \left(Z_\gamma + \frac{Z_\beta}{\sqrt{n}} \right) \sigma \end{aligned} \tag{10.5b}$$

Note that when n, γ, and $1 - \beta$ (or β) are specified, the control limits are inside the μ_L and μ_U values that produce the fraction nonconforming γ. In contrast, when n, δ, and α are specified, the lower control limit falls between μ_L and LSL and the upper control limit falls between μ_U and USL.

It is also possible to choose a sample size for an acceptance control chart so that specified values of δ, α, γ, and β are obtained. By equating the upper control limits (say) for a specified δ and α [equation (10.3a)] and a specified γ and β [equation (10.5a)], we obtain

$$\text{USL} - \left(Z_\delta - \frac{Z_\alpha}{\sqrt{n}} \right) \sigma = \text{USL} - \left(Z_\gamma + \frac{Z_\beta}{\sqrt{n}} \right) \sigma$$

Therefore, a sample size of

$$n = \left(\frac{Z_\alpha + Z_\beta}{Z_\delta - Z_\gamma} \right)^2$$

will yield the required values of δ, α, γ, and β. For example, if $\delta = 0.01$, $\alpha = 0.00135$, $\gamma = 0.05$, and $\beta = 0.20$, we must use a sample of size

$$n = \left(\frac{Z_{0.00135} + Z_{0.20}}{Z_{0.01} - Z_{0.05}} \right)^2 = \left(\frac{3.00 + 0.84}{2.33 - 1.645} \right)^2 = 31.43 \simeq 32$$

on the acceptance control chart. Obviously, to use this approach, n must not be seriously restricted by cost or other factors such as rational subgrouping considerations.

10.3 Control Charts for Multiple-Stream Processes

10.3.1 Multiple-Stream Processes

A **multiple-stream process (MSP)** is a process with data at a point in time consisting of measurements from several individual sources or streams. When the process is in control, the sources or streams are assumed to be identical. Another characteristic of the MSP is that we can monitor and adjust each of the streams individually or in small groups.

The MSP occurs often in practice. For example, a machine may have several heads, with each head producing (we hope) identical units of product. In such situations several possible control procedures may be followed. One possibility is to use separate control charts on each stream. This approach usually results in a prohibitively large number of control charts. If the output streams are highly correlated—say, nearly perfectly correlated—then control charts on only one stream may be adequate. The most common situation is that the streams are only moderately correlated, so monitoring only one of the streams is not appropriate. There are at least two types of situations involving the occurrence of assignable causes in the MSP.

1. The output of one stream (or a few streams) has shifted off target.
2. The output of all streams has shifted off target.

In the first case, we are trying to detect an assignable cause that affects only one stream (or at most a few streams), whereas in the second, we are looking for an assignable cause that impacts *all* streams (such as would result from a change in raw materials).

The standard control chart for the MSP is the group control chart, introduced by Boyd (1950). We will also discuss other approaches to monitoring the MSP. Throughout this section we assume that the process has s streams and that the output quality characteristic from each stream is normally distributed.

10.3.2 Group Control Charts

The **group control chart (GCC)** was introduced by Boyd (1950) and remains the basic procedure for monitoring an MSP. To illustrate the methods of construction and use, suppose that the process has $s = 6$ streams and that each stream has the same target value and inherent variability. Variables measurement is made on the items produced, and the distribution of the measurement is well approximated by the normal. To establish a group control chart, the sampling is performed as if separate control charts were to be set up on each stream. Suppose, for purposes of illustration, that a sample size of $n = 4$ is used. This means that 4 units will be taken from each of the $s = 6$ streams over a short period of time. This will be repeated until about 20 such groups of samples have been taken. At this point we would have $20 \times 6 = 120$ averages of $n = 4$ observations each and 120 corresponding ranges. These averages and ranges would be averaged to produce a grand average $\bar{\bar{x}}$ and an average range $\bar{R}$. The limits on the group control charts would be at

$$\text{UCL} = \bar{\bar{x}} + A_2\bar{R}$$

$$\text{LCL} = \bar{\bar{x}} - A_2\bar{R}$$

for the $\bar{x}$ chart and at

$$\text{UCL} = D_4\bar{R}$$

$$\text{LCL} = D_3\bar{R}$$

for the R chart, with $A_2 = 0.729$, $D_3 = 0$, and $D_4 = 2.282$. Note that the sample size $n = 4$ determines the control chart constants.

When the group control chart is used to control the process, we plot only the largest and smallest of the $s = 6$ means observed at any time period on the $\bar{x}$ chart. If these means are inside the control limits, then all other means will also lie inside the limits. Similarly, only the largest range will be plotted on the range chart. Each plotted point is identified on the chart by the number of the stream that produced it. The process is out of control if a point exceeds a three-sigma limit. Runs tests cannot be applied to these charts, because the conventional runs tests were not developed to test averages or ranges that are the extremes of a group of averages or ranges.

It is useful to examine the stream numbers on the chart. In general, if a stream consistently gives the largest (or smallest) value several times in a row, that may constitute evidence that this stream is different from the others. If the process has s streams and if r is the number of consecutive times that a particular stream is the largest (or smallest) value, then the one-sided in-control average run length for this event is given by Nelson (1986) as

$$\text{ARL}(1)_0 = \frac{s^r - 1}{s - 1} \tag{10.6}$$

if all streams are identical. To illustrate the use of this equation, if $s = 6$ and $r = 4$, then

$$\text{ARL}(1)_0 = \frac{6^4 - 1}{6 - 1} = 259$$

That is, if the process is in control, we will expect to see the same stream producing an extreme value on the group control chart four times in a row only once every 259 samples.

One way to select the value of r to detect the presence of one stream that is different from the others is to use equation (10.6) to find an ARL that is roughly consistent with the ARL of a conventional control chart. The ARL for an in-control process for a single point beyond the upper control limit (say) is 740. Thus, using $r = 4$ for a six-stream process results in an ARL that is too short and that will give too many false alarms. A better choice is $r = 5$, since

$$\text{ARL}(1)_0 = \frac{6^5 - 1}{6 - 1} = 1555$$

Thus, if we have six streams and if the same stream produces an extreme value on the control chart in five consecutive samples, then we have strong evidence that this stream is different from the others.

Using equation (10.6), we can generate some general guidelines for choosing r given the number of streams s. Suitable pairs (s, r) would include (3, 7), (4, 6), (5–6, 5), and (7–10, 4). All of these combinations would give reasonable values of the one-sided ARL when the process is in control.

The two-sided in-control average run length $\text{ARL}(2)_0$ is defined as the expected number of trials until r consecutive largest or r consecutive smallest means come from the same stream while the MSP is in control. Mortell and Runger (1995) and Nelson and Stephenson (1996) used the *Markov chain approach* of Brook and Evans (1972) to compute $\text{ARL}(2)_0$ numerically. A closed-form expression is not possible for $\text{ARL}(2)_0$, but Nelson and Stephenson (1996) give a lower bound on $\text{ARL}(2)_0$ as

$$\text{ARL}(2)_0 = \frac{s^r - 1}{2(r - 1)} \tag{10.7}$$

This approximation agrees closely with the numerical computations from the Markov chain approach.

There are some drawbacks connected with the GCC. These may be summarized as follows:

1. At each point in time where samples are taken, all s streams must be sampled. When s is small, this may be a satisfactory approach. However, some high-speed filling and packaging machines may have $100 \leq s \leq 200$ streams. The GCC is not practical for these situations.
2. There is no information about nonextreme streams at each trial, thus, we cannot utilize past values to form an EWMA or a cusum statistic to improve on the GCC performance.
3. The supplementary run tests fail if more than one stream shifts simultaneously (all up or all down) to the same off-target level, because the shifted streams will most likely alternate having the extreme value, and no single stream shift can be identified.
4. The selection of the run or pattern length r for a specific problem with s streams is a discrete process, and the difference in ARL_0 for $r = r^*$ or $r = r^* + 1$ choice can be substantial, as observed by Mortell and Runger (1995). This problem is evident in the original run rule devised by Nelson (1986), but can be alleviated somewhat by using the modified run rules in Nelson and Stephenson (1996), which are based on an $r - 1$ out of r in a row approach. In fact, it can be easily shown that the difference in the ARL_0 between $r = r^*$ and $r = r^* + 1$ choices equals s^{r^*}, which becomes very large for problems with large numbers of streams.
5. Although using an $r - 1$ out of r in a row run rule allows some flexibility in the choice of the critical value $r = r^*$ and ARL_0, it adds some complexity since it is no longer a simple counting procedure.

10.3.3 Other Approaches

Mortell and Runger (1995) proposed the following model for the MSP to accommodate the practical case of dependent (cross-correlated) streams:

$$x_{tjk} = \mu + A_t + \varepsilon_{tjk} \tag{10.8}$$

where x_{tjk} is the kth measurement on the jth stream at time t, $A_t \sim N(0,\sigma_a^2)$ represents the difference of the mean over all the streams at time t from the MSP target μ, and ε_{tjk} is a $N(0, \sigma^2)$ random variable that represents the difference of the kth measurements on the jth stream from the process mean over all streams at time t. In this representation of MSP data, the total variation is allocated into two sources: σ_a^2 accounting for the variation over time common to all streams, and σ^2 accounting for the variation between the streams at specific time t. From this model, the cross correlation between any pair of streams is given by $\rho = \sigma_a^2/(\sigma_a^2 + \sigma^2)$.

Mortell and Runger proposed monitoring the average at time t of the means across all streams with an individuals control chart to detect an overall assignable cause. Also, they proposed monitoring the range of the stream's means at time t denoted by $R_t = \max(\bar{x}_{tj}) - \min(\bar{x}_{tj})$ or the maximum ***residual*** at time t given by $\max(\bar{x}_{ij}) - \bar{x}_t$, with any individuals control chart to detect a relative assignable cause affecting one or a few streams. These proposed **control charts on residuals** are better than the GCC, especially when the variation in the process means over time is greater than the between-stream variability (i.e., when $\sigma_a > \sigma$). Mortell and Runger compared the performance of these proposed charts with the runs test of the GCC by Nelson (1986) using simulations of various MSPs and concluded that the cusum chart on the range outperforms the Shewhart and EWMA on either the range or the maximum residual. On the other hand, little difference in performance exists between the runs test of Nelson (1986)

and the proposed cusum chart, but it is preferred because of the flexibility in the choice of in-control ARL.

Runger, Alt, and Montgomery (1996a) related control charts for the MSP with correlated data to principal components analysis. (See Chapter 11 for information on principal components.) The first control chart monitors the major principal component variable, which equals the mean of the streams at any time t. The second control chart is based on the Hotelling T^2 statistic (see Chapter 11) of the last $p - 1$ principal components, which is shown to be equivalent to the s^2 chart; thus it is sensitive to relative assignable causes that affect the uniformity across the streams.

Lanning (1998) studied the use of "fractional" and "adaptive" sampling in the multiple-stream process with a large number of identical and independent streams, where simultaneous monitoring of all streams is impractical. In his procedures, only a fraction of the s streams are sampled at each time period, and this fraction is increased when there are indications that an out-of-control condition may be present. (Adaptive sampling schemes are introduced in Section 10.5.) He used the average time to signal (ATS) to compare performance of the competing charts, and focused on the scenarios where assignable causes tend to impact all or most of the process streams together. He concluded that the adaptive fraction approach gives better ATS results than the fixed fraction scheme, and often yields satisfactory results compared to a complete sampling situation. An application of his methodology to a multiple-head filling machine with many filling valves is described in Lanning, Montgomery, and Runger (2002–03).

10.4 SPC with Autocorrelated Process Data

10.4.1 Sources and Effects of Autocorrelation in Process Data

The standard assumptions that are usually cited in justifying the use of control charts are that the data generated by the process when it is in control are normally and independently distributed with mean μ and standard deviation σ. Both μ and σ are considered fixed and unknown. An out-of-control condition is a change or shift in μ or σ (or both) to some different value. Therefore, we could say that when the process is in control the quality characteristic at time t, x_t, is represented by the model

$$x_t = \mu + \varepsilon_t \quad t = 1, 2, \ldots \tag{10.9}$$

where ε_t is normally and independently distributed with mean zero and standard deviation σ. This is often called the **Shewhart process model.**

When these assumptions are satisfied, one may apply conventional control charts and draw conclusions about the state of statistical control of the process. Furthermore, the statistical properties of the control chart, such as the false alarm rate with three-sigma control limits, or the average run length, can be easily determined and used to provide guidance for chart interpretation. Even in situations in which the normality assumption is violated to a slight or moderate degree, these control charts will still work reasonably well.

The most important of the assumptions made concerning control charts is that of independence of the observations, for conventional control charts do not work well if the quality characteristic exhibits even low levels of correlation over time. Specifically, these control charts will give misleading results in the form of too many false alarms if the data are positively correlated. This point has been made by numerous authors, including Berthouex, Hunter, and Pallesen (1978), Alwan and Roberts (1988), Montgomery and Friedman (1989), Alwan (1992), Harris and Ross (1991), Montgomery and Mastrangelo (1991), and Maragah and Woodall (1992).

Unfortunately, the assumption of uncorrelated or independent observations is not even approximately satisfied in some manufacturing processes. Examples include chemical processes

15 Lot-by-Lot Acceptance Sampling for Attributes

Chapter Outline

15.1 THE ACCEPTANCE-SAMPLING PROBLEM
- 15.1.1 Advantages and Disadvantages of Sampling
- 15.1.2 Types of Sampling Plans
- 15.1.3 Lot Formation
- 15.1.4 Random Sampling
- 15.1.5 Guidelines for Using Acceptance Sampling

15.2 SINGLE-SAMPLING PLANS FOR ATTRIBUTES
- 15.2.1 Definition of a Single-Sampling Plan
- 15.2.2 The OC Curve
- 15.2.3 Designing a Single-Sampling Plan with a Specified OC Curve
- 15.2.4 Rectifying Inspection

15.3 DOUBLE, MULTIPLE, AND SEQUENTIAL SAMPLING
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- 15.3.2 Multiple-Sampling Plans
- 15.3.3 Sequential-Sampling Plans

15.4 MILITARY STANDARD 105E (ANSI/ASQC Z1.4, ISO 2859)
- 15.4.1 Description of the Standard
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- 15.4.3 Discussion

15.5 THE DODGE–ROMIG SAMPLING PLANS
- 15.5.1 AOQL Plans
- 15.5.2 LTPD Plans
- 15.5.3 Estimation of Process Average

Supplemental Material for Chapter 15

- S15.1 A Lot Sensitive Compliance (LTPD) Sampling Plan
- S15.2 Consideration of Inspection Error

The supplemental material is on the textbook Website www.wiley.com/college/montgomery.

Chapter Overview and Learning Objectives

This chapter presents lot-by-lot acceptance-sampling plans for attributes. Key topics include the design and operation of single-sampling plans, the use of the operating characteristic curve, and the concepts of rectifying inspection, average outgoing quality, and average total inspection. Similar concepts are briefly introduced for types of sampling plans where more

than one sample may be taken to determine the disposition of a lot (double, multiple, and sequential sampling). Two systems of standard sampling plans are also presented, the military standard plans known as MIL STD 105E and the Dodge–Romig plans. These plans are designed around different philosophies: MIL STD 105E has an acceptable quality level focus, whereas the Dodge–Romig plans are oriented around either the lot tolerance percent defective or the average outgoing quality limit perspective.

After careful study of this chapter you should be able to do the following:

1. Understand the role of acceptance sampling in modern quality control systems
2. Understand the advantages and disadvantages of sampling
3. Understand the difference between attributes and variables sampling plans, and the major types of acceptance-sampling procedures
4. Know how single-, double-, and sequential-sampling plans are used
5. Understand the importance of random sampling
6. Know how to determine the OC curve for a single-sampling plan for attributes
7. Understand the effects of the sampling plan parameters on sampling plan performance
8. Know how to design single-sampling, double-sampling, and sequential-sampling plans for attributes
9. Know how rectifying inspection is used
10. Understand the structure and use of MIL STD 105E and its civilian counterpart plans
11. Understand the structure and use of the Dodge–Romig system of sampling plans

15.1 The Acceptance-Sampling Problem

As we observed in Chapter 1, acceptance sampling is concerned with inspection and decision making regarding products, one of the oldest aspects of quality assurance. In the 1930s and 1940s, acceptance sampling was one of the major components of the field of statistical quality control, and was used primarily for incoming or receiving inspection. In more recent years, it has become typical to work with suppliers to improve their process performance through the use of SPC and designed experiments, and not to rely as much on acceptance sampling as a primary quality assurance tool.

A typical application of acceptance sampling is as follows: A company receives a shipment of product from a supplier. This product is often a component or raw material used in the company's manufacturing process. A sample is taken from the lot, and some quality characteristic of the units in the sample is inspected. On the basis of the information in this sample, a decision is made regarding **lot disposition.** Usually, this decision is either to accept or to reject the lot. Sometimes we refer to this decision as **lot sentencing.** Accepted lots are put into production; rejected lots may be returned to the supplier or may be subjected to some other **lot disposition action.**

Although it is customary to think of acceptance sampling as a receiving inspection activity, there are other uses of sampling methods. For example, frequently a manufacturer will sample and inspect its own product at various stages of production. Lots that are accepted are sent forward for further processing, and rejected lots may be reworked or scrapped.

Three aspects of sampling are important:

1. It is the purpose of acceptance sampling to sentence lots, not to estimate the lot quality. Most acceptance-sampling plans are not designed for estimation purposes.

2. Acceptance-sampling plans do not provide any *direct* form of quality control. Acceptance sampling simply accepts and rejects lots. Even if all lots are of the same quality, sampling will accept some lots and reject others, the accepted lots being no better than the rejected ones. Process controls are used to control and systematically improve quality, but acceptance sampling is not.
3. The most effective use of acceptance sampling is *not* to "inspect quality into the product," but rather as an audit tool to ensure that the output of a process conforms to requirements.

Generally, there are three approaches to lot sentencing: (1) accept with no inspection; (2) 100% inspection—that is, inspect every item in the lot, removing all defective[1] units found (defectives may be returned to the supplier, reworked, replaced with known good items, or discarded); and (3) acceptance sampling. The no-inspection alternative is useful in situations where either the supplier's process is so good that defective units are almost never encountered or where there is no economic justification to look for defective units. For example, if the supplier's process capability ratio is 3 or 4, acceptance sampling is unlikely to discover any defective units. We generally use 100% inspection in situations where the component is extremely critical and passing any defectives would result in an unacceptably high failure cost at subsequent stages, or where the supplier's process capability is inadequate to meet specifications. Acceptance sampling is most likely to be useful in the following situations:

1. When testing is destructive
2. When the cost of 100% inspection is extremely high
3. When 100% inspection is not technologically feasible or would require so much calendar time that production scheduling would be seriously impacted
4. When there are many items to be inspected and the inspection error rate is sufficiently high that 100% inspection might cause a higher percentage of defective units to be passed than would occur with the use of a sampling plan
5. When the supplier has an excellent quality history, and some reduction in inspection from 100% is desired, but the supplier's process capability is sufficiently low as to make no inspection an unsatisfactory alternative
6. When there are potentially serious product liability risks, and although the supplier's process is satisfactory, a program for continuously monitoring the product is necessary

15.1.1 Advantages and Disadvantages of Sampling

When acceptance sampling is contrasted with 100% inspection, it has the following advantages:

1. It is usually less expensive because there is less inspection.
2. There is less handling of the product, hence reduced damage.
3. It is applicable to destructive testing.
4. Fewer personnel are involved in inspection activities.
5. It often greatly reduces the amount of inspection error.
6. The rejection of entire lots as opposed to the simple return of defectives often provides a stronger motivation to the supplier for quality improvements.

[1]In previous chapters, the terms "nonconforming" and "nonconformity" were used instead of defective and defect. This is because the popular meanings of defective and defect differ from their technical meanings and have caused considerable misunderstanding, particularly in product liability litigation. In the field of sampling inspection, however, "defective" and "defect" continue to be used in their technical sense—that is, nonconformance to requirements.

Acceptance sampling also has several disadvantages, however. These include the following:

1. There are risks of accepting "bad" lots and rejecting "good" lots.
2. Less information is usually generated about the product or about the process that manufactured the product.
3. Acceptance sampling requires planning and documentation of the acceptance-sampling procedure whereas 100% inspection does not.

Although this last point is often mentioned as a disadvantage of acceptance sampling, proper design of an acceptance-sampling plan usually requires study of the actual level of quality required by the consumer. This resulting knowledge is often a useful input into the overall quality planning and engineering process. Thus, in many applications, it may not be a significant disadvantage.

We have pointed out that acceptance sampling is a middle ground between the extremes of 100% inspection and no inspection. It often provides a methodology for moving between these extremes as sufficient information is obtained on the control of the manufacturing process that produces the product. Although there is no direct control of quality in the application of an acceptance-sampling plan to an isolated lot, when that plan is applied to a stream of lots from a supplier, it becomes a means of providing protection for both the producer of the lot and the consumer. It also provides for an accumulation of quality history regarding the process that produces the lot, and it may provide feedback that is useful in process control, such as determining when process controls at the supplier's plant are not adequate. Finally, it may place economic or psychological pressure on the supplier to improve the production process.

15.1.2 Types of Sampling Plans

There are a number of different ways to classify acceptance-sampling plans. One major classification is by variables and attributes. **Variables,** of course, are quality characteristics that are measured on a numerical scale. **Attributes** are quality characteristics that are expressed on a "go, no-go" basis. This chapter deals with lot-by-lot acceptance-sampling plans for attributes. Variables sampling plans are the subject of Chapter 16, along with a brief discussion of several special acceptance-sampling procedures.

A **single-sampling plan** is a lot-sentencing procedure in which one sample of n units is selected at random from the lot, and the disposition of the lot is determined based on the information contained in that sample. For example, a single-sampling plan for attributes would consist of a sample size n and an acceptance number c. The procedure would operate as follows: Select n items at random from the lot. If there are c or fewer defectives in the sample, accept the lot, and if there are more than c defective items in the sample, reject the lot. We investigate this type of sampling plan extensively in Section 15.2.

Double-sampling plans are somewhat more complicated. Following an initial sample, a decision based on the information in that sample is made either to (1) accept the lot, (2) reject the lot, or (3) take a second sample. If the second sample is taken, the information from both the first and second sample is combined in order to reach a decision whether to accept or reject the lot. Double-sampling plans are discussed in Section 15.3.

A **multiple-sampling plan** is an extension of the double-sampling concept, in that more than two samples may be required in order to reach a decision regarding the disposition of the lot. Sample sizes in multiple sampling are usually smaller than they are in either single or double sampling. The ultimate extension of multiple sampling is **sequential sampling,** in which units are selected from the lot one at a time, and following inspection of each unit, a decision is made either to accept the lot, reject the lot, or select another unit. Multiple- and sequential-sampling plans are also discussed in Section 15.3.

Single-, double-, multiple-, and sequential-sampling plans can be designed so that they produce equivalent results. That is, these procedures can be designed so that a lot of specified quality has exactly the same probability of acceptance under all four types of sampling plans. Consequently, when selecting the type of sampling procedure, one must consider factors such as the administrative efficiency, the type of information produced by the plan, the average amount of inspection required by the procedure, and the impact that a given procedure may have on the material flow in the manufacturing organization. These issues are discussed in more detail in Section 15.3.

15.1.3 Lot Formation

How the lot is formed can influence the effectiveness of the acceptance-sampling plan. There are a number of important considerations in forming lots for inspection. Some of these are as follows:

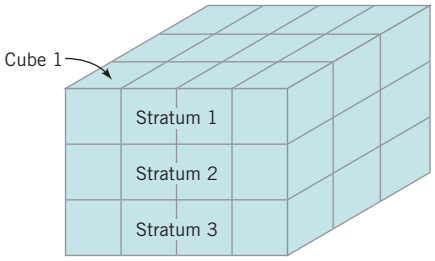
1. **Lots should be homogeneous.** The units in the lot should be produced by the same machines, the same operators, and from common raw materials, at approximately the same time. When lots are nonhomogeneous, such as when the output of two different production lines is mixed, the acceptance-sampling scheme may not function as effectively as it could. Nonhomogeneous lots also make it more difficult to take corrective action to eliminate the source of defective products.
2. **Larger lots are preferred over smaller ones.** It is usually more economically efficient to inspect large lots than small ones.
3. **Lots should be conformable to the materials-handling systems used in both the supplier and consumer facilities.** In addition, the items in the lots should be packaged so as to minimize shipping and handling risks, and so as to make selection of the units in the sample relatively easy.

15.1.4 Random Sampling

The units selected for inspection from the lot should be chosen at random, and they should be representative of all the items in the lot. The random-sampling concept is extremely important in acceptance sampling. Unless random samples are used, bias will be introduced. For example, the supplier may ensure that the units packaged on the top of the lot are of extremely good quality, knowing that the inspector will select the sample from the top layer. "Salting" a lot in this manner is not a common practice, but if it occurs and nonrandom-sampling methods are used, the effectiveness of the inspection process is destroyed.

The technique often suggested for drawing a random sample is to first assign a number to each item in the lot. Then n random numbers are drawn, where the range of these numbers is from 1 to the maximum number of units in the lot. This sequence of random numbers determines which units in the lot will constitute the sample. If products have serial or other code numbers, these numbers can be used to avoid the process of actually assigning numbers to each unit. Another possibility would be to use a three-digit random number to represent the length, width, and depth in a container.

In situations where we cannot assign a number to each unit, utilize serial or code numbers, or randomly determine the location of the sample unit, some other technique must be employed to ensure that the sample is random or representative. Sometimes the inspector may **stratify** the lot. This consists of dividing the lot into strata or layers and then subdividing each strata into cubes, as shown in Fig. 15.1. Units are then selected from within each cube. Although this stratification of the lot is usually an imaginary activity performed by the inspector and does not necessarily ensure random samples, at least it ensures that units are selected from all locations in the lot.



■ FIGURE 15.1 Stratifying a lot.

We cannot overemphasize the importance of random sampling. If judgment methods are used to select the sample, the statistical basis of the acceptance-sampling procedure is lost.

15.1.5 Guidelines for Using Acceptance Sampling

An acceptance-sampling plan is a statement of the sample size to be used and the associated acceptance or rejection criteria for sentencing individual lots. A sampling scheme is defined as a set of procedures consisting of acceptance-sampling plans in which lot sizes, sample sizes, and acceptance or rejection criteria along with the amount of 100% inspection and sampling are related. Finally, a sampling system is a unified collection of one or more acceptance-sampling schemes. In this chapter, we see examples of sampling plans, sampling schemes, and sampling systems.

The major types of acceptance-sampling procedures and their applications are shown in Table 15.1. In general, the selection of an acceptance-sampling procedure depends on both the objective of the sampling organization and the history of the organization whose product is sampled. Furthermore, the application of sampling methodology is not static; that is, there is a natural evolution from one level of sampling effort to another. For example, if we are dealing with a supplier who enjoys an excellent quality history, we might begin with an attributes sampling plan. As our experience with the supplier grows, and its good-quality reputation is proved by the results of our sampling activities, we might transition to a sampling procedure that requires much less inspection, such as skip-lot sampling. Finally, after extensive experience with the supplier, and if its process capability is extremely good, we might stop all acceptance-sampling activities on the product. In another situation, where we have little knowledge of or experience with the supplier's quality-assurance efforts, we might begin with attributes sampling using a plan that assures us that the quality of accepted lots is no worse

■ TABLE 15.1
Acceptance-Sampling Procedures

Objective	Attributes Procedure	Variables Procedure
Assure quality levels for consumer/producer	Select plan for specific OC curve	Select plan for specific OC curve
Maintain quality at a target	AQL system; MIL STD 105E, ANSI/ASQC Z1.4	AQL system; MIL STD 414, ANSI/ASQC Z1.9
Assure average outgoing quality level	AOQL system; Dodge–Romig plans	AOQL system
Reduce inspection, with small sample sizes, good-quality history	Chain sampling	Narrow-limit gauging
Reduce inspection after good-quality history	Skip-lot sampling; double sampling	Skip-lot sampling; double sampling
Assure quality no worse than target	LTPD plan; Dodge–Romig plans	LTPD plan; hypothesis testing

than a specified target value. If this plan proves successful, and if the supplier's performance is satisfactory, we might transition from attributes to variables inspection, particularly as we learn more about the nature of the supplier's process. Finally, we might use the information gathered in variables sampling plans in conjunction with efforts directly at the supplier's manufacturing facility to assist in the installation of process controls. A successful program of process controls at the supplier level might improve the supplier's process capability to the point where inspection could be discontinued.

These examples illustrate that there is a life cycle of application of acceptance-sampling techniques. This was also reflected in the phase diagram, Fig. 1.7, which presented the percentage of application of various quality-assurance techniques as a function of the maturity of the business organization. Typically, we find that organizations with relatively new quality-assurance efforts place a great deal of reliance on acceptance sampling. As their maturity grows and the quality organization develops, they begin to rely less on acceptance sampling and more on statistical process control and experimental design.

Manufacturers try to improve the quality of their products by reducing the number of suppliers from whom they buy their components, and by working more closely with the ones they retain. Once again, the key tool in this effort to improve quality is statistical process control. Acceptance sampling can be an important ingredient of any quality-assurance program; however, remember that it is an activity that you try to avoid doing. It is much more cost effective to use statistically based process monitoring at the appropriate stage of the manufacturing process. Sampling methods can in some cases be a tool that you employ along the road to that ultimate goal.

15.2 Single-Sampling Plans for Attributes

15.2.1 Definition of a Single-Sampling Plan

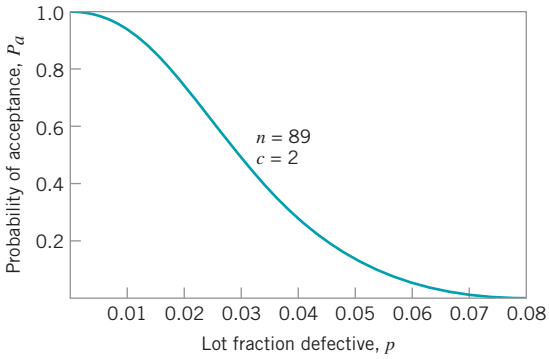
Suppose that a lot of size N has been submitted for inspection. A **single-sampling plan** is defined by the sample size n and the acceptance number c. Thus, if the lot size is $N = 10{,}000$, then the sampling plan

$$n = 89$$
$$c = 2$$

means that from a lot of size 10,000 a random sample of $n = 89$ units is inspected and the number of nonconforming or defective items d observed. If the number of observed defectives d is less than or equal to $c = 2$, the lot will be accepted. If the number of observed defectives d is greater than 2, the lot will be rejected. Since the quality characteristic inspected is an attribute, each unit in the sample is judged to be either conforming or nonconforming. One or several attributes can be inspected in the same sample; generally, a unit that is nonconforming to specifications on one or more attributes is said to be a defective unit. This procedure is called a single-sampling plan because the lot is sentenced based on the information contained in one sample of size n.

15.2.2 The OC Curve

An important measure of the performance of an acceptance-sampling plan is the **operating-characteristic (OC) curve.** This curve plots the probability of accepting the lot versus the lot fraction defective. Thus, the OC curve displays the discriminatory power of the sampling plan. That is, it shows the probability that a lot submitted with a certain fraction defective will be either accepted or rejected. The OC curve of the sampling plan $n = 89$, $c = 2$ is shown in



■ **FIGURE 15.2** OC curve of the single-sampling plan $n = 89$, $c = 2$.

Fig. 15.2. It is easy to demonstrate how the points on this curve are obtained. Suppose that the lot size N is large (theoretically infinite). Under this condition, the distribution of the number of defectives d in a random sample of n items is binomial with parameters n and p, where p is the fraction of defective items in the lot. An equivalent way to conceptualize this is to draw lots of N items at random from a theoretically infinite process, and then to draw random samples of n from these lots. Sampling from the lot in this manner is the equivalent of sampling directly from the process. The probability of observing exactly d defectives is

$$P\{d \text{ defectives}\} = f(d) = \frac{n!}{d!(n-d)!} p^d (1-p)^{n-d} \tag{15.1}$$

The probability of acceptance is simply the probability that d is less than or equal to c, or

$$P_a = P\{d \le c\} = \sum_{d=0}^{c} \frac{n!}{d!(n-d)!} p^d (1-p)^{n-d} \tag{15.2}$$

For example, if the lot fraction defective is $p = 0.01$, $n = 89$, and $c = 2$, then

$$\begin{aligned} P_a = P\{d \le 2\} &= \sum_{d=0}^{2} \frac{89!}{d!(89-d)!} (0.01)^d (0.99)^{89-d} \\ &= \frac{89!}{0!89!}(0.01)^0(0.99)^{89} + \frac{89!}{1!88!}(0.01)^1(0.99)^{88} + \frac{89!}{2!(87)!}(0.01)^2(0.99)^{87} \\ &= 0.9397 \end{aligned}$$

The OC curve is developed by evaluating equation (15.2) for various values of p. Table 15.2 displays the calculated value of several points on the curve.

The OC curve shows the **discriminatory power** of the sampling plan. For example, in the sampling plan $n = 89$, $c = 2$, if the lots are 2% defective, the probability of acceptance is approximately 0.74. This means that if 100 lots from a process that manufactures 2% defective product are submitted to this sampling plan, we will expect to accept 74 of the lots and reject 26 of them.

Effect of n and c on OC Curves. A sampling plan that discriminated perfectly between good and bad lots would have an OC curve that looks like Fig. 15.3. The OC curve runs horizontally at a probability of acceptance $P_a = 1.00$ until a level of lot quality that is considered "bad" is reached, at which point the curve drops vertically to a probability of acceptance $P_a = 0.00$, and then the curve runs horizontally again for all lot fraction defectives

TABLE 15.2
Probabilities of Acceptance for the Single-Sampling Plan $n = 89$, $c = 2$

Fraction Defective, p	Probability of Acceptance, P_a
0.005	0.9897
0.010	0.9397
0.020	0.7366
0.030	0.4985
0.040	0.3042
0.050	0.1721
0.060	0.0919
0.070	0.0468
0.080	0.0230
0.090	0.0109

greater than the undesirable level. If such a sampling plan could be employed, all lots of "bad" quality would be rejected, and all lots of "good" quality would be accepted.

Unfortunately, the **ideal OC curve** in Fig. 15.3 can almost never be obtained in practice. In theory, it could be realized by 100% inspection, if the inspection were error-free. The ideal OC curve shape can be approached, however, by increasing the sample size. Figure 15.4 shows that the OC curve becomes more like the idealized OC curve shape as the sample size increases. (Note that the acceptance number c is kept proportional to n.) Thus, the precision with which a sampling plan differentiates between good and bad lots increases with the size of the sample. The greater is the slope of the OC curve, the greater is the discriminatory power.

Figure 15.5 shows how the OC curve changes as the acceptance number changes. Generally, changing the acceptance number does not dramatically change the slope of the OC curve. As the acceptance number is decreased, the OC curve is shifted to the left. Plans with smaller values of c provide discrimination at lower levels of lot fraction defective than do plans with larger values of c.

Specific Points on the OC Curve. Frequently, the quality engineer's interest focuses on certain points on the OC curve. The supplier or consumer is usually interested in knowing what level of lot or process quality would yield a high probability of acceptance. For example, the supplier might be interested in the 0.95 probability of acceptance point. This would indicate the level of process fallout that could be experienced and still have a 95% chance that the lots would be accepted. Conversely, the consumer might be interested in the other end of the OC curve. That is, what level of lot or process quality will yield a low probability of acceptance?

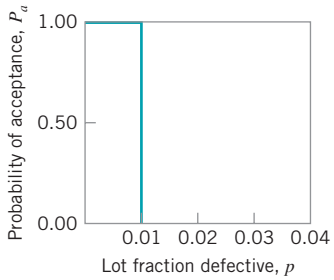


FIGURE 15.3 Ideal OC curve.

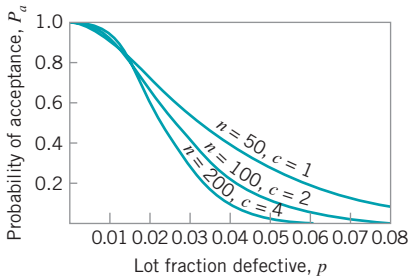


FIGURE 15.4 OC curves for different sample sizes.

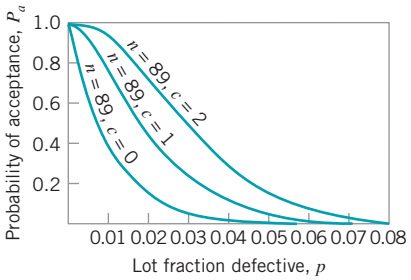


FIGURE 15.5 The effect of changing the acceptance number on the OC curve.

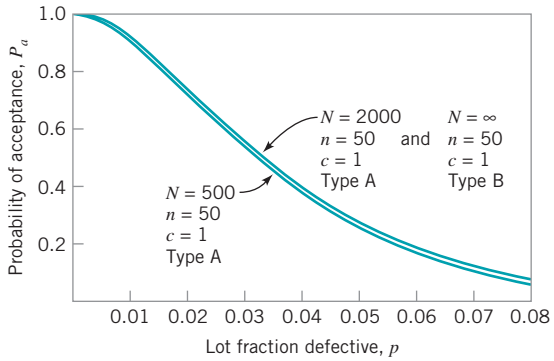
A consumer often establishes a sampling plan for a continuing supply of components or raw material with reference to an **acceptable quality level (AQL).** The AQL represents the poorest level of quality for the supplier's process that the consumer would consider to be acceptable as a process average. Note that the AQL is a property of the supplier's manufacturing process; it is not a property of the sampling plan. The consumer will often design the sampling procedure so that the OC curve gives a high probability of acceptance at the AQL. Furthermore, the AQL is not usually intended to be a specification on the product, nor is it a target value for the supplier's production process. It is simply a standard against which to judge the lots. It is hoped that the supplier's process will operate at a fallout level that is considerably better than the AQL.

The consumer will also be interested in the other end of the OC curve—that is, in the protection that is obtained for individual lots of poor quality. In such a situation, the consumer may establish a **lot tolerance percent defective (LTPD).** The LTPD is the poorest level of quality that the consumer is willing to accept in an individual lot. Note that the lot tolerance percent defective is not a characteristic of the sampling plan, but is a level of lot quality specified by the consumer. Alternate names for the LTPD are the **rejectable quality level (RQL)** and the **limiting quality level (LQL).** It is possible to design acceptance-sampling plans that give specified probabilities of acceptance at the LTPD point. Subsequently, we will see how to design sampling plans that have specified performance at the AQL and LTPD points.

Type-A and Type-B OC Curves. The OC curves that were constructed in the previous examples are called **type-B OC curves.** In the construction of the OC curve it was assumed that the samples came from a large lot or that we were sampling from a stream of lots selected at random from a process. In this situation, the **binomial distribution** is the exact probability distribution for calculating the probability of lot acceptance. Such an OC curve is referred to as a type-B OC curve.

The **type-A OC curve** is used to calculate probabilities of acceptance for an isolated lot of finite size. Suppose that the lot size is N, the sample size is n, and the acceptance number is c. The exact sampling distribution of the number of defective items in the sample is the **hypergeometric distribution.**

Figure 15.6 shows the type-A OC curve for a single-sampling plan with $n = 50$, $c = 1$, where the lot size is $N = 500$. The probabilities of acceptance defining the OC curve were calculated using the hypergeometric distribution. Also shown on this graph is the type-A OC curve for $N = 2000$, $n = 50$, and $c = 1$. Note that the two OC curves are very similar. Generally, as the size of the lot increases, the lot size has a decreasing impact on the OC curve. In fact, if the lot size is at least 10 times the sample size ($n/N \leq 0.10$), the type-A and type-B OC curves are virtually indistinguishable. As an illustration, the type-B OC curve for the sampling plan $n = 50$, $c = 1$ is also shown in Fig. 15.6. Note that it is identical to the type-A OC curve based on a lot size of $N = 2000$.



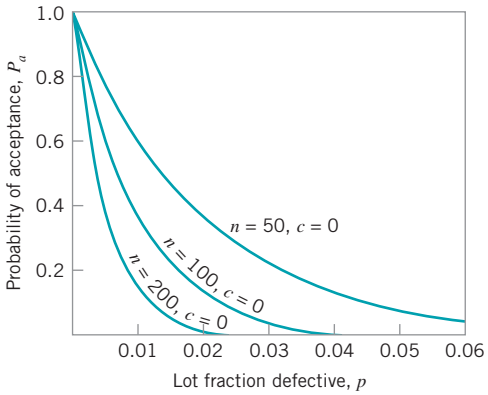
■ **FIGURE 15.6** Type-A and type-B OC curves.

The type-A OC curve will always lie below the type-B OC curve. That is, if a type-B OC curve is used as an approximation for a type-A curve, the probabilities of acceptance calculated for the type-B curve will always be higher than they would have been if the type-A curve had been used instead. However, this difference is only significant when the lot size is small relative to the sample size. Unless otherwise stated, all discussion of OC curves in this text is in terms of the type-B OC curve.

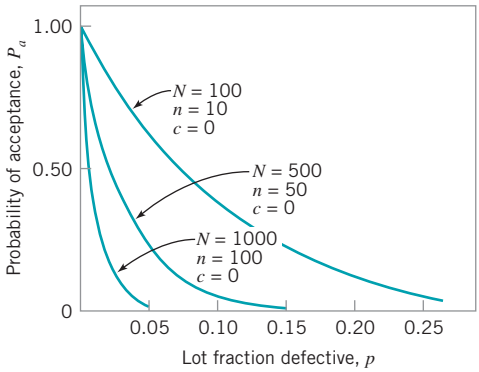
Other Aspects of OC Curve Behavior. Two approaches to designing sampling plans that are encountered in practice have certain implications for the behavior of the OC curve. Since not all of these implications are positive, it is worthwhile to briefly mention these two approaches to sampling plan design. These approaches are the use of sampling plans with zero acceptance numbers ($c = 0$) and the use of sample sizes that are a fixed percentage of the lot size.

Figure 15.7 shows several OC curves for acceptance-sampling plans with $c = 0$. By comparing Fig. 15.7 with Fig. 15.5, it is easy to see that plans with zero acceptance numbers have OC curves that have a very different shape than the OC curves of sampling plans for which $c > 0$. Generally, sampling plans with $c = 0$ have OC curves that are convex throughout their range. As a result of this shape, the probability of acceptance begins to drop very rapidly, even for small values of the lot fraction defective. This is extremely hard on the supplier, and in some circumstances, it may be extremely uneconomical for the consumer. For example, consider the sampling plans in Fig. 15.5. Suppose the acceptable quality level is 1%. This implies that we would like to accept lots that are 1% defective or better. Note that if sampling plan $n = 89$, $c = 1$ is used, the probability of lot acceptance at the AQL is about 0.78. On the other hand, if the plan $n = 89$, $c = 0$ is used, the probability of acceptance at the AQL is approximately 0.41. That is, nearly 60% of the lots of AQL quality will be rejected if we use an acceptance number of zero. If rejected lots are returned to the supplier, then a large number of lots will be unnecessarily returned, perhaps creating production delays at the consumer's manufacturing site. If the consumer screens or 100% inspects all rejected lots, a large number of lots that are of acceptable quality will be screened. This is, at best, an inefficient use of sampling resources. In Chapter 16, we suggest an alternative approach to using zero acceptance numbers called **chain-sampling plans.** Under certain circumstances, chain sampling works considerably better than acceptance-sampling plans with $c = 0$. Also refer to Section S15.1 of the supplemental material for a discussion of **lot-sensitive compliance sampling,** another technique that utilizes zero acceptance numbers.

Figure 15.8 presents the OC curves for sampling plans in which the sample size is a fixed percentage of the lot size. The principal disadvantage of this approach is that the different



■ **FIGURE 15.7** OC curves for single-sampling plan with $c = 0$.



■ **FIGURE 15.8** OC curves for sampling plans where sample size n is 10% of the lot size.

sample sizes offer different levels of protection. It is illogical for the level of protection that the consumer enjoys for a critical part or component to vary as the size of the lot varies. Although sampling procedures such as this one were in wide use before the statistical principles of acceptance sampling were generally known, their use has (unfortunately) not entirely disappeared.

15.2.3 Designing a Single-Sampling Plan with a Specified OC Curve

A common approach to the design of an acceptance-sampling plan is to require that the OC curve pass through two designated points. Note that one point is not enough to fully specify the sampling plan; however, two points are sufficient. In general, it does not matter which two points are specified.

Suppose that we wish to construct a sampling plan such that the probability of acceptance is $1 - \alpha$ for lots with fraction defective p_1, and the probability of acceptance is β for lots with fraction defective p_2. Assuming that binomial sampling (with type-B OC curves) is appropriate, we see that the sample size n and acceptance number c are the solution to

$$1-\alpha = \sum_{d=0}^{c} \frac{n!}{d!(n-d)!} p_1^d (1-p_1)^{n-d}$$
$$\beta = \sum_{d=0}^{c} \frac{n!}{d!(n-d)!} p_2^d (1-p_2)^{n-d} \qquad (15.3)$$

Equation (15.3) was obtained by writing out the two points on the OC curve using the binomial distribution. The two simultaneous equations in equation (15.3) are nonlinear, and there is no simple, direct solution.

The nomograph in Fig. 15.9 can be used for solving these equations. The procedure for using the nomograph is very simple. Two lines are drawn on the nomograph, one connecting p_1 and $1 - \alpha$, and the other connecting p_2 and β. The intersection of these two lines gives the region of the nomograph in which the desired sampling plan is located. To illustrate the use of the nomograph, suppose we wish to construct a sampling plan for which $p_1 = 0.01$, $\alpha = 0.05$, $p_2 = 0.06$, and $\beta = 0.10$. Locating the intersection of the lines connecting ($p_1 = 0.01$, $1 - \alpha = 0.95$) and ($p_2 = 0.06$, $\beta = 0.10$) on the nomograph indicates that the plan $n = 89$, $c = 2$ is very close to passing through these two points on the OC curve. Obviously, since n and c must be integers, this procedure will actually produce several plans that have OC curves that pass close to the desired points. For instance, if the first line is followed either to the c-line just above the intersection point or to the c-line just below it, and the alternate sample sizes are read from the chart, this will produce two plans that pass almost exactly through the p_1, $1 - \alpha$ point, but may deviate somewhat from the p_2, β point. A similar procedure could be followed with the p_2, β-line. The result of following both of these lines would be four plans that pass approximately through the two points specified on the OC curve.

In addition to the graphical procedure that we have described for designing sampling plans with specified OC curves, tabular procedures are also available for the same purpose. Duncan (1986) gives a good description of these techniques.

Although any two points on the OC curve could be used to define the sampling plan, it is customary in many industries to use the AQL and LTPD points for this purpose. When the levels of lot quality specified are $p_1 = \text{AQL}$ and $p_2 = \text{LTPD}$, the corresponding points on the OC curve are usually referred to as the producer's risk point and the consumer's risk point, respectively. Thus, α would be called the producer's risk and β would be called the consumer's risk.

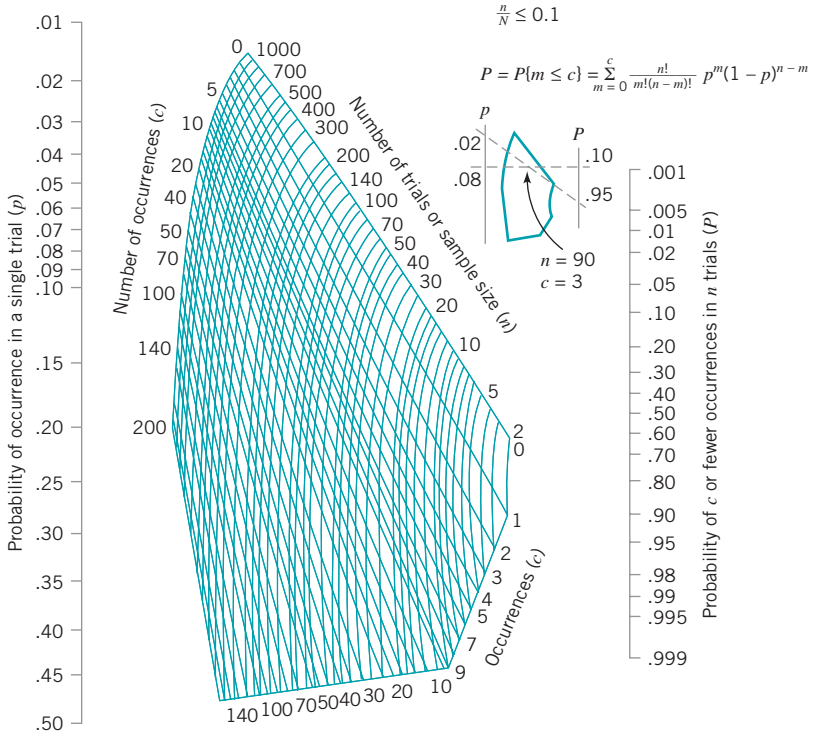


FIGURE 15.9 Binomial nomograph.

15.2.4 Rectifying Inspection

Acceptance-sampling programs usually require corrective action when lots are rejected. This generally takes the form of 100% inspection or **screening** of rejected lots, with all discovered defective items either removed for subsequent rework or return to the supplier, or replaced from a stock of known good items. Such sampling programs are called **rectifying inspection programs,** because the inspection activity affects the final quality of the outgoing product. This is illustrated in Fig. 15.10. Suppose that incoming lots to the inspection activity have fraction defective p_0. Some of these lots will be accepted, and others will be rejected. The rejected lots will be screened, and their final fraction defective will be zero. However, accepted lots have fraction defective p_0. Consequently, the outgoing lots from the inspection activity are a mixture of lots with fraction defective p_0 and fraction defective zero, so the average fraction defective in the stream of outgoing lots is p_1, which is less than p_0. Thus, a rectifying inspection program serves to "correct" lot quality.

Rectifying inspection programs are used in situations where the manufacturer wishes to know the average level of quality that is likely to result at a given stage of the manufacturing

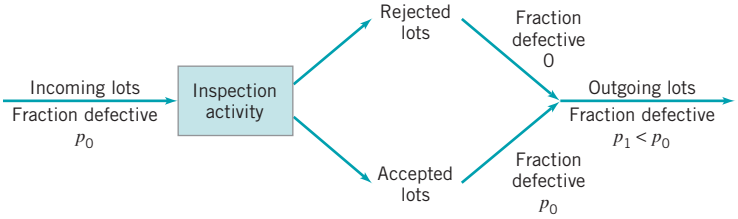


FIGURE 15.10 Rectifying inspection.

operations. Thus, rectifying inspection programs are used either at receiving inspection, in-process inspection of semifinished products, or at final inspection of finished goods. The objective of in-plant usage is to give assurance regarding the average quality of material used in the next stage of the manufacturing operations.

Rejected lots may be handled in a number of ways. The best approach is to return rejected lots to the supplier, and require it to perform the screening and rework activities. This has the psychological effect of making the supplier responsible for poor quality and may exert pressure on the supplier to improve its manufacturing processes or to install better process controls. However, in many situations, because the components or raw materials are required in order to meet production schedules, screening and rework take place at the consumer level. This is not the most desirable situation.

Average outgoing quality is widely used for the evaluation of a rectifying sampling plan. The average outgoing quality is the quality in the lot that results from the application of rectifying inspection. It is the average value of lot quality that would be obtained over a long sequence of lots from a process with fraction defective p. It is simple to develop a formula for average outgoing quality (AOQ). Assume that the lot size is N and that all discovered defectives are replaced with good units. Then in lots of size N, we have

1. n items in the sample that, after inspection, contain no defectives, because all discovered defectives are replaced
2. $N - n$ items that, if the lot is rejected, also contain no defectives
3. $N - n$ items that, if the lot is accepted, contain $p(N - n)$ defectives

Thus, lots in the outgoing stage of inspection have an expected number of defective units equal to $P_a p(N - n)$, which we may express as an *average fraction defective,* called the **average outgoing quality** or

$$\text{AOQ} = \frac{P_a p(N-n)}{N} \tag{15.4}$$

To illustrate the use of equation (15.4), suppose that $N = 10{,}000$, $n = 89$, and $c = 2$, and that the incoming lots are of quality $p = 0.01$. Now at $p = 0.01$, we have $P_a = 0.9397$, and the AOQ is

$$\begin{aligned}\text{AOQ} &= \frac{P_a p(N-n)}{N} \\ &= \frac{(0.9397)(0.01)(10{,}000-89)}{10{,}000} \\ &= 0.0093\end{aligned}$$

That is, the average outgoing quality is 0.93% defective. Note that as the lot size N becomes large relative to the sample size n, we may write equation (15.4) as

$$\text{AOQ} \simeq P_a p \tag{15.5}$$

Average outgoing quality will vary as the fraction defective of the incoming lots varies. The curve that plots average outgoing quality against incoming lot quality is called an *AOQ*

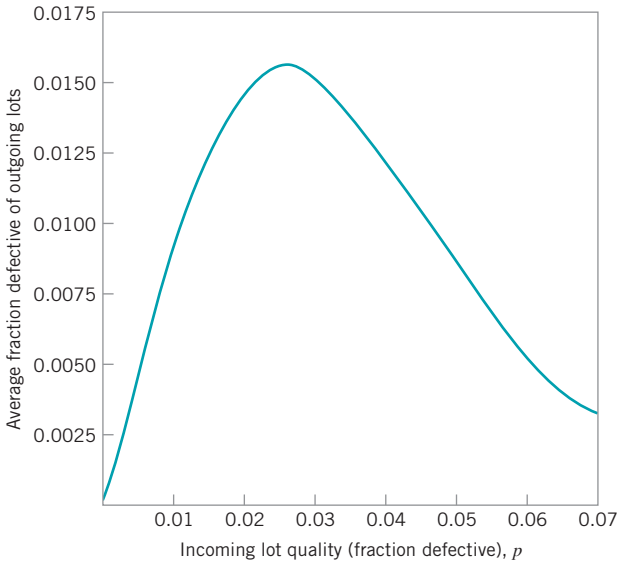


FIGURE 15.11 Average outgoing quality curve for $n = 89$, $c = 2$.

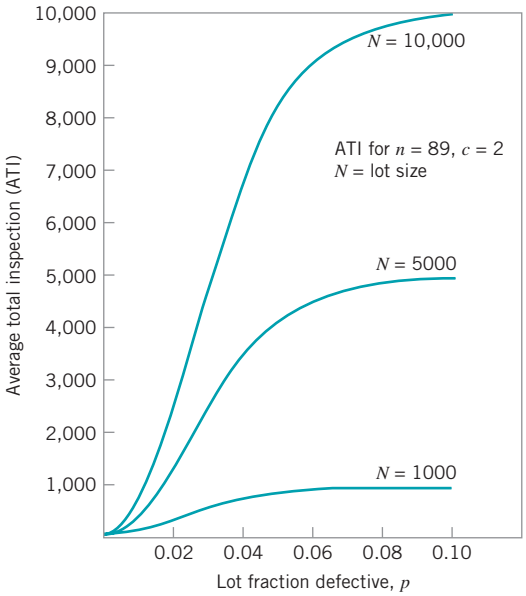
curve. The AOQ curve for the sampling plan $n = 89$, $c = 2$ is shown in Fig. 15.11. From examining this curve we note that when the incoming quality is very good, the average outgoing quality is also very good. In contrast, when the incoming lot quality is very bad, most of the lots are rejected and screened, which leads to a very good level of quality in the outgoing lots. In between these extremes, the AOQ curve rises, passes through a maximum, and descends. The maximum ordinate on the AOQ curve represents the worst possible average quality that would result from the rectifying inspection program, and this point is called the **average outgoing quality limit (AOQL).** From examining Fig. 15.11, the AOQL is seen to be approximately 0.0155. That is, no matter how bad the fraction defective is in the incoming lots, the outgoing lots will never have a worse quality level on the average than 1.55% defective. Let us emphasize that this AOQL is an *average level of quality, across a large stream of lots*. It does not give assurance that an isolated lot will have quality no worse than 1.55% defective.

Another important measure relative to rectifying inspection is the total amount of inspection required by the sampling program. If the lots contain no defective items, no lots will be rejected, and the amount of inspection per lot will be the sample size n. If the items are all defective, every lot will be submitted to 100% inspection, and the amount of inspection per lot will be the lot size N. If the lot quality is $0 < p < 1$, the average amount of inspection per lot will vary between the sample size n and the lot size N. If the lot is of quality p and the probability of lot acceptance is P_a, then the **average total inspection** per lot will be

$$\text{ATI} = n + (1 - P_a)(N - n) \tag{15.6}$$

To illustrate the use of equation (15.6), consider our previous example with $N = 10{,}000$, $n = 89$, $c = 2$, and $p = 0.01$. Then, since $P_a = 0.9397$, we have

$$\begin{aligned}\text{ATI} &= n + (1 - P_a)(N - n) \\ &= 89 + (1 - 0.9397)(10{,}000 - 89) \\ &= 687\end{aligned}$$



■ **FIGURE 15.12** Average total inspection (ATI) curves for sampling plan $n = 89$, $c = 2$, for lot sizes of 1000, 5000, and 10,000.

Remember that this is an average number of units inspected over *many* lots with fraction defective $p = 0.01$.

It is possible to draw a curve of average total inspection as a function of lot quality. Average total inspection curves for the sampling plan $n = 89$, $c = 2$, for lot sizes of 1000, 5000, and 10,000, are shown in Fig. 15.12.

The AOQL of a rectifying inspection plan is a very important characteristic. It is possible to design rectifying inspection programs that have specified values of AOQL. However, specification of the AOQL is not sufficient to determine a unique sampling plan. Therefore, it is relatively common practice to choose the sampling plan that has a specified AOQL and, in addition, yields a minimum ATI at a particular level of lot quality. The level of lot quality usually chosen is the most likely level of incoming lot quality, which is generally called the *process average*. The procedure for generating these plans is relatively straightforward and is illustrated in Duncan (1986). Generally, it is unnecessary to go through this procedure, because tables of sampling plans that minimize ATI for a given AOQL and a specified process average p have been developed by Dodge and Romig. We describe the use of these tables in Section 15.5.

It is also possible to design a rectifying inspection program that gives a specified level of protection at the LTPD point and that minimizes the average total inspection for a specified process average p. The Dodge–Romig sampling inspection tables also provide these LTPD plans. Section 15.5 discusses the use of the Dodge–Romig tables to find plans that offer specified LTPD protection.

15.3 Double, Multiple, and Sequential Sampling

A number of extensions of single-sampling plans for attributes are useful. These include **double-sampling plans, multiple-sampling plans,** and **sequential-sampling plans.** This section discusses the design and application of these sampling plans.

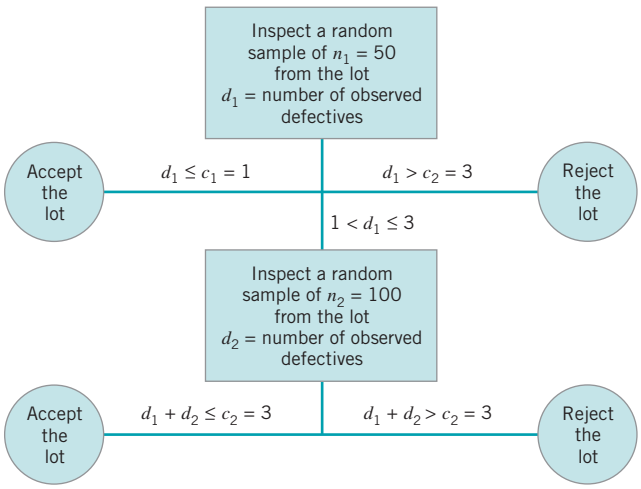
15.3.1 Double-Sampling Plans

A double-sampling plan is a procedure in which, under certain circumstances, a second sample is required before the lot can be sentenced. A double-sampling plan is defined by four parameters:[2]

n_1 = sample size on the first sample
c_1 = acceptance number of the first sample
n_2 = sample size on the second sample
c_2 = acceptance number for both sample

As an example, suppose $n_1 = 50$, $c_1 = 1$, $n_2 = 100$, and $c_2 = 3$. Thus, a random sample of $n_1 = 50$ items is selected from the lot, and the number of defectives in the sample, d_1, is observed. If $d_1 \le c_1 = 1$, the lot is accepted on the first sample. If $d_1 > c_2 = 3$, the lot is rejected on the first sample. If $c_1 < d_1 \le c_2$, a second random sample of size $n_2 = 100$ is drawn from the lot, and the number of defectives in this second sample, d_2, is observed. Now the combined number of observed defectives from both the first and second sample, $d_1 + d_2$, is used to determine the lot sentence. If $d_1 + d_2 \le c_2 = 3$, the lot is accepted. However, if $d_1 + d_2 > c_2 = 3$, the lot is rejected. The operation of this double-sampling plan is illustrated graphically in Fig. 15.13.

The principal advantage of a double-sampling plan with respect to single-sampling is that it may reduce the total amount of required inspection. Suppose that the first sample taken under a double-sampling plan is smaller than the sample that would be required using a single-sampling plan that offers the consumer the same protection. In all cases, then, in which a lot is accepted or rejected on the first sample, the cost of inspection will be lower for double-sampling than it would be for single-sampling. It is also possible to reject a lot without complete inspection of the second sample. (This is called *curtailment* on the second sample.) Consequently, the use of double-sampling can often result in lower total inspection costs. Furthermore, in some situations, a double-sampling plan has the psychological advantage of giving a lot a second chance. This may have some appeal to the supplier. However, there is no real advantage to double sampling in this regard, because single- and double-sampling plans can be chosen so that they have the same OC curves. Thus, both plans would offer the same risks of accepting or rejecting lots of specified quality.



■ **FIGURE 15.13** Operation of the double-sampling plan, $n_1 = 50$, $c_1 = 1$, $n_2 = 100$, $c_2 = 3$.

[2]Some authors prefer the notation n_1, Ac_1, Re_1, n_2, Ac_2, $Re_2 = Ac_2 + 1$. Since the rejection number on the first sample Re_1 is not necessarily equal to Re_2, this gives some additional flexibility in designing double-sampling plans. MIL STD 105E and ANSI/ASQC Z1.4 currently use this notation. However, because assuming that $Re_1 = Re_2$ does not significantly affect the plans obtained, we have chosen to discuss this slightly simpler system.

Double-sampling has two potential disadvantages. First, unless curtailment is used on the second sample, under some circumstances double-sampling may require more total inspection than would be required in a single-sampling plan that offers the same protection. Thus, unless double-sampling is used carefully, its potential economic advantage may be lost. The second disadvantage of double sampling is that it is administratively more complex, which may increase the opportunity for the occurrence of inspection errors. Furthermore, there may be problems in storing and handling raw materials or component parts for which one sample has been taken, but that are awaiting a second sample before a final lot dispositioning decision can be made.

The OC Curve. The performance of a double-sampling plan can be conveniently summarized by means of its operating-characteristic (OC) curve. The OC curve for a double-sampling plan is somewhat more involved than the OC curve for single sampling. In this section, we describe the construction of type-B OC curves for double sampling. A double-sampling plan has a primary OC curve that gives the probability of acceptance as a function of lot or process quality. It also has supplementary OC curves that show the probability of lot acceptance and rejection on the first sample. The OC curve for the probability of rejection on the first sample is simply the OC curve for the single-sampling plan $n = n_1$ and $c = c_2$. Primary and supplementary OC curves for the plan $n_1 = 50$, $c_1 = 1$, $n_2 = 100$, $c_2 = 3$ are shown in Fig. 15.14.

We now illustrate the computation of the OC curve for the plan $n_1 = 50$, $c_1 = 1$, $n_2 = 100$, $c_2 = 3$. If P_a denotes the probability of acceptance on the combined samples, and P_a^{I} and P_a^{II} denote the probability of acceptance on the first and second samples, respectively, then

$$P_a = P_a^{\mathrm{I}} + P_a^{\mathrm{II}}$$

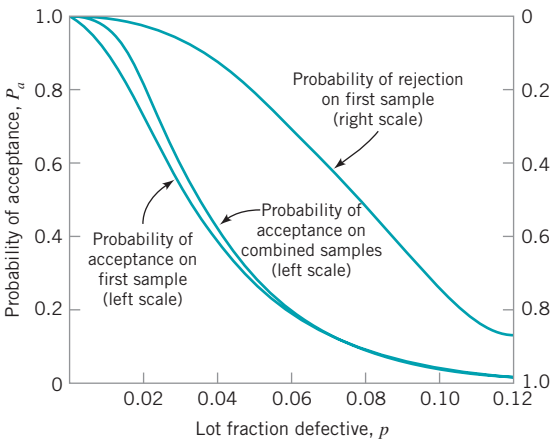
P_a^{I} is just the probability that we will observe $d_1 \le c_1 = 1$ defectives out of a random sample of $n_1 = 50$ items. Thus

$$P_a^{\mathrm{I}} = \sum_{d_1=0}^{1} \frac{50!}{d_1!(50-d_1)!} p^{d_1}(1-p)^{50-d_1}$$

If $p = 0.05$ is the fraction defective in the incoming lot, then

$$P_a^{\mathrm{I}} = \sum_{d_1=0}^{1} \frac{50!}{d_1!(50-d_1)!}(0.05)^{d_1}(0.95)^{50-d_1} = 0.279$$

To obtain the probability of acceptance on the second sample, we must list the number of ways the second sample can be obtained. A second sample is drawn *only* if there are two or three defectives on the first sample—that is, if $c_1 < d_1 \le c_2$.



■ **FIGURE 15.14** OC curves for the double-sampling plan, $n_1 = 50$, $c_1 = 1$, $n_2 = 100$, $c_2 = 3$.

1. $d_1 = 2$ *and* $d_2 = 0$ or 1; that is, we find two defectives on the first sample and one or less defectives on the second sample. The probability of this is

$$\begin{aligned} P\{d_1 = 2, d_2 \le 1\} &= P\{d_1 = 2\} \cdot P\{d_2 \le 1\} \\ &= \frac{50!}{2!48!}(0.05)^2(0.95)^{48} \sum_{d_2=0}^{1} \frac{100!}{d_2!(100-d_2)!}(0.05)^{d_2}(0.95)^{100-d_2} \\ &= (0.261)(0.037) = 0.0097 \end{aligned}$$

2. $d_1 = 3$ *and* $d_2 = 0$; that is, we find three defectives on the first sample and no defectives on the second sample. The probability of this is

$$\begin{aligned} P\{d_1 = 3, d_2 = 0\} &= P\{d_1 = 3\} \cdot P\{d_2 = 0\} \\ &= \frac{50!}{3!(47)!}(0.05)^3(0.95)^{47}\frac{100!}{0!100!}(0.05)^0(0.95)^{100} \\ &= (0.220)(0.0059) = 0.001 \end{aligned}$$

Thus, the probability of acceptance on the second sample is

$$\begin{aligned} P_a^{\text{II}} &= P\{d_1 = 2, d_2 \le 1\} + P\{d_1 = 3, d_2 = 0\} \\ &= 0.0097 + 0.001 = 0.0107 \end{aligned}$$

The probability of acceptance of a lot that has fraction defective $p = 0.05$ is therefore

$$\begin{aligned} P_a &= P_a^{\text{I}} + P_a^{\text{II}} \\ &= 0.279 + 0.0107 = 0.2897 \end{aligned}$$

Other points on the OC curve are calculated similarly.

The Average Sample Number Curve. The average sample number curve of a double-sampling plan is also usually of interest to the quality engineer. In single-sampling, the size of the sample inspected from the lot is always constant, whereas in double-sampling, the size of the sample selected depends on whether or not the second sample is necessary. The probability of drawing a second sample varies with the fraction defective in the incoming lot. With complete inspection of the second sample, the average sample size in double-sampling is equal to the size of the first sample times the probability that there will only be one sample, plus the size of the combined samples times the probability that a second sample will be necessary. Therefore, a general formula for the average sample number in double-sampling, if we assume complete inspection of the second sample, is

$$\begin{aligned} \text{ASN} &= n_1 P_{\text{I}} + (n_1 + n_2)(1 - P_{\text{I}}) \\ &= n_1 + n_2(1 - P_{\text{I}}) \end{aligned} \tag{15.7}$$

where P_{I} is the probability of making a lot-dispositioning decision on the *first* sample. This is

$$P_{\text{I}} = P\{\text{lot is accepted on the first sample}\} + P\{\text{lot is rejected on the first sample}\}$$

If equation (15.7) is evaluated for various values of lot fraction defective p, the plot of ASN versus p is called an **average sample number curve.**

In practice, inspection of the second sample is usually terminated and the lot rejected as soon as the number of observed defective items in the combined sample exceeds the second acceptance number c_2. This is referred to as curtailment of the second sample. The use of curtailed inspection lowers the average sample number required in double-sampling. It is

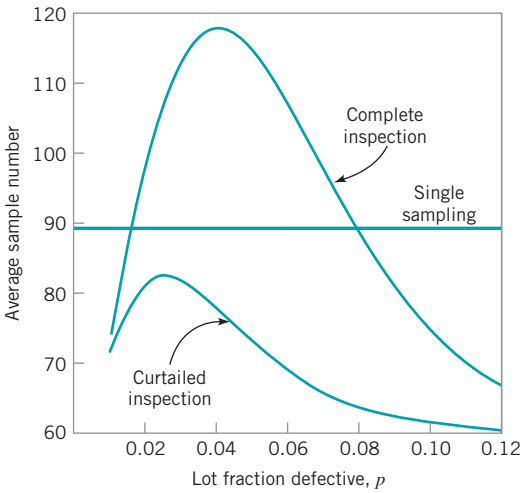
not recommended that curtailment be used in single-sampling, or in the first sample of double-sampling, because it is usually desirable to have complete inspection of a fixed sample size in order to secure an unbiased estimate of the quality of the material supplied by the supplier. If curtailed inspection is used in single-sampling or on the first sample of double-sampling, the estimate of lot or process fallout obtained from these data is biased. For instance, suppose that the acceptance number is one. If the first two items in the sample are defective, and the inspection process is curtailed, the estimate of lot or process fraction defective is 100%. Based on this information, even nonstatistically trained managers or engineers will be very reluctant to believe that the lot is really 100% defective.

The ASN curve formula for a double-sampling plan with curtailment on the second sample is

$$\text{ASN} = n_1 + \sum_{j=c_1+1}^{c_2} P(n_1, j)\left[n_2 P_L(n_2, c_2 - j) + \frac{c_2 - j + 1}{p} P_M(n_2 + 1, c_2 - j + 2)\right] \tag{15.8}$$

In equation (15.8), $P(n_1, j)$ is the probability of observing exactly j defectives in a sample of size n_1, $P_L(n_2, c_2 - j)$ is the probability of observing $c_2 - j$ or fewer defectives in a sample of size n_2, and $P_M(n_2 + 1, c_2 - j + 2)$ is the probability of observing $c_2 - j + 2$ defectives in a sample of size $n_2 + 1$.

Figure 15.15 compares the average sample number curves for complete and curtailed inspection for the double-sampling plan $n_1 = 60$, $c_1 = 2$, $n_2 = 120$, $c_3 = 3$, and the average sample number that would be used in single-sampling with $n = 89$, $c = 2$. Obviously, the sample size in the single-sampling plan is always constant. This double-sampling plan has been selected because it has an OC curve that is nearly identical to the OC curve for the single-sampling plan. That is, both plans offer equivalent protection to the producer and the consumer. Note from inspection of Fig. 15.15 that the ASN curve for double-sampling without curtailment on the second sample is not lower than the sample size used in single-sampling throughout the entire range of lot fraction defective. If lots are of very good quality, they will usually be accepted on the first sample, whereas if lots are of very bad quality, they will usually be rejected on the first sample. This gives an ASN for double sampling that is smaller than the sample size used



■ **FIGURE 15.15** Average sample number curves for single and double sampling.

in single-sampling for lots that are either very good or very bad. However, if lots are of intermediate quality, the second sample will be required in a large number of cases before a lot disposition decision can be made. In this range of lot quality, the ASN performance of double-sampling is worse than single-sampling.

This example points out that it is important to use double-sampling very carefully. Unless care is exercised to ensure that lot or process quality is in the range where double-sampling is most effective, then the economic advantages of double-sampling relative to single-sampling may be lost. It is a good idea to maintain a running estimate of the supplier's lot or process fallout, so that if it shifts into a range where double-sampling is not economically effective, a change to single-sampling (or some other appropriate strategy) can be made. Another way to do this would be to record the proportion of times that the second sample is required in order to make a decision.

Figure 15.15 also shows the ASN curve using curtailment on the second sample. Note that if curtailment is used, the average sample number curve for double sampling always lies below the sample size used in single-sampling.

Designing Double-Sampling Plans with Specified $p_1, 1-\alpha, p_2$, and β. It is often necessary to be able to design a double-sampling plan that has a specified OC curve. Let $(p_1, 1-\alpha)$ and (p_2, β) be the two points of interest on the OC curve. If, in addition, we impose another relationship on the parameters of the sampling plan, then a simple procedure can be used to obtain such plans. The most common constraint is to require that n_2 is a multiple of n_1. Refer to Duncan (1986) for a discussion of these techniques.

Rectifying Inspection. When rectifying inspection is performed with double-sampling, the AOQ curve is given by

$$\text{AOQ} = \frac{\left[P_a^{\text{I}}(N-n_1) + P_a^{\text{II}}(N-n_1-n_2)\right]p}{N} \tag{15.9}$$

assuming that all defective items discovered, either in sampling or 100% inspection, are replaced with good ones. The average total inspection curve is given by

$$\text{ATI} = n_1 P_a^{\text{I}} + (n_1+n_2)P_a^{\text{II}} + N(1-P_a) \tag{15.10}$$

Remember that $P_a = P_a^{\text{I}} + P_a^{\text{II}}$ is the probability of final lot acceptance and that the acceptance probabilities depend on the level of lot or process quality p.

15.3.2 Multiple-Sampling Plans

A multiple-sampling plan is an extension of double-sampling in that more than two samples can be required to sentence a lot. An example of a multiple-sampling plan with five stages follows.

Cumulative-Sample Size	Acceptance Number	Rejection Number
20	0	3
40	1	4
60	3	5
80	5	7
100	8	9

This plan will operate as follows: If, at the completion of any stage of sampling, the number of defective items is less than or equal to the acceptance number, the lot is accepted. If, during any stage, the number of defective items equals or exceeds the rejection number, the lot is rejected; otherwise the next sample is taken. The multiple-sampling procedure continues until the fifth sample is taken, at which time a lot disposition decision must be made. The first sample is usually inspected 100%, although subsequent samples are usually subject to curtailment.

The construction of OC curves for multiple-sampling is a straightforward extension of the approach used in double-sampling. Similarly, it is also possible to compute the average sample number curve of multiple-sampling plans. One may also design a multiple-sampling plan for specified values of p_1, $1 - \alpha$, p_2, and β. For an extensive discussion of these techniques, see Duncan (1986).

The principal advantage of multiple-sampling plans is that the samples required at each stage are usually smaller than those in single- or-double-sampling; thus, some economic efficiency is connected with the use of the procedure. However, multiple-sampling is much more complex to administer.

15.3.3 Sequential-Sampling Plans

Sequential-sampling is an extension of the double-sampling and multiple-sampling concept. In sequential-sampling, we take a sequence of samples from the lot and allow the number of samples to be determined entirely by the results of the sampling process. In practice, sequential-sampling can theoretically continue indefinitely, until the lot is inspected 100%. In practice, sequential-sampling plans are usually truncated after the number inspected is equal to three times the number that would have been inspected using a corresponding single-sampling plan. If the sample size selected at each stage is greater than one, the process is usually called *group* sequential-sampling. If the sample size inspected at each stage is one, the procedure is usually called **item-by-item sequential-sampling.**

Item-by-item sequential-sampling is based on the sequential probability ratio test (SPRT), developed by Wald (1947). The operation of an item-by-item sequential-sampling plan is illustrated in Fig. 15.16. The cumulative observed number of defectives is plotted on the chart. For each point, the abscissa is the total number of items selected up to that time, and the ordinate is the total number of observed defectives. If the plotted points stay within the boundaries of the acceptance and rejection lines, another sample must be drawn. As soon as a point falls on or above the upper line, the lot is rejected. When a sample point falls on or below the lower line, the lot is accepted. The equations for the two limit lines for specified values of p_1, $1 - \alpha$, p_2, and β are

$$X_A = -h_1 + sn \qquad \text{(acceptance line)} \tag{15.11a}$$

$$X_R = h_2 + sn \qquad \text{(rejection line)} \tag{15.11b}$$

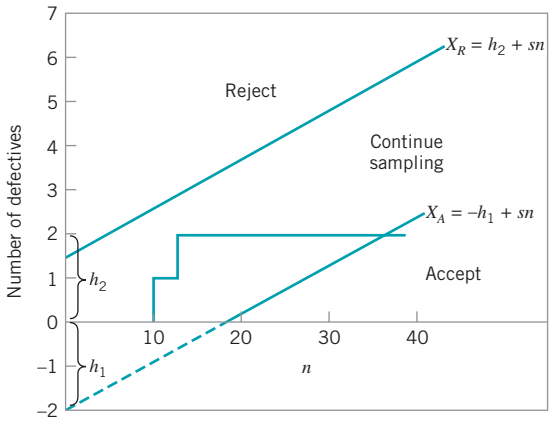
where

$$h_1 = \left(\log \frac{1-\alpha}{\beta}\right) \Big/ k \tag{15.12}$$

$$h_2 = \left(\log \frac{1-\beta}{\alpha}\right) \Big/ k \tag{15.13}$$

$$k = \log \frac{p_2(1-p_1)}{p_1(1-p_2)} \tag{15.14}$$

$$s = \log\left[(1-p_1)/(1-p_2)\right] \Big/ k \tag{15.15}$$



■ **FIGURE 15.16** Graphical performance of sequential sampling.

To illustrate the use of these equations, suppose we wish to find a sequential-sampling plan for which $p_1 = 0.01$, $\alpha = 0.05$, $p_2 = 0.06$, and $\beta = 0.10$. Thus,

$$k = \log \frac{p_2(1-p_1)}{p_1(1-p_2)}$$

$$= \log \frac{(0.06)(0.99)}{(0.01)(0.94)} = 0.80066$$

$$h_1 = \left(\log \frac{1-\alpha}{\beta}\right) \Big/ k$$

$$= \left(\log \frac{0.95}{0.10}\right) \Big/ 0.80066 = 1.22$$

$$h_2 = \left(\log \frac{1-\beta}{\alpha}\right) \Big/ k$$

$$= \left(\log \frac{0.90}{0.05}\right) \Big/ 0.80066 = 1.57$$

$$s = \log\left[(1-p_1)/(1-p_2)\right] \Big/ k$$

$$= \left[\log(0.99/0.94)\right] \Big/ 0.80066 = 0.028$$

Therefore, the limit lines are

$$X_A = -1.22 + 0.028n \qquad \text{(accept)}$$

and

$$X_R = 1.57 + 0.028n \qquad \text{(reject)}$$

Instead of using a graph to determine the lot disposition, the sequential-sampling plan can be displayed in a table such as Table 15.3. The entries in the table are found by substituting values of n into the equations for the acceptance and rejection lines and calculating acceptance and rejection numbers. For example, the calculations for $n = 45$ are

$$X_A = -1.22 + 0.028n$$

$$= -1.22 + 0.028(45) = 0.04 \qquad \text{(accept)}$$

$$X_R = 1.57 + 0.028n$$

$$= 1.57 + 0.028(45) = 2.83 \qquad \text{(reject)}$$

■ TABLE 15.3

Item-by-Item Sequential-Sampling Plan $p_1 = 0.01$, $\alpha = 0.05$, $p_2 = 0.06$, $\beta = 0.10$ (first 46 units only)

Number of Items Inspected, n	Acceptance Number	Rejection Number	Number of Items Inspected, n	Acceptance Number	Rejection Number
1	a	b	24	a	3
2	a	2	25	a	3
3	a	2	26	a	3
4	a	2	27	a	3
5	a	2	28	a	3
6	a	2	29	a	3
7	a	2	30	a	3
8	a	2	31	a	3
9	a	2	32	a	3
10	a	2	33	a	3
11	a	2	34	a	3
12	a	2	35	a	3
13	a	2	36	a	3
14	a	2	37	a	3
15	a	2	38	a	3
16	a	3	39	a	3
17	a	3	40	a	3
18	a	3	41	a	3
19	a	3	42	a	3
20	a	3	43	a	3
21	a	3	44	0	3
22	a	3	45	0	3
23	a	3	46	0	3

"a" means acceptance not possible.
"b" means rejection not possible.

Acceptance and rejection numbers must be integers, so the acceptance number is the next integer less than or equal to X_A, and the rejection number is the next integer greater than or equal to X_R. Thus, for $n = 45$, the acceptance number is 0 and the rejection number is 3. Note that the lot cannot be accepted until at least 44 units have been tested. Table 15.3 shows only the first 46 units. Usually, the plan would be truncated after the inspection of 267 units, which is three times the sample size required for an equivalent single-sampling plan.

The OC Curve and ASN Curve for Sequential Sampling. The OC curve for sequential-sampling can be easily obtained. Two points on the curve are $(p_1, 1 - \alpha)$ and (p_2, β). A third point, near the middle of the curve, is $p = s$ and $P_a = h_2/(h_1 + h_2)$.

The average sample number taken under sequential-sampling is

$$\text{ASN} = P_a\left(\frac{A}{C}\right) + (1 - P_a)\frac{B}{C} \tag{15.16}$$

where

$$A = \log\frac{\beta}{1-\alpha}$$

$$B = \log\frac{1-\beta}{\alpha}$$

and

$$C = p\log\left(\frac{p_2}{p_1}\right) + (1-p)\log\left(\frac{1-p_2}{1-p_1}\right)$$

Rectifying Inspection. The average outgoing quality (AOQ) for sequential-sampling is given approximately by

$$\text{AOQ} \simeq P_a p \tag{15.17}$$

The average total inspection is also easily obtained. Note that the amount of sampling is *A/C* when a lot is accepted and *N* when it is rejected. Therefore, the average total inspection is

$$\text{ATI} = P_a\left(\frac{A}{C}\right) + (1-P_a)N \tag{15.18}$$

15.4 Military Standard 105E (ANSI/ASQC Z1.4, ISO 2859)

15.4.1 Description of the Standard

Standard sampling procedures for inspection by attributes were developed during World War II. MIL STD 105E is the most widely used acceptance-sampling system for attributes in the world today. The original version of the standard, MIL STD 105A, was issued in 1950. Since then, there have been four revisions; the latest version, MIL STD 105E, was issued in 1989.

The sampling plans discussed in previous sections of this chapter are individual sampling plans. A sampling scheme is an overall strategy specifying the way in which sampling plans are to be used. MIL STD 105E is a collection of sampling schemes; therefore, it is an acceptance-sampling system. Our discussion will focus primarily on MIL STD 105E; however, there is a derivative civilian standard, ANSI/ASQC Z1.4, which is quite similar to the military standard. The standard was also adopted by the International Organization for Standardization as ISO 2859.

The standard provides for three types of sampling: single-sampling, double-sampling, and multiple-sampling. For each type of sampling plan, a provision is made for either normal inspection, tightened inspection, or reduced inspection. Normal inspection is used at the start of the inspection activity. Tightened inspection is instituted when the supplier's recent quality history has deteriorated. Acceptance requirements for lots under tightened inspection are more stringent than under normal inspection. Reduced inspection is instituted when the supplier's recent quality history has been exceptionally good. The sample size generally used under reduced inspection is less than that under normal inspection.

The primary focal point of MIL STD 105E is the acceptable quality level (AQL). The standard is indexed with respect to a series of AQLs. When the standard is used for percent defective plans, the AQLs range from 0.10% to 10%. For defects per units plans, there are an additional ten AQLs running up to 1000 defects per 100 units. It should be noted that for the smaller AQL levels, the same sampling plan can be used to control either a fraction defective or a number of defects per unit. The AQLs are arranged in a progression, each AQL being approximately 1.585 times the preceding one.

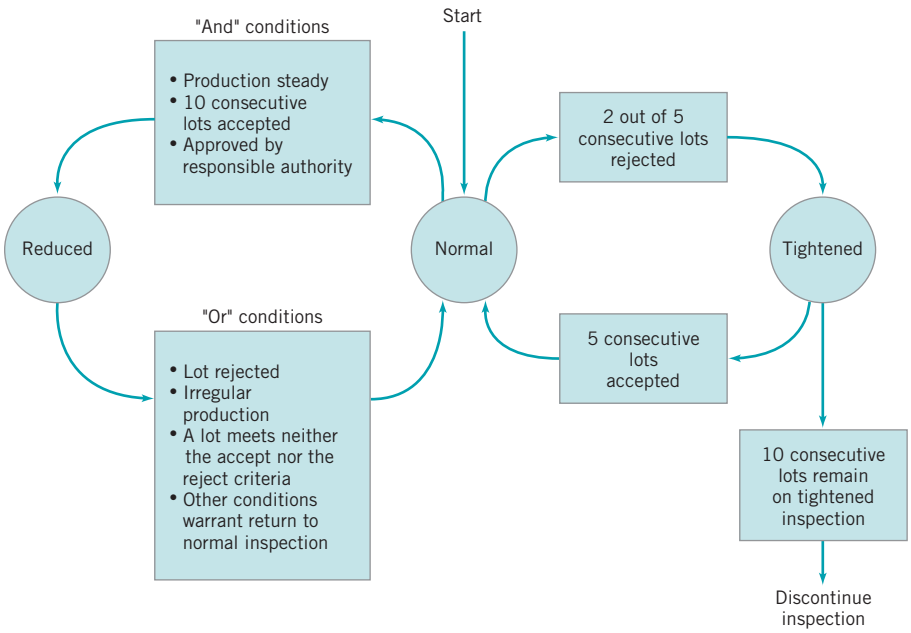
The AQL is generally specified in the contract or by the authority responsible for sampling. Different AQLs may be designated for different types of defects. For example, the standard differentiates critical defects, major defects, and minor defects. It is relatively common practice to choose an AQL of 1% for major defects and an AQL of 2.5% for minor defects. No critical defects would be acceptable.

The sample size used in MIL STD 105E is determined by the lot size and by the choice of inspection level. Three general levels of inspection are provided. Level II is designated as

normal. Level I requires about one-half the amount of inspection as Level II and may be used when less discrimination is needed. Level III requires about twice as much inspection as Level II and should be used when more discrimination is needed. There are also four special inspection levels, S-1, S-2, S-3, and S-4. The special inspection levels use very small samples, and should be employed only when the small sample sizes are necessary and when greater sampling risks can or must be tolerated.

For a specified AQL and inspection level and a given lot size, MIL STD 105E provides a normal sampling plan that is to be used as long as the supplier is producing the product at AQL quality or better. It also provides a procedure for switching to tightened and reduced inspection whenever there is an indication that the supplier's quality has changed. The switching procedures between normal, tightened, and reduced inspection are illustrated graphically in Fig. 15.17 and are described next.

1. **Normal to tightened.** When normal inspection is in effect, tightened inspection is instituted when two out of five consecutive lots have been rejected on original submission.
2. **Tightened to normal.** When tightened inspection is in effect, normal inspection is instituted when five consecutive lots or batches are accepted on original inspection.
3. **Normal to reduced.** When normal inspection is in effect, reduced inspection is instituted provided all four of the following conditions are satisfied.
   - **a.** The preceding ten lots have been on normal inspection, and none of the lots has been rejected on original inspection.
   - **b.** The total number of defectives in the samples from the preceding ten lots is less than or equal to the applicable limit number specified in the standard.
   - **c.** Production is at a steady rate; that is, no difficulty such as machine breakdowns, material shortages, or other problems have recently occurred.
   - **d.** Reduced inspection is considered desirable by the authority responsible for sampling.



■ **FIGURE 15.17** Switching rules for normal, tightened, and reduced inspection, MIL STD 105E.

4. **Reduced to normal.** When reduced inspection is in effect, normal inspection is instituted when any of the following four conditions occur.
   - **a.** A lot or batch is rejected.
   - **b.** When the sampling procedure terminates with neither acceptance nor rejection criteria having been met, the lot or batch will be accepted, but normal inspection is reinstituted starting with the next lot.
   - **c.** Production is irregular or delayed.
   - **d.** Other conditions warrant that normal inspection be instituted.
5. **Discontinuance of inspection.** In the event that ten consecutive lots remain on tightened inspection, inspection under the provision of MIL STD 105E should be terminated, and action should be taken at the supplier level to improve the quality of submitted lots.

15.4.2 Procedure

A step-by-step procedure for using MIL STD 105E is as follows:

1. Choose the AQL.
2. Choose the inspection level.
3. Determine the lot size.
4. Find the appropriate sample size code letter from Table 15.4.
5. Determine the appropriate type of sampling plan to use (single, double, multiple).
6. Enter the appropriate table to find the type of plan to be used.
7. Determine the corresponding normal and reduced inspection plans to be used when required.

Table 15.4 presents the sample size code letters for MIL STD 105E. Tables 15.5, 15.6, and 15.7 present the single-sampling plans for normal inspection, tightened inspection, and reduced inspection, respectively. The standard also contains tables for double-sampling plans and multiple-sampling plans for normal, tightened, and reduced inspection.

TABLE 15.4
Sample Size Code Letters (MIL STD 105E, Table 1)

	Special Inspection Levels				General Inspection Levels		
Lot or Batch Size	**S-1**	**S-2**	**S-3**	**S-4**	**I**	**II**	**III**
2 to 8	A	A	A	A	A	A	B
9 to 15	A	A	A	A	A	B	C
16 to 25	A	A	B	B	B	C	D
26 to 50	A	B	B	C	C	D	E
51 to 90	B	B	C	C	C	E	F
91 to 150	B	B	C	D	D	F	G
151 to 280	B	C	D	E	E	G	H
281 to 500	B	C	D	E	F	H	J
501 to 1200	C	C	E	F	G	J	K
1201 to 3200	C	D	E	G	H	K	L
3201 to 10000	C	D	F	G	J	L	M
10001 to 35000	C	D	F	H	K	M	N
35001 to 150000	D	E	G	J	L	N	P
150001 to 500000	D	E	G	J	M	P	Q
500001 and over	D	E	H	K	N	Q	R

TABLE 15.5

Master Table for Normal Inspection for Single-Sampling (U.S. Dept. of Defense MIL STD 105E, Table II-A)

Sample Size Code Letter	Sample Size	Acceptable Quality Levels (normal inspection)																																					
		0.010		0.015		0.025		0.040		0.065		0.10		0.15		0.25		0.40		0.65		1.0		1.5		2.5		4.0		6.5		10		15		25		40		65		100		150		250		400		650		1000	
		Ac	Re	Ac	Re	Ac	Re	Ac	Re	Ac	Re	Ac	Re	Ac	Re	Ac	Re	Ac	Re	Ac	Re	Ac	Re	Ac	Re	Ac	Re	Ac	Re	Ac	Re	Ac	Re	Ac	Re	Ac	Re	Ac	Re	Ac	Re	Ac	Re	Ac	Re	Ac	Re	Ac	Re	Ac	Re
A	2	⇩		⇩		⇩		⇩		⇩		⇩		⇩		⇩		⇩		⇩		⇩		⇩		⇩		⇩		0	1	⇩		1	2	2	3	3	4	5	6	7	8	10	11	14	15	21	22	30	31
B	3																											0	1	⇧		⇩		1	2	2	3	3	4	5	6	7	8	10	11	14	15	21	22	30	31	44	45
C	5																											⇩		⇧		1	2	2	3	3	4	5	6	7	8	10	11	14	15	21	22	30	31	44	45		
D	8																							0	1	⇧		⇩		1	2	2	3	3	4	5	6	7	8	10	11	14	15	21	22	30	31	44	45				
E	13																					0	1	⇧		⇧		1	2	2	3	3	4	5	6	7	8	10	11	14	15	21	22	30	31	44	45						
F	20																					⇧		⇧		2	3	3	4	5	6	7	8	10	11	14	15	21	22														
G	32																																																		
H	50																																																		
J	80																																																		
K	125																																																		
L	200																																																		
M	315																																																		
N	500																																																		
P	800																																																		
Q	1250																																																		
R	2000																																																		

⇩ = Use first sampling plan below arrow. If sample size equals, or exceeds, lot or batch size, do 100% inspection.

⇧ = Use first sampling plan above arrow.

Ac = Acceptance number.

Re = Rejection number.

TABLE 15.6

Master Table for Tightened Inspection—Single-Sampling (U.S. Dept. of Defense MIL STD 105E, Table II-B)

Sample Size Code Letter	Sample Size	Acceptable Quality Levels (tightened inspection)																														
		0.010	0.015	0.025	0.040	0.065	0.10	0.15	0.25	0.40	0.65	1.0	1.5	2.5	4.0	6.5	10	15	25	40	65	100	150	250	400	650	1000						
		Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re						
A	2	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	1 2	2 3	3 4	5 6	8 9	12 13	18 19	27 28						
B	3	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	0 1	⇩	⇩	1 2	2 3	3 4	5 6	8 9	12 13	18 19	27 28	41 42						
C	5	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	0 1	⇩	⇩	1 2	2 3	3 4	5 6	8 9	12 13	18 19	27 28	41 42	⇧						
D	8	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	0 1	⇩	⇩	1 2	2 3	3 4	5 6	8 9	12 13	18 19	27 28	41 42	⇧							
E	13	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	0 1	⇩	⇩	1 2	2 3	3 4	5 6	8 9	12 13	18 19	27 28	41 42	⇧								
F	20	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	0 1	⇩	⇩	1 2	2 3	3 4	5 6	8 9	12 13	18 19	⇧											
G	32	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	0 1	⇩	⇩	1 2	2 3	3 4	5 6	8 9	12 13	18 19	⇧												
H	50	⇩	⇩	⇩	⇩	⇩	⇩	⇩	⇩	0 1	⇩	⇩	1 2	2 3	3 4	5 6	8 9	12 13	18 19	⇧													
J	80	⇩	⇩	⇩	⇩	⇩	⇩	⇩	0 1	⇩	⇩	1 2	2 3	3 4	5 6	8 9	12 13	18 19	⇧														
K	125	⇩	⇩	⇩	⇩	⇩	⇩	0 1	⇩	⇩	1 2	2 3	3 4	5 6	8 9	12 13	18 19	⇧															
L	200	⇩	⇩	⇩	⇩	⇩	0 1	⇩	⇩	1 2	2 3	3 4	5 6	8 9	12 13	18 19	⇧																
M	315	⇩	⇩	⇩	⇩	0 1	⇩	⇩	1 2	2 3	3 4	5 6	8 9	12 13	18 19	⇧																	
N	500	⇩	⇩	⇩	0 1	⇩	⇩	1 2	2 3	3 4	5 6	8 9	12 11	18 19	⇧																		
P	800	⇩	⇩	0 1	⇩	⇩	1 2	2 3	3 4	5 6	8 9	12 13	18 19	⇧																			
Q	1250	⇩	0 1	⇩	⇩	1 2	2 3	3 4	5 6	8 9	12 13	18 19	⇧																				
R	2000	0 1	⇧	⇩	1 2	2 3	3 4	5 6	8 9	12 13	18 19	⇧																					
S	3150			1 2																													

⇩ = Use first sampling plan below arrow. If sample size equals, or exceeds, lot or batch size, do 100% inspection.

⇧ = Use first sampling plan above arrow.

Ac = Acceptance number.

Re = Rejection number.

TABLE 15.7

Master Table for Reduced Inspection—Single-Sampling (U.S. Dept. of Defense MIL STD 105E, Table II-C)

| Sample Size Code Letter | Sample Size | Acceptable Quality Levels (reduced inspection)† | | |

To illustrate the use of MIL STD 105E, suppose that a product is submitted in lots of size $N = 2000$. The acceptable quality level is 0.65%. We will use the standard to generate normal, tightened, and reduced single-sampling plans for this situation. For lots of size 2000 under general inspection level II, Table 15.4 indicates that the appropriate sample size code letter is K. Therefore, from Table 15.5, for single-sampling plans under normal inspection, the normal inspection plan is $n = 125$, $c = 2$. Table 15.6 indicates that the corresponding tightened inspection plan is $n = 125$, $c = 1$. Note that in switching from normal to tightened inspection, the sample size remains the same, but the acceptance number is reduced by one. This general strategy is used throughout MIL STD 105E for a transition to tightened inspection. If the normal inspection acceptance number is 1, 2, or 3, the acceptance number for the corresponding tightened inspection plan is reduced by one. If the normal inspection acceptance number is 5, 7, 10, or 14, the reduction in acceptance number for tightened inspection is two. For a normal acceptance number of 21, the reduction is three. Table 15.7 indicates that under reduced inspection, the sample size for this example would be $n = 50$, the acceptance number would be $c = 1$, and the rejection number would be $r = 3$. Thus, if two defectives were encountered, the lot would be accepted, but the next lot would be inspected under normal inspection.

In examining the tables, note that if a vertical arrow is encountered, the first sampling plan above or below the arrow should be used. When this occurs, the sample size code letter and the sample size change. For example, if a single-sampling plan is indexed by an AQL of 1.5% and a sample size code letter of F, the code letter changes to G and the sample size changes from 20 to 32.

15.4.3 Discussion

MIL STD 105E presents the OC curves for single-sampling plans. These are all type-B OC curves. The OC curves for the matching double- and multiple-sampling plans are roughly comparable with those for the corresponding single-sampling plans. Figure 15.18 presents an example of these curves for code letter K. The OC curves presented in the standard are for the intitial sampling plan only. They are not the OC curves for the overall inspection program,[3] including shifts to and from tightened or reduced inspection.

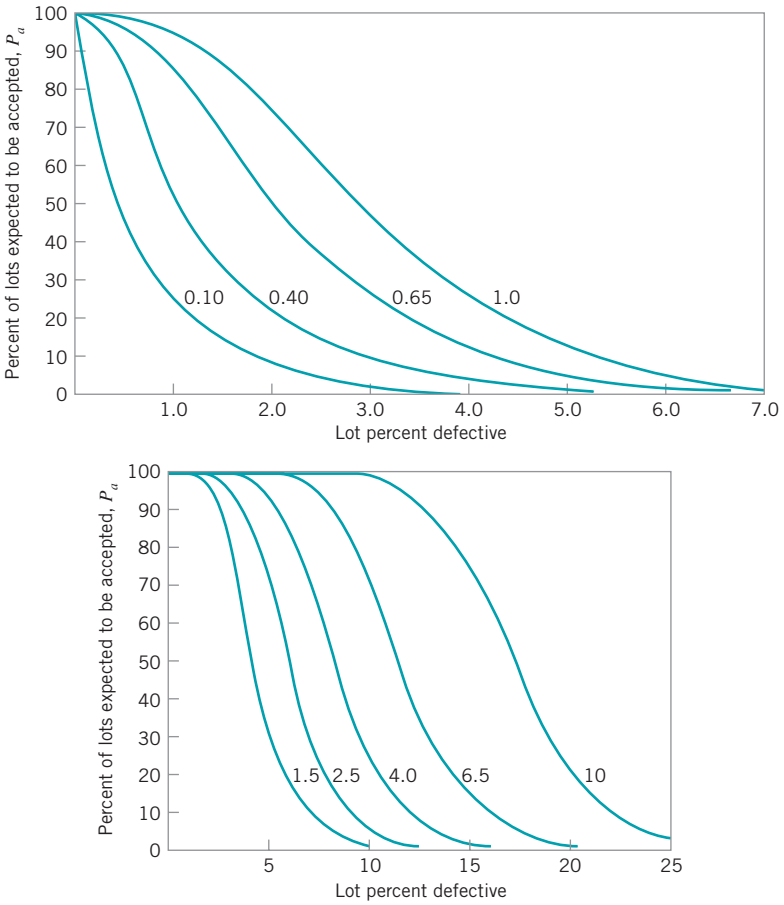
Average sample number curves for double and multiple sampling are given, assuming that no curtailment is used. These curves are useful in evaluating the average sample sizes that may be expected to occur under the various sampling plans for a given lot or process quality.

There are several points about MIL STD 105E that should be emphasized. These include the following. First, MIL STD 105E is AQL-oriented. It focuses attention on the producer's risk end of the OC curve. The only control over the discriminatory power of the sampling plan (i.e., the steepness of the OC curve) is through the choice of inspection level.

Second, the sample sizes selected for use in MIL STD 105E are 2, 3, 5, 8, 13, 20, 32, 50, 80, 125, 200, 315, 500, 800, 1250, and 2000. Thus, not all sample sizes are possible. Note that there are some rather significant gaps, such as between 125 and 200, and between 200 and 315.

Third, the sample sizes in MIL STD 105E are related to the lot sizes. To see the nature of this relationship, calculate the midpoint of each lot size range, and plot the logarithm of the sample size for that lot size range against the logarithm of the lot size range midpoint. Such a plot will follow roughly a straight line up to $n = 80$, and thereafter another straight

[3]ANSI/ASQC Z1.4 presents the scheme performance of the standard, giving scheme OC curves and the corresponding percentage points.



■ **FIGURE 15.18** OC curves for sample size code letter K, MIL STD 105E.

line with a shallower slope. Thus, the sample size will increase as the lot size increases. However, the ratio of sample size to lot size will decrease rapidly. This gives significant economy in inspection costs per unit when the supplier submits large lots. For a given AQL, the effect of this increase in sample size as the lot size increases is to increase the probability of acceptance for submitted lots of AQL quality. The probability of acceptance at a given AQL will vary with increasing sample size from about 0.91 to about 0.99. This feature of the standard was and still is subject to some controversy. The argument in favor of the approach in MIL STD 105E is that rejection of a large lot has more serious consequences for the supplier than rejection of a small lot, and if the probability of acceptance at the AQL increases with sample size, this reduces the risk of false rejection of a large lot. Furthermore, the large sample also gives a more discriminating OC curve, which means that the protection that the consumer receives against accepting an isolated bad lot will also be increased.

Fourth, the switching rules from normal to tightened inspection and from tightened to normal inspection are also subject to some criticism. In particular, some engineers dislike the switching rules because there is often a considerable amount of misswitching from normal to tightened or normal to reduced inspection when the process is actually producing lots of AQL quality. Also, there is a significant probability that production would even be discontinued, even though there has been no actual quality deterioration.

Fifth, a **flagrant** and **common abuse** of MIL STD 105E is failure to use the switching rules at all. When this is done, it results in ineffective and deceptive inspection and a substantial increase in the consumer's risk. It is not recommended that MIL STD 105E be implemented without use of the switching rules from normal to tightened and normal to reduced inspection.

A civilian standard, ANSI/ASQC Z1.4 or ISO 2859, is the counterpart of MIL STD 105E. It seems appropriate to conclude our discussion of MIL STD 105E with a comparison of the military and civilian standards. ANSI/ASQC Z1.4 was adopted in 1981 and differs from MIL STD 105E in the following five ways:

1. The terminology "nonconformity," "nonconformance," and "percent nonconforming" is used.
2. The switching rules were changed slightly to provide an option for reduced inspection without the use of limit numbers.
3. Several tables that show measures of scheme performance (*including* the switching rules) were introduced. Some of these performance measures include AOQL, limiting quality for which $P_a = 0.10$ and $P_a = 0.05$, ASN, and operating-characteristic curves.
4. A section was added describing proper use of individual sampling plans when extracted from the system.
5. A figure illustrating the switching rules was added.

These revisions modernize the terminology and emphasize the system concept of the civilian standard. All tables, numbers, and procedures used in MIL STD 105E are retained in ANSI/ASQC Z1.4 and ISO 2859.

15.5 The Dodge–Romig Sampling Plans

H. F. Dodge and H. G. Romig developed a set of sampling inspection tables for lot-by-lot inspection of product by attributes using two types of sampling plans: plans for lot tolerance percent defective (LTPD) protection and plans that provide a specified average outgoing quality limit (AOQL). For each of these approaches to sampling plan design, there are tables for single- and double-sampling.

Sampling plans that emphasize LTPD protection, such as the Dodge–Romig plans, are often preferred to AQL-oriented sampling plans, such as those in MIL STD 105E, particularly for critical components and parts. Many manufacturers believe that they have relied too much on AQLs in the past, and they are now emphasizing other measures of performance, such as defective parts per million (ppm). Consider the following:

AQL	Defective Parts per Million
10%	100,000
1%	10,000
0.1%	1,000
0.01%	100
0.001%	10
0.0001%	1

Thus, even very small AQLs imply large numbers of defective ppm. In complex products, the effect of this can be devastating. For example, suppose that a printed circuit

board contains 100 elements, each manufactured by a process operating at 0.5% defective. If the AQLs for these elements are 0.5% and if all elements on the printed circuit board must operate for the card to function properly, then the probability that a board works is

$$P(\text{function properly}) = (0.995)^{100} = 0.6058$$

Thus, there is an obvious need for sampling plans that emphasize LTPD protection, even when the process average fallout is low. The Dodge–Romig plans are often useful in these situations.

The Dodge–Romig AOQL plans are designed so that the average total inspection for a given AOQL and a specified process average p will be minimized. Similarly, the LTPD plans are designed so that the average total inspection is a minimum. This makes the Dodge–Romig plans very useful for in-plant inspection of semifinished product.

The Dodge–Romig plans apply only to programs that submit rejected lots to 100% inspection. Unless rectifying inspection is used, the AOQL concept is meaningless. Furthermore, to use the plans, we must know the process average—that is, the average fraction nonconforming of the incoming product. When a supplier is relatively new, we usually do not know its process fallout. Sometimes this may be estimated from a preliminary sample or from data provided by the supplier. Alternatively, the largest possible process average in the table can be used until enough information has been generated to provide a more accurate estimate of the supplier's process fallout. Obtaining a more accurate estimate of the incoming fraction nonconforming or process average will allow a more appropriate sampling plan to be adopted. It is not uncommon to find that sampling inspection begins with one plan, and after sufficient information is generated to reestimate the supplier's process fallout, a new plan is adopted. We discuss estimation of the process average in more detail in Section 15.5.3.

15.5.1 AOQL Plans

The Dodge–Romig (1959) tables give AOQL sampling plans for AOQL values of 0.1%, 0.25%, 0.5%, 0.75%, 1%, 1.5%, 2%, 2.5%, 3%, 4%, 5%, 7%, and 10%. For each of these AOQL values, six classes of values for the process average are specified. Tables are provided for both single- and double-sampling. These plans have been designed so that the average total inspection at the given AOQL and process average is approximately a minimum.

An example of the Dodge–Romig sampling plans is shown in Table 15.8.[4] To illustrate the use of the Dodge–Romig AOQL tables, suppose that we are inspecting LSI memory elements for a personal computer and that the elements are shipped in lots of size $N = 5000$. The supplier's process average fallout is 1% nonconforming. We wish to find a single-sampling plan with an AOQL = 3%. From Table 15.8, we find that the plan is

$$n = 65 \qquad c = 3$$

Table 15.8 also indicates that the LTPD for this sampling plan is 10.3%. This is the point on the OC curve for which $P_a = 0.10$. Therefore, the sampling plan $n = 65$, $c = 3$ gives an AOQL of 3% nonconforming and provides assurance that 90% of incoming lots that are as bad as 10.3% defective will be rejected. Assuming that incoming quality is equal to the process average and

[4]Tables 15.8 and 15.9 are adapted from H. F. Dodge and H. G. Romig, *Sampling Inspection Tables, Single and Double Sampling*, 2nd ed., John Wiley, New York, 1959, with the permission of the publisher.

TABLE 15.8

Dodge–Romig Inspection Table—Single-Sampling Plans for AOQL = 3.0%

	Process Average																	
	0–0.06%			0.07–0.60%			0.61–1.20%			1.21–1.80%			1.81–2.40%			2.41–3.00%		
			LTPD			LTPD			LTPD			LTPD			LTPD			LTPD
Lot Size	*n*	*c*	%	*n*	*c*	%	*n*	*c*	%	*n*	*c*	%	*n*	*c*	%	*n*	*c*	%
1–10	All	0	—	All	0	—	All	0	—	All	0	—	All	0	—	All	0	—
11–50	10	0	19.0	10	0	19.0	10	0	19.0	10	0	19.0	10	0	19.0	10	0	19.0
51–100	11	0	18.0	11	0	18.0	11	0	18.0	11	0	18.0	11	0	18.0	22	1	16.4
101–200	12	0	17.0	12	0	17.0	12	0	17.0	25	1	15.1	25	1	15.1	25	1	15.1
201–300	12	0	17.0	12	0	17.0	26	1	14.6	26	1	14.6	26	1	14.6	40	2	12.8
301–400	12	0	17.1	12	0	17.1	26	1	14.7	26	1	14.7	41	2	12.7	41	2	12.7
401–500	12	0	17.2	27	1	14.1	27	1	14.1	42	2	12.4	42	2	12.4	42	2	12.4
501–600	12	0	17.3	27	1	14.2	27	1	14.2	42	2	12.4	42	2	12.4	60	3	10.8
601–800	12	0	17.3	27	1	14.2	27	1	14.2	43	2	12.1	60	3	10.9	60	3	10.9
801–1000	12	0	17.4	27	1	14.2	44	2	11.8	44	2	11.8	60	3	11.0	80	4	9.8
1,001–2,000	12	0	17.5	28	1	13.8	45	2	11.7	65	3	10.2	80	4	9.8	100	5	9.1
2,001–3,000	12	0	17.5	28	1	13.8	45	2	11.7	65	3	10.2	100	5	9.1	140	7	8.2
3,001–4,000	12	0	17.5	28	1	13.8	65	3	10.3	85	4	9.5	125	6	8.4	165	8	7.8
4,001–5,000	28	1	13.8	28	1	13.8	65	3	10.3	85	4	9.5	125	6	8.4	210	10	7.4
5,001–7,000	28	1	13.8	45	2	11.8	65	3	10.3	105	5	8.8	145	7	8.1	235	11	7.1
7,001–10,000	28	1	13.9	46	2	11.6	65	3	10.3	105	5	8.8	170	8	7.6	280	13	6.8
10,001–20,000	28	1	13.9	46	2	11.7	85	4	9.5	125	6	8.4	215	10	7.2	380	17	6.2
20,001–50,000	28	1	13.9	65	3	10.3	105	5	8.8	170	8	7.6	310	14	6.5	560	24	5.7
50,001–100,000	28	1	13.9	65	3	10.3	125	6	8.4	215	10	7.2	385	17	6.2	690	29	5.4

TABLE 15.9

Dodge–Romig Single-Sampling Table for Lot Tolerance Percent Defective (LTPD) = 1.0%

	Process Average																			
	0–0.01%				0.011%–0.10%				0.11–0.20%				0.21–0.30%				0.31–0.40%				0.41–0.50%			
				AOQL				AOQL				AOQL				AOQL				AOQL				AOQL
Lot Size	*n*	*c*	%		*n*	*c*	%		*n*	*c*	%		*n*	*c*	%		*n*	*c*	%		*n*	*c*	%	
1–120	All	0	0		All	0	0		All	0	0		All	0	0		All	0	0		All	0	0	
121–150	120	0	0.06		120	0	0.06		120	0	0.06		120	0	0.06		120	0	0.06		120	0	0	0.06
151–200	140	0	0.08		140	0	0.08		140	0	0.08		140	0	0.08		140	0	0.08		140	0	0	0.08
201–300	165	0	0.10		165	0	0.10		165	0	0.10		165	0	0.10		165	0	0.10		165	0	0	0.10
301–400	175	0	0.12		175	0	0.12		175	0	0.12		175	0	0.12		175	0	0.12		175	0	0	0.12
401–500	180	0	0.13		180	0	0.13		180	0	0.13		180	0	0.13		180	0	0.13		180	0	0	0.13
501–600	190	0	0.13		190	0	0.13		190	0	0.13		190	0	0.13		190	0	0.13		305	1	0	0.14
601–800	200	0	0.14		200	0	0.14		200	0	0.14		330	1	0.15		330	1	0.15		330	1	0	0.15
801–1000	205	0	0.14		205	0	0.14		205	0	0.14		335	1	0.17		335	1	0.17		335	1	0	0.17
1,001–2,000	220	0	0.15		220	0	0.15		360	1	0.19		490	2	0.21		490	2	0.21		610	3	0	0.22
2,001–3,000	220	0	0.15		375	1	0.20		505	2	0.23		630	3	0.24		745	4	0.26		870	5	0	0.26
3,001–4,000	225	0	0.15		380	1	0.20		510	2	0.23		645	3	0.25		880	5	0.28		1,000	6	0	0.29
4,001–5,000	225	0	0.16		380	1	0.20		520	2	0.24		770	4	0.28		895	5	0.29		1,120	7	0	0.31
5,001–7,000	230	0	0.16		385	1	0.21		655	3	0.27		780	4	0.29		1,020	6	0.32		1,260	8	0	0.34
7,001–10,000	230	0	0.16		520	2	0.25		660	3	0.28		910	5	0.32		1,150	7	0.34		1,500	10	0	0.37
10,001–20,000	390	1	0.21		525	2	0.26		785	4	0.31		1,040	6	0.35		1,400	9	0.39		1,980	14	0	0.43
20,001–50,000	390	1	0.21		530	2	0.26		920	5	0.34		1,300	8	0.39		1,890	13	0.44		2,570	19	0	0.48
50,001–100,000	390	1	0.21		670	3	0.29		1,040	6	0.36		1,420	9	0.41		2,120	15	0.47		3,150	23	0	0.50

that the probability of lot acceptance at this level of quality is $P_a = 0.9957$, we find that the average total inspection for this plan is

$$\begin{aligned} \text{ATI} &= n \quad 1 - P_a \quad N - n \\ &= 65 \quad 1 - 0.9957 \quad 5000 - 65 \quad = 86.22 \end{aligned}$$

Thus, we will inspect approximately 86 units, on the average, in order to sentence a lot.

15.5.2 LTPD Plans

The Dodge–Romig LTPD tables are designed so that the probability of lot acceptance at the LTPD is 0.1. Tables are provided for LTPD values of 0.5%, 1%, 2%, 3%, 4%, 5%, 7%, and 10%. Table 15.9 for an LTPD of 1% is representative of these Dodge–Romig tables.

To illustrate the use of these tables, suppose that LSI memory elements for a personal computer are shipped from the supplier in lots of size $N = 5000$. The supplier's process average fallout is 0.25% nonconforming, and we wish to use a single-sampling plan with an LTPD of 1%. From inspection of Table 15-9, the sampling plan that should be used is

$$n = 770 \qquad c = 4$$

If we assume that rejected lots are screened 100% and that defective items are replaced with good ones, the AOQL for this plan is approximately 0.28%.

Note from inspection of the Dodge–Romig LTPD tables that values of the process average cover the interval from zero to one-half the LTPD. Provision for larger process averages is unnecessary, since 100% inspection is more economically efficient than inspection-sampling when the process average exceeds one-half the desired LTPD.

15.5.3 Estimation of Process Average

As we have observed, selection of a Dodge–Romig plan depends on knowledge of the supplier's process average fallout or percent nonconforming. An estimate of the process average can be obtained using a fraction defective control chart, based on the first 25 lots submitted by the supplier. If double-sampling is used, only the results from the first sample should be included in the computations. Any lot fraction defective that exceeds the upper control limit will be discarded, provided it has an assignable cause, and a new process average is calculated. Until results from 25 lots have been accumulated, the recommended procedure is to use the largest process average in the appropriate table.

Important Terms and Concepts

100% inspection
Acceptable quality level (AQL)
Acceptance-sampling plan
ANSI/ASQC Z1.4, ISO 2859
AOQL plans
Attributes data
Average outgoing quality
Average outgoing quality limit
Average sample number curve
Average total inspection
Dodge–Romig sampling plans
Double-sampling plan
Ideal OC curve
Lot disposition actions
Lot sentencing
Lot tolerance percent defective (LTPD)
LTPD plans
MIL STD 105E
Multiple-sampling plan
Normal, tightened, and reduced inspection
Operating-characteristic (OC) curve
Random sampling
Rectifying inspection
Sample size code letters

Sequential-sampling plan
Single-sampling plan
Switching rules in MIL STD 105E
Type-A and Type-B OC curves
Variables data

Exercises



The Student Resource Manual presents comprehensive annotated solutions to the odd-numbered exercises included in the Answers to Selected Exercises section in the back of this book.

15.1. Draw the type-B OC curve for the single-sampling plan $n = 50$, $c = 1$.

15.2. Draw the type-B OC curve for the single-sampling plan $n = 100$, $c = 2$.

15.3. Suppose that a product is shipped in lots of size $N = 5000$. The receiving inspection procedure used is single-sampling with $n = 50$ and $c = 1$.
(a) Draw the type-A OC curve for the plan.
(b) Draw the type-B OC curve for this plan and compare it to the type-A OC curve found in part (a).
(c) Which curve is appropriate for this situation?

15.4. Find a single-sampling plan for which $p_1 = 0.01$, $\alpha = 0.05$, $p_2 = 0.10$, and $\beta = 0.10$.

15.5. Find a single-sampling plan for which $p_1 = 0.05$, $\alpha = 0.05$, $p_2 = 0.15$, and $\beta = 0.10$.

15.6. Find a single-sampling plan for which $p_1 = 0.02$, $\alpha = 0.01$, $p_2 = 0.06$, and $\beta = 0.10$.

15.7. A company uses the following acceptance-sampling procedure. A sample equal to 10% of the lot is taken. If 2% or less of the items in the sample are defective, the lot is accepted; otherwise, it is rejected. If submitted lots vary in size from 5000 to 10,000 units, what can you say about the protection by this plan? If 0.05 is the desired LTPD, does this scheme offer reasonable protection to the consumer?

15.8. A company uses a sample size equal to the square root of the lot size. If 1% or less of the items in the sample are defective, the lot is accepted; otherwise, it is rejected. Submitted lots vary in size from 1000 to 5000 units. Comment on the effectiveness of this procedure.

15.9. Consider the single-sampling plan found in Exercise 15-4. Suppose that lots of $N = 2000$ are submitted. Draw the ATI curve for this plan. Draw the AOQ curve and find the AOQL.

15.10. Suppose that a single-sampling plan with $n = 150$ and $c = 2$ is being used for receiving inspection where the supplier ships the product in lots of size $N = 3000$.
(a) Draw the OC curve for this plan.
(b) Draw the AOQ curve and find the AOQL.
(c) Draw the ATI curve for this plan.

15.11. Suppose that a supplier ships components in lots of size 5000. A single-sampling plan with $n = 50$ and $c = 2$ is being used for receiving inspection. Rejected lots are screened, and all defective items are reworked and returned to the lot.
(a) Draw the OC curve for this plan.
(b) Find the level of lot quality that will be rejected 90% of the time.
(c) Management has objected to the use of the above sampling procedure and wants to use a plan with an acceptance number $c = 0$, arguing that this is more consistent with their zero-defects program. What do you think of this?
(d) Design a single-sampling plan with $c = 0$ that will give a 0.90 probability of rejection of lots having the quality level found in part (b). Note that the two plans are now matched at the LTPD point. Draw the OC curve for this plan and compare it to the one for $n = 50$, $c = 2$ in part (a).
(e) Suppose that incoming lots are 0.5% nonconforming. What is the probability of rejecting these lots under both plans? Calculate the ATI at this point for both plans. Which plan do you prefer? Why?

15.12. Draw the primary and supplementary OC curves for a double-sampling plan with $n_1 = 50$, $c_1 = 2$, $n_2 = 100$, $c_2 = 6$. If the incoming lots have fraction nonconforming $p = 0.05$, what is the probability of acceptance on the first sample? What is the probability of final acceptance? Calculate the probability of rejection on the first sample.

15.13. (a) Derive an item-by-item sequential-sampling plan for which $p_1 = 0.01$, $\alpha = 0.05$, $p_2 = 0.10$, and $\beta = 0.10$.
(b) Draw the OC curve for this plan.

14.14. (a) Derive an item-by-item sequential-sampling plan for which $p_1 = 0.02$, $\alpha = 0.05$, $p_2 = 0.15$, and $\beta = 0.10$.
(b) Draw the OC curve for this plan.

15.15. Consider rectifying inspection for single-sampling. Develop an AOQ equation assuming that all defective items are removed but *not* replaced with good ones.

15.16. A supplier ships a component in lots of size $N = 3000$. The AQL has been established for this product at 1%. Find the normal, tightened, and reduced

single-sampling plans for this situation from MIL STD 105E, assuming that general inspection level II is appropriate.

15.17. Repeat Exercise 15.16, using general inspection level I. Discuss the differences in the various sampling plans.

15.18. A product is supplied in lots of size $N = 10{,}000$. The AQL has been specified at 0.10%. Find the normal, tightened, and reduced single-sampling plans from MIL STD 105E, assuming general inspection level II.

15.19. MIL STD 105E is being used to inspect incoming lots of size $N = 5000$. Single sampling, general inspection level II, and an AQL of 0.65% are being used.

(a) Find the normal, tightened, and reduced inspection plans.

(b) Draw the OC curves of the normal, tightened, and reduced inspection plans on the same graph.

15.20. A product is shipped in lots of size $N = 2000$. Find a Dodge–Romig single-sampling plan for which the LTPD = 1%, assuming that the process average is 0.25% defective. Draw the OC curve and the ATI curve for this plan. What is the AOQL for this sampling plan?

15.21. We wish to find a single-sampling plan for a situation where lots are shipped from a supplier. The supplier's process operates at a fallout level of 0.50% defective. We want the AOQL from the inspection activity to be 3%.

(a) Find the appropriate Dodge–Romig plan.

(b) Draw the OC curve and the ATI curve for this plan. How much inspection will be necessary, on the average, if the supplier's process operates close to the average fallout level?

(c) What is the LTPD protection for this plan?

15.22. A supplier ships a product in lots of size $N = 8000$. We wish to have an AOQL of 3%, and we are going to use single sampling. We do not know the supplier's process fallout but suspect that it is at most 1% defective.

(a) Find the appropriate Dodge–Romig plan.

(b) Find the ATI for this plan, assuming that incoming lots are 1% defective.

(c) Suppose that our estimate of the supplier's process average is incorrect and that it is really 0.25% defective. What sampling plan should we have used? What reduction in ATI would have been realized if we had used the correct plan?